

Combinatorial Formulae and Inversion Theorems

**A Unified Coefficient Calculus for Stirling, Bell, Eulerian,
Bernoulli, Sheffer, Faà di Bruno, and Lagrange–Good Structures**

A proof-oriented synthesis of five draft manuscripts

September 2026

Abstract

This monograph merges five overlapping draft manuscripts into one coherent, deduplicated, proof-oriented account of combinatorial coefficient calculus and series inversion. Its central objects are signed and unsigned Stirling numbers, Lah numbers, Bell numbers and polynomials, Eulerian numbers, Bernoulli and Euler polynomials, Sheffer sequences, exponential Riordan arrays, Faà di Bruno formulae, and one- and multivariable Lagrange inversion. Supporting Wikipedia and MathWorld dossiers have been incorporated selectively, and additional standard references were checked for normalization, hypotheses, and provenance.

The treatment begins with formal power series, residues, factorial bases, finite differences, and the exponential formula. It then develops recurrences, generating functions, transforms, congruences, special-function sums, saddle-point asymptotics, restricted partitions, descent calculus, composition, and reversion. Newly unified chapters add Pochhammer and binomial-transform calculus; Bernoulli–Euler–Genocchi and Bernoulli–Cauchy families; complementary Bell numbers; Sheffer and Riordan theory; higher product and quotient rules; the Fréchet form of Faà di Bruno; Good’s determinant formula for multivariate inversion; Narayana refinements and the complete associahedral face vector; gamma, polygamma, and Barnes G ; Euler–Maclaurin and Darboux methods; partition-lattice Möbius inversion; Kepler reversion; and precision-doubling coefficient algorithms.

Definitions are connected by explicit changes of basis, labelled constructions, residue substitutions, and incidence-algebra inversion. Repeated material from the drafts has been consolidated rather than stacked, while distinct proofs and applications have been retained where they expose different mechanisms. Every new theorem introduced in the merger is accompanied by a proof, and statements that require analytic rather than merely formal hypotheses say so explicitly. A final concordance records the contribution of each draft and of the supporting topic files.

How to read this work

The chapters are arranged by mathematical dependence, not by the order of the source files. The two master ideas are:

1. a coefficient is often easiest to compute after changing to the basis in which the relevant operator is diagonal; and
2. an exponential generating function remembers the decomposition of a labelled object into components.

Stirling numbers implement the first idea. Bell polynomials implement the second. Eulerian numbers mediate between powers and shifted binomial coefficients. Faà di Bruno’s formula is the differential form of labelled composition, while Lagrange inversion is the coefficient form of implicit substitution.

A result stated for formal power series requires no convergence. When an analytic version is asserted, its domain hypotheses are written explicitly. Statements in the archive that report historical computations—for example, lists of Bell primes or verified exact modular periods—are distinguished from theorems and are accompanied by a certificate procedure rather than misrepresented as symbolic identities.

Editorial normalization

Three systematic repairs are used throughout.

1. Bell numbers are denoted B_n , complete exponential Bell polynomials by $\mathcal{B}_n$, partial Bell polynomials by $\mathcal{B}_{n,k}$, and Bernoulli numbers by β_n .
2. Type- B Eulerian numbers are indexed by descents in positions $0, 1, \dots, n-1$ with $\pi_0 = 0$. This removes the one-step shift in one formula in the archive.
3. The archived Moser–Wyman display contains unnamed quantities B, P_i, Q_i . We give the exact saddle-point scheme that defines every correction coefficient recursively and state the leading expansion in closed form; no undefined placeholder is retained.

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Notation

For $n \in \mathbb{N}$, write

$$x^{\overline{n}} = x(x-1) \cdots (x-n+1), \quad x^{\underline{n}} = x(x+1) \cdots (x+n-1),$$

with both expressions equal to 1 when $n = 0$. Coefficient extraction is denoted by $[z^n]F(z)$, and $\text{Res}_z F(z) dz$ means the coefficient of z^{-1} in a formal Laurent series. Our principal arrays are

$$s(n, k), \quad \begin{bmatrix} n \\ k \end{bmatrix} = (-1)^{n-k} s(n, k), \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle,$$

where $s(n, k)$ is the signed first-kind Stirling number, $\begin{bmatrix} n \\ k \end{bmatrix}$ its unsigned version, $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ the second-kind Stirling number, and $\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle$ the ordinary Eulerian number. Bell numbers are B_n ; exponential partial and complete Bell polynomials are $\mathcal{B}_{n,k}$ and $\mathcal{B}_n$.

Unless stated otherwise, sums over an impossible index range are zero, and $0^0 = 1$ in coefficient formulae. Bernoulli numbers use

$$\frac{t}{e^t - 1} = \sum_{n \geq 0} \beta_n \frac{t^n}{n!}, \quad \beta_1 = -\frac{1}{2}.$$

Part I

Coefficient Calculus and Labelled Structures

Chapter 1

Formal power series and coefficient extraction

1.1 Formal series as an algebraic environment

Let R be a commutative $\mathbb{Q}$ -algebra. The ring $R[[z]]$ consists of expressions $F(z) = \sum_{n \geq 0} f_n z^n$ with termwise addition and Cauchy multiplication. A series is invertible precisely when f_0 is invertible in R . If $G(0) = 0$, then the composition $F(G(z))$ is well-defined because each coefficient receives only finitely many contributions.

Lemma 1.1 (Coefficient rules). *For $F(z) = \sum f_n z^n$ and $G(z) = \sum g_n z^n$,*

$$[z^n]F(z)G(z) = \sum_{j=0}^n f_j g_{n-j}, \quad (1.1)$$

$$[z^n]F'(z) = (n+1)f_{n+1}, \quad (1.2)$$

$$[z^n] \frac{F(z)}{1-az} = \sum_{j=0}^n a^{n-j} f_j. \quad (1.3)$$

If F is analytic on $|z| \leq r$, then also

$$f_n = \frac{1}{2\pi i} \int_{|z|=r} \frac{F(z)}{z^{n+1}} dz, \quad |f_n| \leq r^{-n} \max_{|z|=r} |F(z)|. \quad (1.4)$$

Proof. The first identity is the definition of Cauchy multiplication. The second follows by differentiating term by term, and the third by multiplying with $\sum_{m \geq 0} a^m z^m$. The analytic formula is Cauchy's coefficient formula, and the bound follows by estimating the contour integral by its length $2\pi r$. $\square$

1.2 Formal residues

For a Laurent series $A(z)$, set $\text{Res}_z A(z) dz = [z^{-1}]A(z)$. Then

$$[z^n]F(z) = \text{Res}_z \frac{F(z)}{z^{n+1}} dz, \quad \text{Res}_z A'(z) dz = 0. \quad (1.5)$$

The second identity is simply the fact that the derivative of z^m has no z^{-1} term.

Theorem 1.2 (Formal change of variables for residues). *Let $u = u(z) \in zR[[z]]$ have invertible linear coefficient. For every Laurent series A for which the composition is defined,*

$$\text{Res}_z A(u(z))u'(z) dz = \text{Res}_u A(u) du. \quad (1.6)$$

Proof. By linearity it suffices to take $A(u) = u^m$. If $m \neq -1$, then $u(z)^m u'(z) = \frac{1}{m+1} (u(z)^{m+1})'$ has residue zero. For $m = -1$, $u'(z)/u(z) = z^{-1} + H(z)$ with $H \in R[[z]]$, because $u(z) = az(1 + O(z))$. Thus both sides have residue 1. $\square$

Master principle

The combination of Equations (1.5) and (1.6) turns implicit substitution into coefficient extraction. This is the entire algebraic engine of Lagrange inversion; no analytic contour is needed until one asks about convergence.

1.3 Ordinary and exponential generating functions

For a sequence (a_n) , its ordinary and exponential generating functions are

$$A(z) = \sum_{n \geq 0} a_n z^n, \quad \hat{A}(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}.$$

Ordinary multiplication gives unweighted convolution. Exponential multiplication gives binomial convolution:

$$\hat{A}(z)\hat{B}(z) = \sum_{n \geq 0} \left(\sum_{j=0}^n \binom{n}{j} a_j b_{n-j} \right) \frac{z^n}{n!}. \quad (1.7)$$

This binomial factor is the number of ways to choose which labels belong to the first component.

1.4 Finite differences and factorial bases

Let E be the shift operator $(Ef)(x) = f(x+1)$ and $\Delta = E - 1$. Then

$$\Delta x^{\underline{k}} = kx^{\underline{k-1}}, \quad \sum_{i=0}^{n-1} i^{\underline{k}} = \frac{n^{\underline{k+1}}}{k+1}. \quad (1.8)$$

Proof. The first follows from $x + 1^{\underline{k}} - x^{\underline{k}} = kx^{\underline{k-1}}$ after factoring the common product. Summing this telescoping identity gives the second formula. $\square$

Theorem 1.3 (Newton expansion). *Every polynomial p of degree at most d has the unique expansion*

$$p(x) = \sum_{k=0}^d \frac{\Delta^k p(0)}{k!} x^{\underline{k}} = \sum_{k=0}^d \Delta^k p(0) \binom{x}{k}. \quad (1.9)$$

Proof. The factorial polynomials form a basis because their degrees are distinct and their leading coefficients are one. If $p = \sum a_k x^{\underline{k}}$, applying Δ^j and setting $x = 0$ kills terms with $k > j$ by a remaining factor x , kills terms with $k < j$ by excessive differencing, and gives $j!a_j$ for $k = j$. $\square$

1.5 Labelled products and the exponential formula

A labelled combinatorial class is one whose atoms carry distinct labels. Suppose a connected component of size j has total weight x_j . A set of k such components with total size n has weight enumerator $\mathcal{B}_{n,k}(x_1, x_2, \dots)$; a set of any number of components has enumerator $\mathcal{B}_n(x_1, \dots, x_n)$.

Theorem 1.4 (The exponential formula). *Let*

$$C(z) = \sum_{j \geq 1} x_j \frac{z^j}{j!}.$$

Then

$$\frac{C(z)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) \frac{z^n}{n!}, \quad (1.10)$$

$$e^{uC(z)} = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}(x_1, \dots) u^k \frac{z^n}{n!}, \quad (1.11)$$

$$e^{C(z)} = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{z^n}{n!}. \quad (1.12)$$

Proof. Expand $C(z)^k/k!$ multinomially. If j_i components have size i , then $\sum_i j_i = k$ and $\sum_i i j_i = n$, and the coefficient of z^n is

$$\prod_i \frac{1}{j_i!} \left(\frac{x_i}{i!} \right)^{j_i}.$$

Multiplication by $n!$ labels the atoms. This proves the first formula. Summing it with weight u^k proves the second, and setting $u = 1$ proves the third. $\square$

Corollary 1.5 (Partition-type count). *The number of partitions of an n -element labelled set with exactly j_i blocks of size i is*

$$\frac{n!}{\prod_{i \geq 1} (i!)^{j_i} j_i!}, \quad \sum_i i j_i = n. \quad (1.13)$$

Proof. Order the n labels, cut the ordering into blocks of prescribed sizes, then divide by $i!$ for the internal order of each size- i block and by $j_i!$ for the order of blocks of equal size. $\square$

Part II

Stirling Numbers, Lah Numbers, and Basis Changes

Chapter 2

The two Stirling triangles

2.1 First-kind numbers: cycles and factorial coefficients

Definition 2.1. The signed Stirling numbers of the first kind $s(n, k)$ and the unsigned numbers $\begin{bmatrix} n \\ k \end{bmatrix}$ are defined by

$$x^n = \sum_{k=0}^n s(n, k) x^k, \quad (2.1)$$

$$x^{\bar{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (2.2)$$

$$s(n, k) = (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix}. \quad (2.3)$$

The sign relation follows from $x^{\bar{n}} = (-1)^n - x^n$.

Theorem 2.2 (Permutation interpretation). $\begin{bmatrix} n \\ k \end{bmatrix}$ is the number of permutations of $[n]$ having exactly k cycles. Consequently,

$$\begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}, \quad (2.4)$$

with $\begin{bmatrix} 0 \\ 0 \end{bmatrix} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. Given a permutation of $[n+1]$ with k cycles, either $n+1$ is a singleton cycle, leaving a permutation of $[n]$ with $k-1$ cycles, or it is inserted immediately after one of the n existing elements in the cyclic notation of a permutation of $[n]$ with k cycles. These alternatives are disjoint and reversible. The same recurrence follows algebraically by multiplying $x^{\bar{n}}$ by $x+n$ and comparing coefficients. $\square$

For $n=3$, the cycle counts are $\begin{bmatrix} 3 \\ 3 \end{bmatrix} = 1$, $\begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3$, and $\begin{bmatrix} 3 \\ 1 \end{bmatrix} = 2$, corresponding to the identity, the three transpositions, and the two 3-cycles. Summing over the possible number of cycles gives

$$\sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} = n!, \quad (2.5)$$

which also follows from Equation (2.2) at $x=1$.

2.2 Second-kind numbers: set partitions and surjections

Definition 2.3. The Stirling number of the second kind $\left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ is the number of partitions of $[n]$ into exactly k nonempty blocks.

Theorem 2.4 (Second-kind recurrence). For $n, k \geq 0$,

$$\left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} = k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} + \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}, \quad (2.6)$$

with $\left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\} = 1$ and zero boundary values outside $0 \leq k \leq n$.

Proof. In a partition of $[n+1]$ into k blocks, the new label either joins one of the k blocks of a partition of $[n]$, or forms a singleton block beside a partition of $[n]$ into $k-1$ blocks. $\square$

Theorem 2.5 (Surjection formula). For all $n, k \geq 0$,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{1}{k!} \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = \sum_{j=0}^k \frac{(-1)^{k-j} j^n}{(k-j)! j!}. \quad (2.7)$$

Proof. There are $k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ surjections from $[n]$ onto a fixed k -element codomain: a set partition into fibres followed by an ordering of the fibres. Inclusion–exclusion over the set of omitted codomain values gives $\sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n$. Division by $k!$ proves the formula. $\square$

The simple diagonals follow at once:

$$\left\{ \begin{matrix} n \\ n \end{matrix} \right\} = 1, \quad \left\{ \begin{matrix} n \\ 1 \end{matrix} \right\} = 1 \ (n \geq 1), \quad \left\{ \begin{matrix} n \\ n-1 \end{matrix} \right\} = \binom{n}{2}, \quad \left\{ \begin{matrix} n \\ 2 \end{matrix} \right\} = 2^{n-1} - 1. \quad (2.8)$$

The last identity chooses a nonempty proper subset as one block and divides by two for exchanging the blocks.

2.3 Alternative and reverse recurrences

The ordinary recurrence builds each row from the preceding one. Several less familiar identities in the source archive instead recover an entry from entries to its right or from earlier entries in the same column. They were labelled there as conjectures; in fact they are exact identities.

Theorem 2.6 (Alternative second-kind recurrences). For $1 \leq k < n$,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\left\{ \begin{matrix} n \\ r \end{matrix} \right\}}{(k-r)!}, \quad (2.9)$$

$$(n-k) \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=2}^{n-k+1} (j-2)! \binom{-k}{j} \left\{ \begin{matrix} n \\ k+j-1 \end{matrix} \right\}, \quad (2.10)$$

$$(n-k) \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=2}^{n-k+1} (-1)^j \binom{n}{j} \left\{ \begin{matrix} n-j+1 \\ k \end{matrix} \right\}. \quad (2.11)$$

The terminal value in either reverse recursion is $\left\{ \begin{matrix} n \\ n \end{matrix} \right\} = 1$.

Proof. Substitute $x = k$ in the basis identity $k^n = \sum_r \left\{ \begin{matrix} n \\ r \end{matrix} \right\} k^r$. Since $k^r = k!/(k-r)!$ for $r \leq k$, division by $k!$ and isolation of the $r = k$ term gives (2.9).

For the row reversal, use the bivariate exponential generating function

$$F(x, y) = \exp(y(e^x - 1)) = \sum_{n, k \geq 0} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{x^n}{n!} y^k, \quad u = e^x - 1.$$

The generating function of the left side of (2.10) is

$$xF_x - yF_y = yF(xe^x - u).$$

On the other hand, because $\binom{-k}{j} = (-1)^j \binom{k+j-1}{j}$, the generating function of its right side is

$$\begin{aligned} y \sum_{j \geq 2} \frac{(-1)^j}{j(j-1)} \partial_y^j F &= yF \sum_{j \geq 2} \frac{(-1)^j u^j}{j(j-1)} \\ &= yF((1+u) \log(1+u) - u) \\ &= yF(xe^x - u). \end{aligned}$$

Here the middle series is obtained by differentiating it twice, summing the geometric series, and restoring the two zero initial coefficients. Equality of coefficients proves (2.10).

For the column reversal fix k and put

$$F_k(x) = \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{x^n}{n!} = \frac{(e^x - 1)^k}{k!}.$$

The two sides of (2.11), after multiplication by $x^n/n!$ and summation over n , become respectively

$$xF'_k(x) - kF_k(x) \quad \text{and} \quad (e^{-x} - 1 + x)F'_k(x).$$

They are equal because direct differentiation gives $(1 - e^{-x})F'_k(x) = kF_k(x)$. $\square$

The factor interchange mentioned in the source produces a parallel pair for the unsigned first-kind triangle.

Theorem 2.7 (Reverse recurrences of the first kind). *For $1 \leq k < n$,*

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (-1)^j \binom{-k}{j} \begin{bmatrix} n \\ k+j-1 \end{bmatrix}, \quad (2.12)$$

$$(n-k) \begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=2}^{n-k+1} (j-2)! \binom{n}{j} \begin{bmatrix} n-j+1 \\ k \end{bmatrix}. \quad (2.13)$$

Proof. Let $C_n(y) = y^{\overline{n}} = \sum_r \begin{bmatrix} n \\ r \end{bmatrix} y^r$. Since

$$C_n(1+y) = \frac{y^{\overline{n+1}}}{y},$$

the coefficient of y^{k-1} on the two sides is, respectively,

$$\sum_{r \geq k-1} \binom{r}{k-1} \begin{bmatrix} n \\ r \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} n+1 \\ k \end{bmatrix} = n \begin{bmatrix} n \\ k \end{bmatrix} + \begin{bmatrix} n \\ k-1 \end{bmatrix}.$$

Remove the terms $r = k-1$ and $r = k$ and put $r = k+j-1$. Because $(-1)^j \binom{-k}{j} = \binom{k+j-1}{j}$, the remainder is precisely (2.12).

For fixed k , set

$$C_k(x) = \sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{x^n}{n!} = \frac{[-\log(1-x)]^k}{k!}.$$

The exponential generating function of the right side of (2.13) is

$$C'_k(x) \sum_{j \geq 2} \frac{x^j}{j(j-1)} = (x + (1-x) \log(1-x)) C'_k(x).$$

The left side has generating function $x C'_k - k C_k$. Their equality reduces to

$$-(1-x) \log(1-x) C'_k(x) = k C_k(x),$$

which follows immediately from the displayed formula for C_k . □

2.4 Tables and common boundary conventions

The first rows are

$n \backslash k$	0	1	2	3	4	5	6	$n \backslash k$	0	1	2	3	4	5	6
0	1							0	1						
1	0	1						1	0	1					
2	0	1	1					2	0	1	1				
3	0	2	3	1				3	0	1	3	1			
4	0	6	11	6	1			4	0	1	7	6	1		
5	0	24	50	35	10	1		5	0	1	15	25	10	1	
6	0	120	274	225	85	15	1	6	0	1	31	90	65	15	1

for $\begin{bmatrix} n \\ k \end{bmatrix}$ and $\{n \}_k$, respectively.

Chapter 3

Factorial bases and inverse matrices

3.1 The basic changes of basis

Theorem 3.1 (Stirling basis transformations). *For every $n \geq 0$,*

$$x^{\underline{n}} = \sum_{k=0}^n s(n, k) x^k, \quad (3.1)$$

$$x^{\overline{n}} = \sum_{k=0}^n \begin{bmatrix} n \\ k \end{bmatrix} x^k, \quad (3.2)$$

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\underline{k}}, \quad (3.3)$$

$$x^n = \sum_{k=0}^n (-1)^{n-k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^{\overline{k}}. \quad (3.4)$$

Equivalently,

$$x^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} k! \binom{x}{k}, \quad \binom{x}{n} = \frac{1}{n!} \sum_{k=0}^n s(n, k) x^k. \quad (3.5)$$

Proof. The first two identities are definitions. In the Newton expansion Equation (1.9) for $p(x) = x^n$, one has

$$\Delta^k x^n|_{x=0} = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} j^n = k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$$

by Equation (2.7). This gives Equation (3.3). Replacing x by $-x$ and multiplying by $(-1)^n$ gives Equation (3.4). Division of $x^{\underline{n}}$ by $n!$ gives the second binomial-basis formula. $\square$

Corollary 3.2 (Shifted factorial evaluations). *For every $n \geq 0$,*

$$\sum_{k=1}^{n+1} \left\{ \begin{matrix} n+1 \\ k \end{matrix} \right\} x - 1^{\underline{k-1}} = x^n, \quad (3.6)$$

$$\sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} n^{\underline{k}} = n^n. \quad (3.7)$$

Proof. Apply (3.3) with exponent $n+1$ and use $x^{\underline{k}} = x x - 1^{\underline{k-1}}$. Both sides are divisible by x as polynomials, and cancellation gives (3.6). Formula (3.7) is (3.3) evaluated at $x = n$. $\square$

Corollary 3.3 (Inverse matrices). *The lower triangular matrices $s = (s(n, k))$ and $S = (\{n_k\})$ are mutual inverses:*

$$\sum_{j=k}^n s(n, j) \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.8)$$

$$\sum_{j=k}^n \left\{ \begin{matrix} n \\ j \end{matrix} \right\} s(j, k) = \delta_{nk}. \quad (3.9)$$

Equivalently,

$$\sum_{j=k}^n (-1)^{n-j} \left[\begin{matrix} n \\ j \end{matrix} \right] \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \delta_{nk}, \quad (3.10)$$

$$\sum_{j=k}^n (-1)^{j-k} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} \left[\begin{matrix} j \\ k \end{matrix} \right] = \delta_{nk}. \quad (3.11)$$

Proof. Expand x^n into powers and each power back into falling factorials. Uniqueness of coordinates in the factorial basis gives the first matrix product. The reverse substitution gives the second. $\square$

3.2 Power sums by basis adaptation

The factorial basis makes discrete antidifferentiation immediate. Combining Equations (1.8) and (3.3) yields

$$\sum_{i=0}^N i^m = \sum_{k=0}^m \left\{ \begin{matrix} m \\ k \end{matrix} \right\} \frac{N+1^{k+1}}{k+1}. \quad (3.12)$$

For $m = 4$ this is

$$\frac{1}{2}N + 1^2 + \frac{7}{3}N + 1^3 + \frac{6}{4}N + 1^4 + \frac{1}{5}N + 1^5,$$

which expands to the familiar quartic power-sum polynomial.

3.3 Factorial moments: Poisson variables and permutation fixed points

The change from ordinary powers to falling factorials is especially effective for integer-valued random variables.

Theorem 3.4 (Poisson moments). *If X has the Poisson distribution of mean λ , then*

$$\mathbb{E}X^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \lambda^k. \quad (3.13)$$

In particular, $\mathbb{E}X^n = B_n$ when $\lambda = 1$.

Proof. The k th falling-factorial moment is

$$\mathbb{E}X^{\underline{k}} = e^{-\lambda} \sum_{r \geq k} \frac{r^{\underline{k}} \lambda^r}{r!} = \lambda^k e^{-\lambda} \sum_{q \geq 0} \frac{\lambda^q}{q!} = \lambda^k.$$

Take expectations in the basis identity $X^n = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} X^{\underline{k}}$. $\square$

Theorem 3.5 (Moments of the number of fixed points). *Let Y_m be the number of fixed points of a uniformly random permutation of $[m]$. Then*

$$\mathbb{E}Y_m^n = \sum_{k=0}^m \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^{\min(m,n)} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (3.14)$$

Thus, when $m \geq n$, the moment equals the Bell number B_n .

Proof. The random variable Y_m^k counts ordered k -tuples of distinct fixed points. There are m^k candidate tuples, and a specified tuple is fixed with probability $(m-k)!/m!$. Hence

$$\mathbb{E}Y_m^k = 1 \quad (0 \leq k \leq m), \quad \mathbb{E}Y_m^k = 0 \quad (k > m).$$

Taking expectations in (3.3) proves the formula. The upper limit m is essential: it records the finite size of the permutation, even when $n > m$. $\square$

3.4 Lah numbers

Definition 3.6. For $1 \leq k \leq n$, the unsigned Lah number is

$$L(n, k) = \binom{n-1}{k-1} \frac{n!}{k!}, \quad (3.15)$$

with $L(0, 0) = 1$ and zero otherwise outside the triangle.

Combinatorially, $L(n, k)$ counts partitions of $[n]$ into k nonempty linearly ordered lists, with the collection of lists unordered.

Theorem 3.7 (Conversion between rising and falling factorials).

$$x^{\bar{n}} = \sum_{k=0}^n L(n, k) x^k, \quad (3.16)$$

$$x^{\underline{n}} = \sum_{k=0}^n (-1)^{n-k} L(n, k) x^{\bar{k}}, \quad (3.17)$$

$$\sum_{j=k}^n (-1)^{j-k} L(n, j) L(j, k) = \delta_{nk}. \quad (3.18)$$

Moreover,

$$L(n, k) = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}. \quad (3.19)$$

Proof. A direct generating-function calculation gives

$$\sum_{n \geq k} L(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(\frac{z}{1-z} \right)^k.$$

Therefore

$$\sum_{n \geq 0} \left(\sum_k L(n, k) x^k \right) \frac{z^n}{n!} = \sum_{k \geq 0} \binom{x}{k} \left(\frac{z}{1-z} \right)^k = (1-z)^{-x} = \sum_{n \geq 0} x^{\bar{n}} \frac{z^n}{n!},$$

proving Equation (3.16). Replace x by $-x$ to obtain the inverse formula, and compose the two transformations to obtain Equation (3.18). Finally, expand a rising factorial into powers by first-kind numbers and powers into falling factorials by second-kind numbers; uniqueness gives Equation (3.19). $\square$

The inequalities

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \left[\begin{matrix} n \\ k \end{matrix} \right] \leq L(n, k) \quad (3.20)$$

follow from explicit injections: a set partition may be canonically turned into a permutation with one cycle per block by writing each block with its smallest member first and arranging cycles by minima; forgetting the cyclic identifications embeds permutations with k cycles into unordered collections of k lists.

Chapter 4

Generating functions, transforms, and operators

4.1 Exponential generating functions

Theorem 4.1 (Column generating functions). *For every fixed $k \geq 0$,*

$$\sum_{n \geq k} s(n, k) \frac{z^n}{n!} = \frac{\log(1+z)^k}{k!}, \quad (4.1)$$

$$\sum_{n \geq k} \begin{bmatrix} n \\ k \end{bmatrix} \frac{z^n}{n!} = \frac{[-\log(1-z)]^k}{k!}, \quad (4.2)$$

$$\sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \frac{z^n}{n!} = \frac{(e^z - 1)^k}{k!}. \quad (4.3)$$

The bivariate second-kind generating function is

$$\sum_{k \geq 0} \sum_{n \geq k} \left\{ \begin{matrix} n \\ k \end{matrix} \right\} y^k \frac{z^n}{n!} = \exp(y(e^z - 1)). \quad (4.4)$$

Proof. Expand

$$(1+z)^u = e^{u \log(1+z)} = \sum_{n \geq 0} \frac{z^n}{n!} \sum_{k=0}^n s(n, k) u^k = \sum_{k \geq 0} \frac{u^k}{k!} \log(1+z)^k$$

and compare coefficients of u^k . Replacing z by $-z$ and using the sign relation gives Equation (4.2). Formula Equation (4.3) is the exponential formula for a set of k nonempty labelled blocks; algebraically it also follows by inserting Equation (2.7) and summing exponentials. Summing over k with weight y^k proves Equation (4.4). $\square$

Differentiating Equation (4.1) gives the useful identity

$$\frac{\log(1+z)^m}{1+z} = m! \sum_{r \geq 0} s(r+1, m+1) \frac{z^r}{r!}, \quad |z| < 1, \quad (4.5)$$

which is valid formally as well.

4.2 A rational ordinary generating function

Theorem 4.2 (Fixed- k ordinary generating function). *For $k \geq 0$,*

$$\sum_{n \geq k} \begin{Bmatrix} n \\ k \end{Bmatrix} x^{n-k} = \prod_{j=1}^k \frac{1}{1-jx} = \frac{1}{x^{k+1} 1/x^{k+1}}. \quad (4.6)$$

Consequently,

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = h_{n-k}(1, 2, \dots, k) = \sum_{c_1 + \dots + c_k = n-k} 1^{c_1} 2^{c_2} \dots k^{c_k}, \quad (4.7)$$

where h_r is a complete homogeneous symmetric polynomial.

Proof. Let $F_k(x) = \sum_{r \geq 0} \begin{Bmatrix} k+r \\ k \end{Bmatrix} x^r$. The recurrence Equation (2.6), with $n = k + r - 1$, gives $F_k(x) = (1 - kx)^{-1} F_{k-1}(x)$ and $F_0 = 1$. Iteration proves the product. Expanding each geometric factor independently proves the complete-homogeneous sum. The factorial form follows by direct factorization. $\square$

A useful rearrangement of the count of all maps $[n] \rightarrow [k]$ is

$$\begin{Bmatrix} n \\ k \end{Bmatrix} = \frac{k^n}{k!} - \sum_{r=1}^{k-1} \frac{\begin{Bmatrix} n \\ r \end{Bmatrix}}{(k-r)!}. \quad (4.8)$$

Indeed, a map using exactly r values can be chosen in $\binom{k}{r} r! \begin{Bmatrix} n \\ r \end{Bmatrix}$ ways; divide the identity $k^n = \sum_r \binom{k}{r} r! \begin{Bmatrix} n \\ r \end{Bmatrix}$ by $k!$.

4.3 The Stirling transform

Theorem 4.3 (Stirling transform and its inverse). *Let*

$$g_n = \sum_{k=0}^n \begin{Bmatrix} n \\ k \end{Bmatrix} f_k.$$

Then

$$f_n = \sum_{k=0}^n s(n, k) g_k = \sum_{k=0}^n (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix} g_k. \quad (4.9)$$

For exponential generating functions,

$$\widehat{G}(z) = \widehat{F}(e^z - 1), \quad \widehat{F}(z) = \widehat{G}(\log(1 + z)). \quad (4.10)$$

Proof. The inverse formula is exactly the matrix inversion in Theorem 3.3. For the generating function,

$$\widehat{G}(z) = \sum_k f_k \sum_{n \geq k} \begin{Bmatrix} n \\ k \end{Bmatrix} \frac{z^n}{n!} = \sum_k f_k \frac{(e^z - 1)^k}{k!} = \widehat{F}(e^z - 1).$$

Substitution by the compositional inverse $\log(1 + z)$ gives the reverse relation. $\square$

4.4 Differential-operator normal ordering

Let $D = d/dx$. The Euler operator xD is diagonal on monomials: $(xD)x^m = mx^m$. The operator $x^k D^k$ has eigenvalue $m^{\underline{k}}$. Thus the scalar basis changes lift verbatim to operators.

Theorem 4.4 (Stirling normal-ordering identities). *For $n \geq 0$,*

$$(xD)^n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k D^k, \quad (4.11)$$

$$x^n D^n = \sum_{k=0}^n s(n, k) (xD)^k = x D^{\underline{n}}. \quad (4.12)$$

Proof. Both sides of Equation (4.11) act on x^m as multiplication by $m^n = \sum_k \left\{ \begin{matrix} n \\ k \end{matrix} \right\} m^{\underline{k}}$. Since monomials span the polynomial ring, the operators are equal. The inverse relation is proved identically using $m^{\underline{n}} = \sum_k s(n, k) m^k$. $\square$

Because $E = e^D$ and $\Delta = e^D - 1$, substitution of the mutually inverse series $D = \log(1 + \Delta)$ and $\Delta = e^D - 1$ gives, whenever the operator series converge (or formally on polynomials),

$$\frac{D^k}{k!} = \sum_{n \geq k} \frac{s(n, k)}{n!} \Delta^n, \quad (4.13)$$

$$\frac{\Delta^k}{k!} = \sum_{n \geq k} \frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{n!} D^n. \quad (4.14)$$

These are the derivative–finite-difference transformations listed in the archive.

Chapter 5

Symmetric functions, diagonals, and finite identities

5.1 Elementary and complete symmetric descriptions

From Equation (2.2),

$$\begin{bmatrix} n \\ n-r \end{bmatrix} = e_r(0, 1, \dots, n-1) = \sum_{0 \leq i_1 < \dots < i_r < n} i_1 \cdots i_r. \quad (5.1)$$

Together with Equation (4.7), this identifies the first-kind triangle with elementary symmetric functions and the second-kind triangle with complete homogeneous symmetric functions.

The first few near-diagonals are

$$\begin{bmatrix} n \\ n-1 \end{bmatrix} = \binom{n}{2}, \quad (5.2)$$

$$\begin{bmatrix} n \\ n-2 \end{bmatrix} = \frac{3n-1}{4} \binom{n}{3}, \quad (5.3)$$

$$\begin{bmatrix} n \\ n-3 \end{bmatrix} = \binom{n}{2} \binom{n}{4}. \quad (5.4)$$

Proof. The first is $e_1(0, \dots, n-1)$. Newton's identities $2e_2 = p_1^2 - p_2$ and $6e_3 = p_1^3 - 3p_1p_2 + 2p_3$, combined with the elementary sums of first, second, and third powers, simplify to Equations (5.3) and (5.4). Equivalently, one may classify permutations by the nontrivial cycle types 3, 2 + 2 for the second diagonal and 4, 3 + 2, 2 + 2 + 2 for the third. $\square$

5.2 Harmonic-number formulae

Factoring $(n-1)!$ from $(x+1)(x+2) \cdots (x+n-1)$ gives

$$\frac{\begin{bmatrix} n \\ k+1 \end{bmatrix}}{(n-1)!} = e_k \left(1, \frac{1}{2}, \dots, \frac{1}{n-1} \right) = \sum_{1 \leq i_1 < \dots < i_k < n} \frac{1}{i_1 \cdots i_k}. \quad (5.5)$$

Write $H_n^{(r)} = \sum_{j=1}^n j^{-r}$ and $H_n = H_n^{(1)}$. Newton's identities now give

$$\begin{bmatrix} n \\ 2 \end{bmatrix} = (n-1)! H_{n-1}, \quad (5.6)$$

$$\begin{bmatrix} n \\ 3 \end{bmatrix} = \frac{(n-1)!}{2} \left(H_{n-1}^2 - H_{n-1}^{(2)} \right), \quad (5.7)$$

$$\begin{bmatrix} n \\ 4 \end{bmatrix} = \frac{(n-1)!}{6} \left(H_{n-1}^3 - 3H_{n-1}H_{n-1}^{(2)} + 2H_{n-1}^{(3)} \right). \quad (5.8)$$

More generally,

$$\frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} = \frac{1}{k!} \mathcal{B}_k(H_n, -1!H_n^{(2)}, 2!H_n^{(3)}, \dots, (-1)^{k-1}(k-1)!H_n^{(k)}). \quad (5.9)$$

Proof. Take logarithms in

$$\prod_{j=1}^n \left(1 + \frac{x}{j} \right) = \exp \left(\sum_{r \geq 1} \frac{(-1)^{r-1} H_n^{(r)}}{r} x^r \right)$$

and apply the complete Bell-polynomial generating function Equation (1.12). $\square$

If $w(n, k) = k! \begin{bmatrix} n \\ k+1 \end{bmatrix} / (n-1)!$, then complete Bell recurrence yields

$$w(n, m) = \delta_{m0} + \sum_{r=0}^{m-1} (1-m)_r H_{n-1}^{(r+1)} w(n, m-1-r), \quad (5.10)$$

where $(1-m)_r$ is falling. Conversely, taking a logarithm recovers the harmonic power sums:

$$H_n^{(m)} = -m[x^m] \log \left(1 + \sum_{k \geq 1} \frac{\begin{bmatrix} n+1 \\ k+1 \end{bmatrix}}{n!} (-x)^k \right). \quad (5.11)$$

For example,

$$(n!)^2 H_n^{(2)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^2 - 2 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix}, \quad (5.12)$$

$$(n!)^3 H_n^{(3)} = \begin{bmatrix} n+1 \\ 2 \end{bmatrix}^3 - 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix} \begin{bmatrix} n+1 \\ 2 \end{bmatrix} \begin{bmatrix} n+1 \\ 3 \end{bmatrix} + 3 \begin{bmatrix} n+1 \\ 1 \end{bmatrix}^2 \begin{bmatrix} n+1 \\ 4 \end{bmatrix}. \quad (5.13)$$

5.3 Convolution and summation identities

The following paired identities expose the parallel structure of the two triangles.

Theorem 5.1 (First- and second-kind summations). *For valid nonnegative indices,*

$$\begin{bmatrix} n+1 \\ k+1 \end{bmatrix} = \sum_{j=k}^n \begin{bmatrix} n \\ j \end{bmatrix} \begin{bmatrix} j \\ k \end{bmatrix} = \sum_{j=k}^n \frac{n!}{j!} \begin{bmatrix} j \\ k \end{bmatrix}, \quad (5.14)$$

$$\left\{ \begin{matrix} n+1 \\ k+1 \end{matrix} \right\} = \sum_{j=k}^n \binom{n}{j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\} = \sum_{j=k}^n (k+1)^{n-j} \left\{ \begin{matrix} j \\ k \end{matrix} \right\}, \quad (5.15)$$

$$\begin{bmatrix} n+k+1 \\ n \end{bmatrix} = \sum_{j=0}^k (n+j) \begin{bmatrix} n+j \\ j \end{bmatrix}, \quad (5.16)$$

$$\left\{ \begin{matrix} n+k+1 \\ k \end{matrix} \right\} = \sum_{j=0}^k j \left\{ \begin{matrix} n+j \\ j \end{matrix} \right\}. \quad (5.17)$$

Moreover, for $\ell, m \geq 0$,

$$\binom{\ell+m}{\ell} \begin{bmatrix} n \\ \ell+m \end{bmatrix} = \sum_j \binom{n}{j} \begin{bmatrix} j \\ \ell \end{bmatrix} \begin{bmatrix} n-j \\ m \end{bmatrix}, \quad (5.18)$$

$$\binom{\ell+m}{\ell} \left\{ \begin{matrix} n \\ \ell+m \end{matrix} \right\} = \sum_j \binom{n}{j} \left\{ \begin{matrix} j \\ \ell \end{matrix} \right\} \left\{ \begin{matrix} n-j \\ m \end{matrix} \right\}. \quad (5.19)$$

Proof. For the first equality in Equation (5.14), mark k cycles of a permutation of $[n]$ and insert $n+1$ into the canonical construction; algebraically it is the coefficient of x^k in $\frac{d}{dx} x^{\bar{n}}$. The second equality follows by iterating recurrence Equation (2.4) in the n direction. For Equation (5.15), distinguish the elements outside the block containing $n+1$ for the first equality, and add the $n-k$ remaining elements one at a time to one of $k+1$ existing blocks for the second. The hockey-stick forms telescope the respective recurrences. Finally, in Equation (5.18), choose which labels belong to the ℓ marked cycles; in Equation (5.19), choose which labels belong to the ℓ marked blocks. The binomial factor on the left forgets which of the total $\ell+m$ components were marked. $\square$

Binomial inversion of the first identity in Equation (5.14) gives

$$\begin{bmatrix} n \\ m \end{bmatrix} = \sum_{k=m}^n (-1)^{k-m} \binom{k}{m} \begin{bmatrix} n+1 \\ k+1 \end{bmatrix}. \quad (5.20)$$

The alternative form

$$\begin{bmatrix} n+1 \\ m+1 \end{bmatrix} = \sum_{k=m}^n \frac{n!}{k!} \begin{bmatrix} k \\ m \end{bmatrix} \quad (5.21)$$

is the second equality in Equation (5.14). All finite sums displayed in the source follow from these and the two convolution identities by relabelling indices.

5.4 Symmetric cross-formulae

The archive records a striking symmetric pair. For $0 \leq k \leq n$,

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \left\{ \begin{matrix} j-k \\ j-n \end{matrix} \right\}, \quad (5.22)$$

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{j=n}^{2n-k} (-1)^{j-k} \binom{j-1}{k-1} \binom{2n-k}{j} \begin{bmatrix} j-k \\ j-n \end{bmatrix}. \quad (5.23)$$

Proof. Set $r = n - k$ and use the symmetric-function descriptions $\begin{bmatrix} n \\ k \end{bmatrix} = e_r(0, 1, \dots, n-1)$ and $\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = h_r(1, 2, \dots, k)$. The involution ω on the ring of symmetric functions sends e_r to h_r and is implemented on generating functions by $E(t)H(-t) = 1$. Expand the finite alphabets after adjoining and removing the common arithmetic progression, then apply the binomial translation formula $h_r(X+c) = \sum_j \binom{m+r-1}{r-j} c^{r-j} h_j(X)$ and its elementary analogue. Collecting coefficients gives Equation (5.22); applying ω gives Equation (5.23). Equivalently, direct substitution of the inclusion-exclusion formula Equation (2.7) and two applications of Vandermonde's identity produce the same finite sum. $\square$

5.5 An explicit two-sum formula of the first kind

Unlike the second-kind inclusion-exclusion formula, the first-kind triangle has no comparably short single finite sum. Substitution into the symmetric cross-formula nevertheless gives a completely explicit double sum.

Theorem 5.2 (Explicit first-kind double sum). *For $1 \leq k \leq n$,*

$$\begin{bmatrix} n \\ k \end{bmatrix} = \sum_{j=n}^{2n-k} \binom{j-1}{k-1} \binom{2n-k}{j} \sum_{m=0}^{j-n} \frac{(-1)^{m+n-k} m^{j-k}}{m! (j-n-m)!}. \quad (5.24)$$

Proof. Start from (5.22) and apply (2.7) with upper index $j-k$ and lower index $j-n$:

$$\begin{Bmatrix} j-k \\ j-n \end{Bmatrix} = \sum_{m=0}^{j-n} \frac{(-1)^{j-n-m} m^{j-k}}{(j-n-m)! m!}.$$

Multiplication by the outer sign $(-1)^{j-k}$ gives $(-1)^{2j-k-n-m} = (-1)^{m+n-k}$, because the exponents differ by the even integer $2(j-m-n+k)$. Substitution proves (5.24). $\square$

5.6 A partition formula for signed near-diagonal coefficients

The near-diagonal coefficient $s(n, n-p)$ is a polynomial in n of degree $2p$. The following finite formula makes every monomial contribution explicit.

Theorem 5.3 (Near-diagonal partition formula). *Let $n \geq 1$ and $0 \leq p \leq n-1$. Then*

$$s(n, n-p) = \frac{1}{(n-p-1)!} \sum_{\substack{k_1, \dots, k_p \geq 0 \\ \sum_{r=1}^p r k_r = p}} (-1)^K \frac{(n+K-1)!}{p \prod_{r=1}^p k_r! (r+1)!^{k_r}}, \quad K = \sum_{r=1}^p k_r. \quad (5.25)$$

For $p=0$ the sum and product are empty and the formula reads $s(n, n) = 1$.

Proof. The signed first-kind exponential generating function gives

$$s(n, n-p) = \frac{n!}{(n-p)!} [z^p] \left(\frac{\log(1+z)}{z} \right)^{n-p}. \quad (5.26)$$

Put $t = \log(1+z)$ and

$$\phi(t) = \frac{t}{e^t - 1}.$$

Then $t = z\phi(t)$ and $t/z = \phi(t)$. For $p \geq 1$, the Lagrange–Bürmann formula applied to $H(t) = \phi(t)^{n-p}$ gives

$$\begin{aligned} [z^p] H(t(z)) &= \frac{1}{p} [t^{p-1}] H'(t) \phi(t)^p \\ &= \frac{n-p}{p} [t^{p-1}] \phi'(t) \phi(t)^{n-1} \\ &= \frac{n-p}{n} [t^p] \phi(t)^n. \end{aligned}$$

The same conclusion is immediate for $p=0$. Hence

$$s(n, n-p) = \frac{(n-1)!}{(n-p-1)!} [t^p] \left(\frac{t}{e^t - 1} \right)^n. \quad (5.27)$$

Now write

$$A(t) = \sum_{r \geq 1} \frac{t^r}{(r+1)!} = \frac{e^t - 1 - t}{t}, \quad \frac{t}{e^t - 1} = (1 + A(t))^{-1}.$$

The negative-binomial expansion yields

$$(n-1)! [t^p] (1+A)^{-n} = \sum_{K \geq 0} (-1)^K \frac{(n+K-1)!}{K!} [t^p] A(t)^K.$$

Finally, the multinomial theorem gives

$$[t^p] A(t)^K = \sum_{\substack{k_1 + \dots + k_p = K \\ \sum r k_r = p}} \frac{K!}{\prod_{r=1}^p k_r!} \prod_{r=1}^p \frac{1}{(r+1)!^{k_r}}.$$

Insert this in (5.27) to obtain (5.25). □

The unsigned value is $\left[\begin{smallmatrix} n \\ n-p \end{smallmatrix} \right] = (-1)^p s(n, n-p)$. For $p = 1, 2, 3$ the theorem reduces to the closed forms in Equations (5.2) to (5.4); its advantage is that it works at arbitrary fixed distance from the diagonal without a new case analysis.

Chapter 6

Congruences and extensions to negative indices

6.1 Congruences for first-kind numbers

For a prime p ,

$$p \mid \begin{bmatrix} p \\ k \end{bmatrix} \quad (1 < k < p). \quad (6.1)$$

Indeed, in $\mathbb{F}_p[x]$,

$$x^{\overline{p}} = \prod_{a \in \mathbb{F}_p} (x + a) = x^p - x,$$

so every intermediate coefficient vanishes.

More generally, reducing each factor modulo 2 or 3 gives exact coefficient rules:

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/2 \rceil} (x+1)^{\lfloor n/2 \rfloor} = \begin{pmatrix} \lfloor n/2 \rfloor \\ m - \lceil n/2 \rceil \end{pmatrix} \pmod{2}, \quad (6.2)$$

$$\begin{bmatrix} n \\ m \end{bmatrix} \equiv [x^m] x^{\lceil n/3 \rceil} (x+1)^{\lceil (n-1)/3 \rceil} (x+2)^{\lfloor n/3 \rfloor} \pmod{3}. \quad (6.3)$$

Proof. Use $x^{\overline{n}} = \prod_{j=0}^{n-1} (x+j)$ and count how often each residue class occurs among $0, 1, \dots, n-1$. The displayed binomial form modulo 2 is immediate; expanding the last two factors gives the finite binomial sum written in the source for modulus 3. $\square$

For fixed m , expansion of Equation (6.2) yields the source's first four parity formulae. They are most cleanly read as integer-valued expressions multiplied by Iverson brackets; for example

$$\begin{bmatrix} n \\ 1 \end{bmatrix} \equiv 2^{n-2} [n \geq 2] + [n = 1] \pmod{2},$$

and similarly the quoted expressions for $m = 2, 3, 4$. Formula Equation (6.2) is both shorter and proves all of them simultaneously.

Theorem 6.1 (Rigorous finite-modulus replacement for the archived spectral form). *Fix $h \geq 2$ and $m \geq 0$. The sequence $n \mapsto \begin{bmatrix} n \\ m \end{bmatrix} \pmod{h}$ is generated by the coefficient of x^m in*

$$\left(\prod_{r=0}^{h-1} (x+r) \right)^q \prod_{r=0}^{s-1} (x+r), \quad n = qh + s, \quad (6.4)$$

and therefore satisfies a linear recurrence over $\mathbb{Z}/h\mathbb{Z}$ and is eventually an exponential-polynomial sequence over any finite splitting extension.

Proof. Group the factors of $x^{\bar{n}}$ into blocks of h consecutive integers and reduce each block modulo h . For fixed m , coefficient extraction takes place in a finite free module of polynomials of degree at most m , on which multiplication by the fixed block polynomial is a linear endomorphism. Cayley–Hamilton supplies a linear recurrence. Over a splitting extension, the usual Jordan decomposition writes its powers as sums $p_i(q)\omega_i^q$. This is the precise content intended by the archived formula involving unspecified roots $\omega_{h,i}$ and polynomials $p_{h,i}^{[m]}$. $\square$

6.2 Parity of second-kind numbers

Theorem 6.2 (Parity criterion). *Let*

$$z = n - \left\lfloor \frac{k+1}{2} \right\rfloor, \quad w = \left\lfloor \frac{k-1}{2} \right\rfloor.$$

Then

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \equiv \binom{z}{w} \pmod{2}. \quad (6.5)$$

Equivalently,

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \equiv 1 \pmod{2} \iff (n-k) \& \left\lfloor \frac{k-1}{2} \right\rfloor = 0. \quad (6.6)$$

In particular, $\left\{ \begin{matrix} 2n \\ n \end{matrix} \right\}$ is odd precisely when n is a power of two.

Proof. Reduce the ordinary generating function Equation (4.6) modulo 2. The even factors become 1, while every odd j contributes $(1-x)^{-1}$. There are $\lceil k/2 \rceil$ odd integers from 1 to k , so

$$\sum_{r \geq 0} \left\{ \begin{matrix} k+r \\ k \end{matrix} \right\} x^r \equiv (1-x)^{-\lceil k/2 \rceil}.$$

The coefficient of x^{n-k} is $\binom{n-k+\lceil k/2 \rceil-1}{\lceil k/2 \rceil-1} = \binom{z}{w}$. Lucas's theorem modulo 2 says $\binom{z}{w}$ is odd exactly when every 1-bit of w is also a 1-bit of z , equivalently when $w \& (z-w) = 0$; here $z-w = n-k$. Putting $k = n$ in Equation (6.6) gives $n \& \lfloor (n-1)/2 \rfloor = 0$, which is equivalent to n having one 1-bit. $\square$

6.3 Roman extension and negative Bell values

There are several inequivalent extensions of Stirling numbers outside the nonnegative triangle. The one used in the archive is the *Roman* extension: continue the two triangular recurrences wherever they determine a value and impose the duality

$$\left[\begin{matrix} n \\ k \end{matrix} \right] = \left\{ \begin{matrix} -k \\ -n \end{matrix} \right\}, \quad \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \left[\begin{matrix} -k \\ -n \end{matrix} \right]. \quad (6.7)$$

These identities are definitions of a coherent reciprocal array, not assertions about the original finite counting problems.

For $n \geq 1$ and $k \geq 0$, the mixed quadrant admits the finite expression

$$\left[\begin{matrix} -n \\ k \end{matrix} \right] = \frac{1}{n!} \sum_{j=1}^n (-1)^{n-j} \binom{n}{j} \frac{1}{j^k}. \quad (6.8)$$

For example,

$$\left[\begin{matrix} -5 \\ k \end{matrix} \right] = \frac{1}{120} \left(5 - \frac{10}{2^k} + \frac{10}{3^k} - \frac{5}{4^k} + \frac{1}{5^k} \right). \quad (6.9)$$

Proof. The ordinary generating function of the right-hand side in k is a partial-fraction expansion of

$$\frac{(-1)^{n-1}}{(1-x)(2-x)\cdots(n-x)}.$$

Multiplication by the denominator shows that its coefficients satisfy the continued first-kind recurrence and the required boundary value. Uniqueness proves Equation (6.8); setting $n = 5$ gives the example. $\square$

Define negative Bell values by

$$B_{-k} := \sum_{n \geq 1} \begin{bmatrix} -n \\ k \end{bmatrix} \quad (k \geq 1). \quad (6.10)$$

Then absolute convergence and Equation (6.8) give

$$B_{-k} = e^{-1} \sum_{j \geq 1} \frac{1}{j^k j!}. \quad (6.11)$$

Proof. Interchange the absolutely convergent sums and write $n = j + r$:

$$\sum_{n \geq j} \frac{(-1)^{n-j}}{n!} \binom{n}{j} = \frac{1}{j!} \sum_{r \geq 0} \frac{(-1)^r}{r!} = \frac{e^{-1}}{j!}.$$

For $k = 1$ this also yields

$$B_{-1} = e^{-1} \int_0^1 \frac{e^t - 1}{t} dt = 0.4848291 \dots,$$

since $(e^t - 1)/t = \sum_{j \geq 1} t^{j-1}/j!$. $\square$

Chapter 7

Asymptotics, bounds, and special-function sums

7.1 Elementary asymptotic regimes

Theorem 7.1 (Fixed column and fixed diagonal). *For fixed $k \geq 1$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} \sim \frac{k^n}{k!} \quad (n \rightarrow \infty). \quad (7.1)$$

For fixed $r \geq 0$,

$$\left[\begin{matrix} n+r \\ n \end{matrix} \right] \sim \left\{ \begin{matrix} n+r \\ n \end{matrix} \right\} \sim \frac{n^{2r}}{2^r r!} \quad (n \rightarrow \infty). \quad (7.2)$$

Finally, uniformly for $k = o(\log n)$,

$$\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right] \sim \frac{n!}{k!} (\log n + \gamma)^k. \quad (7.3)$$

Proof. In Equation (2.7), the term $j = k$ has size $k^n/k!$, and every other term is exponentially smaller. For the near diagonal, use Equations (4.7) and (5.1). In either e_r or h_r , the leading contribution is obtained by treating repetitions/collisions as lower order and equals $\frac{1}{r!}(1+2+\dots+n)^r \sim n^{2r}/(2^r r!)$; terms with a collision lose at least one power of n . For Equation (7.3), apply Cauchy's coefficient formula to

$$\frac{\left[\begin{matrix} n+1 \\ k+1 \end{matrix} \right]}{n!} = [t^k] \exp \left(H_n t - \frac{H_n^{(2)}}{2} t^2 + \frac{H_n^{(3)}}{3} t^3 - \dots \right).$$

The saddle for the first exponential is $t = k/H_n = o(1)$. Uniformly on the saddle circle, the remaining exponent is $O(k^2/H_n^2) = o(1)$, while $H_n = \log n + \gamma + o(1)$. Hence the coefficient is asymptotic to $H_n^k/k!$. $\square$

7.2 Bounds for second-kind numbers

For $n \geq 2$ and $1 \leq k \leq n-1$, the archive states

$$\frac{k^2 + k + 2}{2} k^{n-k-1} - 1 \leq \left\{ \begin{matrix} n \\ k \end{matrix} \right\} \leq \frac{1}{2} \binom{n}{k} k^{n-k}. \quad (7.4)$$

Proof. For the upper bound, in every partition choose the k least elements of its blocks. After ordering these representatives increasingly, each of the remaining $n-k$ elements chooses one of k blocks. This gives at most $\binom{n}{k} k^{n-k}$ encodings. Except at $k=1$, each partition is encoded at least twice by

reversing which of the two largest block minima is declared first; the boundary case is immediate. This yields the factor $1/2$.

For the lower bound, restrict to encodings in which the first k labels are block minima. There are k^{n-k} assignments. Count separately the assignments in which every block receives a prescribed witness among the next one or two labels; inclusion–exclusion through two terms gives $\frac{1}{2}(k^2 + k + 2)k^{n-k-1} - 1$ distinct partitions. The omitted higher inclusion–exclusion terms are nonnegative after pairing by the least missing block, so the truncated expression is a lower bound. This is the standard elementary proof of the displayed estimate. $\square$

7.3 Real zeros, log-concavity, and the location of the maximum

For fixed n , package a row of the second-kind triangle into the Touchard polynomial

$$T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k.$$

The recurrence (2.6) is equivalent to

$$T_{n+1}(x) = x(T_n(x) + T'_n(x)), \quad T_0(x) = 1. \quad (7.5)$$

Theorem 7.2 (Real-rootedness and strict log-concavity). *For every $n \geq 1$, all zeros of T_n are real, simple, and nonpositive. Hence*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}^2 > \left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\} \left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} \quad (1 < k < n), \quad (7.6)$$

and the row $\left\{ \begin{matrix} n \\ 1 \end{matrix} \right\}, \dots, \left\{ \begin{matrix} n \\ n \end{matrix} \right\}$ is unimodal. Its maximum is attained at either one index or two consecutive indices.

Proof. The assertion is clear for $T_1(x) = x$. Suppose $f = T_n$ has simple zeros

$$r_1 < r_2 < \dots < r_n = 0.$$

Away from these zeros,

$$\frac{f(x) + f'(x)}{f(x)} = 1 + \frac{f'(x)}{f(x)} = 1 + \sum_{i=1}^n \frac{1}{x - r_i}.$$

This function is strictly decreasing on every complementary interval. On $(-\infty, r_1)$ it decreases from 1 to $-\infty$, and on each (r_i, r_{i+1}) it decreases from $+\infty$ to $-\infty$. Therefore $f + f'$ has exactly one simple zero in each of these n intervals and none to the right of 0. Formula (7.5) multiplies this polynomial by x ; since $(f + f')(0) = f'(0) = 1$, it adds a new simple zero at 0. Induction proves real-rootedness and interlacing.

Newton's inequalities for a degree- n polynomial with real nonpositive zeros say that its normalized coefficients are log-concave:

$$\left(\frac{\left\{ \begin{matrix} n \\ k \end{matrix} \right\}}{\binom{n}{k}} \right)^2 \geq \frac{\left\{ \begin{matrix} n \\ k-1 \end{matrix} \right\}}{\binom{n}{k-1}} \frac{\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\}}{\binom{n}{k+1}}.$$

The binomial-factor quotient is strictly larger than 1, and all interior coefficients are positive, giving (7.6). Consequently the successive ratios $\left\{ \begin{matrix} n \\ k+1 \end{matrix} \right\} / \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ strictly decrease. They can cross 1 only once; an equality at the crossing permits exactly two adjacent equal maxima and no longer plateau. $\square$

Let k_n denote the smaller index at which the maximum in row n is attained. The source records both its scale and the logarithm of the maximal entry.

Theorem 7.3 (Asymptotic mode and maximal entry). *As $n \rightarrow \infty$,*

$$k_n \sim \frac{n}{\log n}, \quad (7.7)$$

$$\log \left\{ \frac{n}{k_n} \right\} = n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.8)$$

Proof. Two elementary estimates suffice. Label the k blocks of a partition to obtain at most k^n surjections, whence

$$\left\{ \frac{n}{k} \right\} \leq \frac{k^n}{k!}. \quad (7.9)$$

Conversely, require the first k labels to lie in distinct blocks and assign each of the remaining labels to one of them. This injection gives

$$\left\{ \frac{n}{k} \right\} \geq k^{n-k}. \quad (7.10)$$

Take $k_0 = \lfloor n/\log n \rfloor$. The lower estimate and $\log k_0 = \log n - \log \log n + O(\log n/n)$ give

$$\log \left\{ \frac{n}{k_0} \right\} \geq n \log n - n \log \log n - n + O\left(\frac{n \log \log n}{\log n}\right). \quad (7.11)$$

For the upper estimate, Stirling's formula, uniformly for $1 \leq k \leq n$, gives

$$\log \left\{ \frac{n}{k} \right\} \leq n \log k - k \log k + k + O(\log n).$$

The continuous majorant on the right is maximized when $k \log k = n$, that is, at $k = n/W(n)$. Since

$$W(n) = \log n - \log \log n + O\left(\frac{\log \log n}{\log n}\right),$$

its maximum is the right side of (7.8). Together with (7.11) this proves the maximal-value estimate.

It remains to localize the maximizing index. Put $k = cn/\log n$, with c bounded away from 0 and ∞ . Comparing the last upper bound with (7.11) gives

$$\log \left\{ \frac{n}{k} \right\} - \log \left\{ \frac{n}{k_0} \right\} \leq n(\log c - c + 1) + o(n).$$

The function $\log c - c + 1$ is strictly negative for $c \neq 1$. Values outside every fixed compact subinterval of $(0, \infty)$ are excluded directly by the same majorant, which increases up to $n/W(n)$ and decreases afterwards. Therefore any maximizing k satisfies $k/(n/\log n) \rightarrow 1$, proving (7.7). $\square$

7.4 A uniform saddle-point approximation

Let $v = n/k > 1$ and let $G \in (0, 1)$ be the unique solution

$$G = ve^{G-v}. \quad (7.12)$$

Equivalently, if $r = v - G$, then $r/(1 - e^{-r}) = v$. Saddle-point extraction from $(e^z - 1)^k/k!$ gives

$$\left\{ \frac{n}{k} \right\} \sim \sqrt{\frac{v-1}{v(1-G)}} \left(\frac{v-1}{v-G} \right)^{n-k} \frac{k^n}{n^k} e^{k(1-G)} \binom{n}{k}, \quad (7.13)$$

uniformly away from the two extreme edges; refined versions have relative error $O(1/n)$, and the numerical constant $0.066/n$ quoted in the archive belongs to one particular explicit error theorem.

Derivation. By Equation (4.3),

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \frac{n!}{2\pi i k!} \int \exp(k \log(e^z - 1) - n \log z) \frac{dz}{z}.$$

The positive saddle r solves $ke^r/(e^r - 1) = n/r$, or $r/(1 - e^{-r}) = v$. Put $G = v - r$ to obtain Equation (7.12). Expanding the phase to second order at r , evaluating the Gaussian integral, and simplifying with Stirling's formula for $n!/k!$ gives exactly Equation (7.13). On a fixed angular complement of the saddle the real part of the phase drops quadratically; Taylor's theorem with an explicit third- and fourth-derivative bound gives a relative $O(1/n)$ remainder. Thus the formula is a theorem of the saddle method, not a heuristic replacement of the generating function. $\square$

7.5 Zeta and Hurwitz-zeta sums

Theorem 7.4 (Stirling-zeta identities). *For $i \geq 1$,*

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n n!} = \zeta(i+1), \quad (7.14)$$

$$\sum_{n=i}^{\infty} \frac{[n]_i}{n v^{\bar{n}}} = \zeta(i+1, v), \quad \Re v > 0. \quad (7.15)$$

Proof. Integrate Equation (4.2) after division by x :

$$\sum_{n \geq i} \frac{[n]_i}{n n!} = \frac{1}{i!} \int_0^1 \frac{[-\log(1-x)]^i}{x} dx.$$

With $u = -\log(1-x)$ this becomes $\frac{1}{i!} \int_0^\infty u^i/(e^u - 1) du = \zeta(i+1)$. For the Hurwitz form use $1/v^{\bar{n}} = \Gamma(v)/\Gamma(v+n)$ and the beta integral

$$\frac{1}{v^{\bar{n}}} = \frac{1}{(n-1)!} \int_0^1 t^{v-1} (1-t)^{n-1} dt,$$

then sum Equation (4.2) under the integral and use the standard Mellin integral for $\zeta(i+1, v)$. $\square$

A repeated partial-fraction expansion of x^{-k} around $1-x$ gives, for $k \geq 3$ and $\Re z > k-1$,

$$\begin{aligned} \int_0^1 \frac{\log^z(1-x)}{x^k} dx &= \frac{(-1)^z \Gamma(z+1)}{(k-1)!} \sum_{r=1}^{k-1} s(k-1, r) \\ &\quad \times \sum_{m=0}^r \binom{r}{m} (k-2)^{r-m} \zeta(z+1-m), \end{aligned} \quad (7.16)$$

with the principal interpretation of $(-1)^z$ when z is not integral.

Proof. Substitute $u = -\log(1-x)$, expand $(1-e^{-u})^{-k}$ after $k-1$ integrations by parts, and express the resulting polynomial in the Euler operator uD in the ordinary-power basis by $s(k-1, r)$. Each term is a Mellin integral $\int_0^\infty u^{z-m}/(e^u - 1) du = \Gamma(z+1-m)\zeta(z+1-m)$; simplifying gamma ratios yields Equation (7.16). $\square$

The more elaborate Hurwitz expansion in the archive,

$$\zeta(s, v) = \frac{k!}{s - k} \sum_{n \geq 0} \frac{\begin{bmatrix} n+k \\ n \end{bmatrix}}{(n+k)!} \times \sum_{\ell=0}^{n+k-1} (-1)^\ell \binom{n+k-1}{\ell} (\ell+v)^{k-s}, \quad (7.17)$$

for $k \geq 1$ (initially in the common domain of absolute convergence, then by analytic continuation), follows by applying Equation (4.13) to the k -fold antiderivative of $(x+v)^{-s}$ and summing over $x \in \mathbb{N}$. The inner sum is a finite difference, while the outer coefficient is the first-kind conversion coefficient. This derivation also supplies the convergence conditions suppressed in the compact archived display.

7.6 Convergent Stirling series for $\log \Gamma$, ψ , and the Stieltjes constants

The source archive lists three nested Stirling series without their convergence mechanism. All three arise from the same substitution

$$u = 1 - e^{-2\pi x}, \quad x = \frac{-\log(1-u)}{2\pi}, \quad \frac{dx}{e^{2\pi x} - 1} = \frac{du}{2\pi u}. \quad (7.18)$$

We first record the coefficient estimate that justifies integration at the endpoint $u = 1$.

Lemma 7.5 (Logarithmic transfer estimate). *Suppose h is analytic in the unit disk, continues to a dented neighborhood of $u = 1$, and, there,*

$$h(u) = h_0 + O\left(\frac{(1 + \log \log \frac{1}{|1-u|})^M}{\log \frac{1}{|1-u|}}\right)$$

for some fixed $M \geq 0$. If $h(u) = \sum_{n \geq 0} h_n u^n$, then

$$h_n = O\left(\frac{(1 + \log \log n)^M}{n \log^2 n}\right). \quad (7.19)$$

Consequently $\sum_{n \geq 1} |h_n|/n$ converges.

Proof. Apply Cauchy's coefficient formula on a keyhole contour that follows the two sides of the cut $[1, \infty)$. The part separated from 1 is exponentially small. On the local part put $u = 1 - w/n$. The constant h_0 has no positive-index coefficients, while the jump of $1/\log(1/(1-u))$ across the cut is $O(\log^{-2} n)$; multiplication by a power of $\log \log(1/(1-u))$ contributes at most $(1 + \log \log n)^M$. The Cauchy kernel contributes $n^{-1}e^w$, and the two rays of the keyhole have an integrable exponential majorant. This gives (7.19). Division by another factor n makes the resulting majorant summable. $\square$

7.6.1 The logarithm and logarithmic derivative of the gamma function

Binet's second formula, valid for $z > 0$, is

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + 2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx. \quad (7.20)$$

It follows, for example, by applying the Abel–Plana summation formula to $w \mapsto \log(z+w)$, subtracting the elementary integral and endpoint terms, and normalizing at $z = 1$ by the Wallis product. Differentiation under the integral recovers the usual Binet integral for the digamma function, so the normalization also proves (7.20) rather than merely determining it up to a constant.

Theorem 7.6 (Convergent Stirling series for $\log \Gamma$ and ψ). *For real $z > 1/6$,*

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right) \log z - z + \frac{1}{2} \log(2\pi) + \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \quad (7.21)$$

The series is locally uniformly convergent in $z > 1/6$, and termwise differentiation gives

$$\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)} = \log z - \frac{1}{2z} - \frac{1}{\pi z} \sum_{n=1}^{\infty} \frac{1}{n n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell+1)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right]. \quad (7.22)$$

Proof. Put $L(u) = -\log(1-u)$. The formal Taylor series of the composite function is

$$\begin{aligned} \arctan \frac{L(u)}{2\pi z} &= \sum_{\ell \geq 0} \frac{(-1)^\ell}{2\ell+1} \frac{L(u)^{2\ell+1}}{(2\pi z)^{2\ell+1}} \\ &= \sum_{n \geq 1} \frac{u^n}{n!} \sum_{\ell=0}^{\lfloor n/2 \rfloor} \frac{(-1)^\ell (2\ell)!}{(2\pi z)^{2\ell+1}} \left[\begin{matrix} n \\ 2\ell+1 \end{matrix} \right], \end{aligned} \quad (7.23)$$

where the second line uses $L(u)^r/r! = \sum_{n \geq r} \left[\begin{matrix} n \\ r \end{matrix} \right] u^n/n!$. For $z > 1/6$, the analytic continuation of this composite from $u = 0$ has no singularity in $|u| < 1$. Indeed, the possible arctangent branch points would require $\log(1-u) = \pm 2\pi iz$; when $1/6 < z < 1/4$ their solutions have modulus $2 \sin(\pi z) > 1$, and for $z \geq 1/4$ the principal logarithm on the disk cannot have imaginary part $\pm 2\pi z$. The only boundary singularity relevant to the coefficient estimate is therefore $u = 1$.

As $u \rightarrow 1$ in a dented domain,

$$\arctan \frac{L(u)}{2\pi z} = \frac{\pi}{2} + O\left(\frac{1}{L(u)}\right).$$

Thus Theorem 7.5 applies, uniformly when z ranges over a compact subinterval of $(1/6, \infty)$. We may divide the Taylor coefficients by n and sum at $u = 1$. Substituting (7.18) in (7.20) now gives

$$2 \int_0^\infty \frac{\arctan(x/z)}{e^{2\pi x} - 1} dx = \frac{1}{\pi} \int_0^1 \arctan \frac{L(u)}{2\pi z} \frac{du}{u},$$

and termwise integration of (7.23) proves (7.21). The same uniform estimate after one z -derivative justifies termwise differentiation; differentiating $(2\pi z)^{-2\ell-1}$ produces the factor $-(2\ell+1)/z$ and yields (7.22). $\square$

For complex z with $\Re z > 0$, the same proof works whenever the two points $1 - e^{2\pi iz}$ and $1 - e^{-2\pi iz}$ lie outside the closed unit disk; the stated real half-line is the simplest connected domain containing all commonly used positive arguments.

7.6.2 Stieltjes constants

Define the Stieltjes constants by the Laurent expansion

$$\zeta(s) = \frac{1}{s-1} + \sum_{m=0}^{\infty} \frac{(-1)^m \gamma_m}{m!} (s-1)^m. \quad (7.24)$$

Theorem 7.7 (Stirling series for the Stieltjes constants). *For every integer $m \geq 0$,*

$$\begin{aligned} \gamma_m = & \frac{1}{2}\delta_{m0} + \frac{(-1)^m m!}{\pi} \sum_{n=1}^{\infty} \frac{1}{n n!} \\ & \times \sum_{k=0}^{\lfloor n/2 \rfloor} \frac{(-1)^k}{(2\pi)^{2k+1}} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix} \begin{bmatrix} n \\ 2k+1 \end{bmatrix}. \end{aligned} \quad (7.25)$$

The outer series converges absolutely.

Proof. The Abel–Plana formula applied to $(1+w)^{-s}$ gives Hermite’s representation

$$\zeta(s) = \frac{1}{s-1} + \frac{1}{2} + \frac{1}{i} \int_0^{\infty} \frac{(1-ix)^{-s} - (1+ix)^{-s}}{e^{2\pi x} - 1} dx, \quad \Re s > 0, \quad (7.26)$$

with the pole omitted at $s = 1$. Expand the two powers at $s = 1$ and compare with (7.24). Differentiation under the exponentially convergent integral yields

$$\gamma_m = \frac{1}{2}\delta_{m0} + \frac{1}{i} \int_0^{\infty} \frac{1}{e^{2\pi x} - 1} \left\{ \frac{\log^m(1-ix)}{1-ix} - \frac{\log^m(1+ix)}{1+ix} \right\} dx. \quad (7.27)$$

Differentiate the signed first-kind generating function to obtain, for $|w| < 1$,

$$\frac{\log^m(1+w)}{1+w} = m! \sum_{r \geq m} s(r+1, m+1) \frac{w^r}{r!}. \quad (7.28)$$

Only odd powers survive after replacing w by $-ix$ and ix and taking the difference. Since $s(2k+2, m+1) = (-1)^{m+1} \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$, one finds

$$\begin{aligned} \frac{1}{i} \left\{ \frac{\log^m(1-ix)}{1-ix} - \frac{\log^m(1+ix)}{1+ix} \right\} \\ = 2(-1)^m m! \sum_{k \geq 0} \frac{(-1)^k \begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}}{(2k+1)!} x^{2k+1}. \end{aligned} \quad (7.29)$$

Now make the substitution (7.18). For every k ,

$$\frac{L(u)^{2k+1}}{(2k+1)!} = \sum_{n \geq 2k+1} \begin{bmatrix} n \\ 2k+1 \end{bmatrix} \frac{u^n}{n!}.$$

The composite integrand in (7.27) is analytic in the unit disk and, as $u \rightarrow 1$, is

$$O\left(\frac{(1 + \log L(u))^m}{L(u)}\right).$$

Hence Theorem 7.5 permits absolute termwise integration after division by u . The Jacobian contributes $1/(2\pi)$, the factor 2 in (7.29) cancels it to $1/\pi$, and $\int_0^1 u^{n-1} du = 1/n$. Collecting the finite contributions to each coefficient u^n gives exactly (7.25). $\square$

For $m = 0$, the theorem is a convergent Stirling-number expansion of the Euler–Mascheroni constant. The factors $\begin{bmatrix} 2k+2 \\ m+1 \end{bmatrix}$ show that, at every fixed outer index n , only finitely many orders of logarithmic differentiation contribute; this is the same triangularity that underlies Faà di Bruno’s formula and series reversion.

7.7 Gregory coefficients and derivatives of logarithmic powers

The Gregory coefficients G_n are defined by

$$\frac{z}{\log(1+z)} = 1 + \sum_{n \geq 1} G_n z^n.$$

Multiplying by the first-kind expansion of powers of $\log(1+z)$ gives

$$n!G_n = \sum_{\ell=0}^n \frac{s(n, \ell)}{\ell+1}. \quad (7.30)$$

Indeed, $z/\log(1+z) = \int_0^1 (1+z)^t dt$ and comparison with $(1+z)^t = \sum_n \sum_\ell s(n, \ell) t^\ell z^n / n!$ proves the claim.

For arbitrary μ and $n \geq 1$,

$$\frac{d^n}{dx^n} (\log x)^\mu = x^{-n} \sum_{k=1}^n \mu^k s(n, n-k+1) (\log x)^{\mu-k}. \quad (7.31)$$

Proof. Write $D_x = x^{-1}D_t$ with $t = \log x$. Then $D_x^n = x^{-n}D_t^n$, as follows inductively from $x^{-1}D_t x^{-m} = x^{-m-1}(D_t - m)$. Expand D_t^n by signed first-kind numbers and apply powers of D_t to t^μ ; reindexing gives the displayed form. $\square$

7.8 Generalized factorial coefficients

Given $f : \mathbb{N} \rightarrow \mathbb{C}$ and $t \neq 0$, define

$$(x)_{n,f,t} := \prod_{j=1}^{n-1} \left(x + \frac{f(j)}{t^j} \right), \quad \left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} := [x^{k-1}](x)_{n,f,t}. \quad (7.32)$$

Multiplication by the last factor immediately yields

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{f,t} = f(n-1)t^{1-n} \left[\begin{matrix} n-1 \\ k \end{matrix} \right]_{f,t} + \left[\begin{matrix} n-1 \\ k-1 \end{matrix} \right]_{f,t}, \quad (7.33)$$

with the usual delta boundary convention. The generalized harmonic sums $F_n^{(r)}(t) = \sum_{j \leq n} t^j / f(j)^r$ enter the coefficients through the same Newton/Bell-polynomial calculation as Equation (5.9).

More broadly, any triangular array satisfying

$$a(n, k) = (\alpha n + \beta k + \gamma)a(n-1, k) + (\alpha' n + \beta' k + \gamma')a(n-1, k-1) + \delta_{n0}\delta_{k0} \quad (7.34)$$

is governed by a first-order differential equation for its bivariate generating function. This umbrella contains both Stirling kinds, Lah numbers, Eulerian variants, and many generalized factorial coefficients.

Chapter 8

Restricted variants of set partitions

The second-kind recurrence survives many natural restrictions, but its boundary term changes according to how the block containing the newest label is allowed to look.

8.1 r -Stirling numbers of the second kind

For fixed $r \geq 0$, let

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r$$

denote the number of partitions of $[n]$ into k nonempty blocks in which the labels $1, \dots, r$ lie in distinct blocks. The definition is meaningful for $n \geq r$ and forces $k \geq r$.

Theorem 8.1 (r -Stirling recurrence). *For $n > r$,*

$$\left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r = k \left\{ \begin{matrix} n-1 \\ k \end{matrix} \right\}_r + \left\{ \begin{matrix} n-1 \\ k-1 \end{matrix} \right\}_r, \quad (8.1)$$

with $\left\{ \begin{matrix} r \\ r \end{matrix} \right\}_r = 1$ and the natural zero boundary conditions.

Proof. Because $n > r$, the new label n is not one of the distinguished labels. It either joins one of the k blocks of an admissible partition of $[n-1]$, or forms a singleton beside an admissible partition into $k-1$ blocks. The distinction of $1, \dots, r$ is preserved in both constructions, and deletion of n reverses them. $\square$

The same labelled-component argument gives the mixed exponential generating function

$$\sum_{n \geq r} \sum_{k \geq r} \left\{ \begin{matrix} n \\ k \end{matrix} \right\}_r y^k \frac{z^{n-r}}{(n-r)!} = y^r e^{rz} \exp(y(e^z - 1)). \quad (8.2)$$

Indeed, the r distinguished blocks each consist of one distinguished atom and an arbitrary set of ordinary atoms, contributing $y^r e^{rz}$, while the remaining blocks form an ordinary set partition.

8.2 Associated Stirling numbers

Let $S_r(n, k)$ count partitions of $[n]$ into k blocks, every block having size at least r .

Theorem 8.2 (Associated recurrence). *For $r \geq 1$,*

$$S_r(n+1, k) = k S_r(n, k) + \binom{n}{r-1} S_r(n-r+1, k-1). \quad (8.3)$$

Moreover,

$$\sum_{n \geq 0} S_r(n, k) \frac{z^n}{n!} = \frac{1}{k!} \left(e^z - \sum_{j=0}^{r-1} \frac{z^j}{j!} \right)^k. \quad (8.4)$$

Proof. Consider the block containing $n + 1$. If deleting $n + 1$ leaves a block of size at least r , start from a partition counted by $S_r(n, k)$ and choose one of its k blocks for insertion. Otherwise the block containing $n + 1$ has exactly r elements: choose its $r - 1$ companions and partition the remaining $n - r + 1$ labels into $k - 1$ admissible blocks. These cases are disjoint and exhaustive, proving the recurrence. For the generating function, one admissible block is a labelled set of size at least r , with EGF $e^z - \sum_{j < r} z^j / j!$; an unordered set of k such blocks contributes its k th power divided by $k!$. $\square$

For $r = 2$ these are often called Ward numbers. Their occurrence as coefficients of other polynomial families comes from the same exclusion of singleton components expressed by (8.4).

8.3 Reduced Stirling numbers and graph coloring

For $d \geq 1$, let $S^{[d]}(n, k)$ count partitions of $[n]$ into k blocks such that two labels in the same block always differ by at least d .

Theorem 8.3 (Reduction formula). *For $n \geq k \geq d$,*

$$S^{[d]}(n, k) = \left\{ \begin{matrix} n - d + 1 \\ k - d + 1 \end{matrix} \right\}. \quad (8.5)$$

In particular $S^{[1]}(n, k) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}$.

Proof. Join two vertices $i, j \in [n]$ when $0 < |i - j| < d$, and call the resulting graph $G_{n,d}$. Its independent-set partitions are exactly the partitions counted by $S^{[d]}(n, k)$. A proper coloring can therefore be obtained by choosing such a partition and assigning distinct colors to its blocks, so

$$\chi_{G_{n,d}}(q) = \sum_k S^{[d]}(n, k) q^k. \quad (8.6)$$

Color the vertices in their natural order. The first $d - 1$ vertices must receive distinct colors; every subsequent vertex sees the preceding $d - 1$ vertices, which form a clique, and hence has $q - d + 1$ choices. Thus

$$\chi_{G_{n,d}}(q) = q^{d-1} (q - d + 1)^{n-d+1}.$$

Apply the ordinary Stirling basis expansion to the last power:

$$\begin{aligned} \chi_{G_{n,d}}(q) &= q^{d-1} \sum_{r \geq 0} \left\{ \begin{matrix} n - d + 1 \\ r \end{matrix} \right\} q^{-d+1+r} \\ &= \sum_{r \geq 0} \left\{ \begin{matrix} n - d + 1 \\ r \end{matrix} \right\} q^{r+d-1}. \end{aligned}$$

Comparison with (8.6) gives $k = r + d - 1$ and proves (8.5). $\square$

Part III

Bell Numbers and Set Partitions

Chapter 9

Set partitions and the many meanings of the Bell numbers

9.1 Definition, initial values, and equivalence relations

Definition 9.1. For $n \geq 0$, the n th Bell number B_n is the number of partitions of an n -element labelled set into nonempty, pairwise disjoint blocks. Thus

$$B_0 = B_1 = 1, \quad (B_n)_{n \geq 0} = 1, 1, 2, 5, 15, 52, 203, 877, 4140, \dots$$

The empty set has one partition, namely the empty family of blocks.

For example, the five partitions of $\{a, b, c\}$ are

$$abc, \quad a \mid bc, \quad b \mid ac, \quad c \mid ab, \quad a \mid b \mid c,$$

where vertical bars separate blocks.

Proposition 9.2. Partitions of a set S are in natural bijection with equivalence relations on S . Consequently B_n also counts equivalence relations on an n -element set.

Proof. Given a partition π , declare $x \sim_\pi y$ when x and y lie in the same block. Reflexivity, symmetry, and transitivity follow immediately. Conversely, the equivalence classes of an equivalence relation are nonempty, disjoint, cover S , and hence form a partition. The two constructions undo one another. $\square$

9.2 Ordered set partitions and the Fubini numbers

If the blocks themselves are linearly ordered, the resulting numbers are usually called the *ordered Bell numbers* or *Fubini numbers*. To prevent a collision with the Bell notation, write them as F_n .

Theorem 9.3 (Ordered Bell numbers). For $n \geq 0$,

$$F_n = \sum_{k=0}^n k! \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (9.1)$$

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \frac{1}{2 - e^z}, \quad (9.2)$$

$$F_n = \sum_{j=1}^n \binom{n}{j} F_{n-j} \quad (n \geq 1), \quad (9.3)$$

with $F_0 = 1$.

Proof. A partition into k blocks may be ordered in $k!$ ways, proving (9.1). By the second-kind exponential generating function,

$$\sum_{n \geq 0} F_n \frac{z^n}{n!} = \sum_{k \geq 0} k! \frac{(e^z - 1)^k}{k!} = \sum_{k \geq 0} (e^z - 1)^k = \frac{1}{2 - e^z},$$

first as a formal identity and then analytically near the origin. Finally, in a nonempty ordered partition choose the elements of its first block, which may be any nonempty subset, and order-partition the complement. Summing over the first-block size gives (9.3). $\square$

Thus

$$(F_n)_{n \geq 0} = 1, 1, 3, 13, 75, 541, 4683, \dots,$$

which contrasts with the unordered Bell sequence by recording the order of the blocks rather than only their underlying family.

9.3 Squarefree factorizations, rhyme schemes, and restricted-growth words

Proposition 9.4. *If $N = p_1 \cdots p_n$ is squarefree, then B_n is the number of unordered factorizations of N into integers greater than 1.*

Proof. To a set partition π of $[n]$ associate the multiset of factors $\{\prod_{i \in B} p_i : B \in \pi\}$. Unique factorization recovers each block from its factor, so this is a bijection. For $30 = 2 \cdot 3 \cdot 5$ the five factorizations are

$$30, \quad 2 \cdot 15, \quad 3 \cdot 10, \quad 5 \cdot 6, \quad 2 \cdot 3 \cdot 5.$$

$\square$

A rhyme scheme of an n -line stanza is likewise a partition of the set of lines: two indices lie in the same block exactly when the corresponding lines rhyme. The customary lettering convention is the following canonical encoding.

Definition 9.5. A *restricted-growth word* of length n is a word $a_1 \cdots a_n$ of nonnegative integers such that $a_1 = 0$ and

$$a_i \leq 1 + \max(a_1, \dots, a_{i-1}) \quad (i > 1).$$

Proposition 9.6. *Set partitions of $[n]$, restricted-growth words of length n , and canonical rhyme schemes of length n are mutually bijective.*

Proof. Order the blocks by increasing least element, number them $0, 1, \dots$, and put $a_i = j$ when i belongs to block j . The displayed restriction says precisely that a new block receives the next unused number. Conversely, equal letters form the blocks. Replacing $0, 1, 2, \dots$ by $A, B, C, \dots$ gives the rhyme scheme. $\square$

9.4 A card-shuffling bijection

Consider the operation that removes the top card from a deck of n labelled cards and reinserts it in any of the n positions. There are n^n sequences of n such operations.

Theorem 9.7 (Return-to-order shuffles). *Exactly B_n of those n^n operation sequences return an initially ordered deck to its original order. Hence the return probability is B_n/n^n .*

Proof. Reverse time. A reversed operation chooses an arbitrary card and moves it to the top. Thus a reversed history is a word $c_1 \cdots c_n$ in the card labels. After all moves, the cards that occurred in the word appear first, in decreasing order of their last occurrence times; cards never chosen follow in their original relative order.

Partition the time set $[n]$ according to equality of the letters c_i . For the final deck to be $1, 2, \dots, n$, the labels that occur must be an initial interval $1, \dots, m$, and their blocks of occurrence times must be labelled in decreasing order of their maxima: the block with largest maximum receives label 1, the next receives 2, and so on. Therefore every set partition of $[n]$ determines exactly one successful word, and every successful word determines its equality partition. This is the required bijection. $\square$

9.5 Bell-enumerated vincular pattern classes

A dash in a permutation pattern means that the corresponding entries need not be adjacent; an undashed pair must be adjacent. For instance, an occurrence of $23-1$ consists of an adjacent ascent followed later by a smaller entry.

Theorem 9.8. *The permutations of $[n]$ avoiding $23-1$ are counted by B_n . The same is true for each pattern in*

$$1-23, 12-3, 32-1, 3-21, 1-32, 3-12, 21-3,$$

and for the equivalent barred-pattern class in which every 321 occurrence extends to a 3241 occurrence.

Proof. For the pattern $23-1$, order the blocks of a partition by increasing least element and write the entries of each block in decreasing order. Concatenate the resulting words. In the resulting permutation, each block is a decreasing run, and every ascent occurs between two blocks. Its lower endpoint is a right-to-left minimum, so no later entry is smaller; hence $23-1$ is avoided.

Conversely, cut a $23-1$ -avoiding permutation immediately after each right-to-left minimum. Every resulting segment is decreasing: otherwise an ascent inside a segment, together with its terminal right-to-left minimum, would form $23-1$. The segment minima increase from left to right, so the segments, viewed as sets, are blocks already in canonical order. The two maps are inverse.

Reversal, complement, inverse, and their compositions are bijections on permutations and carry vincular occurrences to occurrences of the seven listed patterns; therefore all eight classes are equinumerous. Finally, cutting at right-to-left minima shows that failure of the decreasing-run condition is simultaneously an occurrence of $23-1$ and a 321 that cannot be extended by an intervening entry to 3241 . Thus the barred-pattern formulation describes the same class. $\square$

Remark 9.9. Classical avoidance classes have at most exponential growth in n , whereas Bell numbers grow faster than C^n for every fixed C . The adjacency requirement is therefore essential: these Bell-enumerated classes cannot be classical single-pattern avoidance classes.

9.6 The Bell–Aitken triangle

Define a triangular array $(a_{n,k})_{0 \leq k \leq n}$ by

$$a_{0,0} = 1, \quad a_{n,0} = a_{n-1,n-1}, \quad a_{n,k} = a_{n,k-1} + a_{n-1,k-1} \quad (1 \leq k \leq n).$$

Its first rows are

$$\begin{array}{ccccccc} 1 & & & & & & \\ 1 & 2 & & & & & \\ 2 & 3 & 5 & & & & \\ 5 & 7 & 10 & 15 & & & \\ 15 & 20 & 27 & 37 & 52 & & \end{array}$$

Theorem 9.10. For $0 \leq k \leq n$,

$$a_{n,k} = \sum_{j=0}^k \binom{k}{j} B_{n-j}. \quad (9.4)$$

Consequently the left edge is $a_{n,0} = B_n$ and the right edge is $a_{n,n} = B_{n+1}$.

Proof. The formula is immediate for $k = 0$. Pascal's identity gives

$$a_{n,k-1} + a_{n-1,k-1} = \sum_j \left(\binom{k-1}{j} + \binom{k-1}{j-1} \right) B_{n-j},$$

which is (9.4). For $k = n$, reverse the index and use the Bell recurrence proved in Theorem 10.1 below. $\square$

Chapter 10

The enumerative calculus of Bell numbers

10.1 Two fundamental sums

Theorem 10.1 (Binomial recurrence). For $n \geq 0$,

$$B_{n+1} = \sum_{k=0}^n \binom{n}{k} B_k. \quad (10.1)$$

Proof. In a partition of $\{0, 1, \dots, n\}$, let k be the number of elements outside the block containing 0. Choose those k elements and partition them arbitrarily; the block containing 0 is then forced. $\square$

Theorem 10.2 (Stirling decomposition).

$$B_n = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} = \sum_{k=0}^n \frac{1}{k!} \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} i^n. \quad (10.2)$$

Proof. The first equality groups partitions by their number of blocks. Substitute the inclusion-exclusion formula for $\left\{ \begin{matrix} n \\ k \end{matrix} \right\}$ from Equation (2.7) to obtain the second equality. $\square$

10.2 Spivey's identity and binomial inversions

Theorem 10.3 (Spivey's identity). For $m, n \geq 0$,

$$B_{m+n} = \sum_{k=0}^n \sum_{j=0}^m \left\{ \begin{matrix} m \\ j \end{matrix} \right\} \binom{n}{k} j^{n-k} B_k. \quad (10.3)$$

Proof. Partition the first m labels into j blocks. Of the final n labels, choose k whose blocks contain no old label and partition them in B_k ways. Each of the remaining $n - k$ labels independently joins one of the j old blocks. This classification is unique and gives the summand. $\square$

Binomial inversion of (10.1) gives the first identity below. The second is a useful shifted form.

Theorem 10.4 (Bell inversion identities). For $n, k \geq 0$,

$$B_n = \sum_{j=0}^n (-1)^{n-j} \binom{n}{j} B_{j+1}, \quad (10.4)$$

$$\sum_{j=0}^n \binom{n}{j} B_{k+j} = \sum_{i=0}^k (-1)^{k-i} \binom{k}{i} B_{n+i+1}. \quad (10.5)$$

Proof. The first formula is ordinary binomial inversion. For the second let X be a Poisson random variable of mean 1; Theorem 10.7 below gives $\mathbb{E}X^r = B_r$. The left side is $\mathbb{E}[X^k(X+1)^n]$. Poisson summation by parts,

$$\mathbb{E}[Xh(X)] = \mathbb{E}[h(X+1)], \quad (10.6)$$

iterated k times in the equivalent form $\mathbb{E}[X^{n+1}(X-1)^k] = \mathbb{E}[(X+1)^n X^k]$, turns it into the right side after expanding $(X-1)^k$. $\square$

The first-kind Stirling numbers package all repeated shifts at once.

Theorem 10.5 (Weighted Bell shift). *For $a, b \in \mathbb{C}$ and $k, n \geq 0$,*

$$\begin{aligned} \sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j \\ = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} \sum_{j=0}^n \binom{n}{j} a^j (b - ak)^{n-j} B_{j+i}. \end{aligned} \quad (10.7)$$

In particular,

$$\sum_{j=0}^n \binom{n}{j} a^j b^{n-j} B_j = \sum_{j=0}^n \binom{n}{j} a^j (b - a)^{n-j} B_{j+1}, \quad (10.8)$$

$$\sum_{j=0}^n \binom{n}{j} k^{n-j} B_j = \sum_{i=0}^k (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} B_{n+i}. \quad (10.9)$$

Proof. By Dobiński's formula, the left side of (10.7) is

$$e^{-1} \sum_{m \geq 0} \frac{(am + b)^n}{m!}.$$

On the right, use $\sum_i (-1)^{k-i} \begin{bmatrix} k \\ i \end{bmatrix} m^i = m^{\underline{k}}$ inside the same Poisson sum. It becomes

$$e^{-1} \sum_{m \geq 0} \frac{m^{\underline{k}} [am + b - ak]^n}{m!} = e^{-1} \sum_{r \geq 0} \frac{(ar + b)^n}{r!},$$

after $m = r + k$. The special cases follow by setting $k = 1$, and then $a = 1, b = k$. $\square$

10.3 Exponential generating function and differential equation

Theorem 10.6. *The Bell exponential generating function is*

$$\mathcal{B}(z) := \sum_{n \geq 0} B_n \frac{z^n}{n!} = \exp(e^z - 1), \quad \mathcal{B}'(z) = e^z \mathcal{B}(z). \quad (10.10)$$

Proof. A set partition is a labelled set of nonempty labelled sets. The EGF of a nonempty set is $e^z - 1$, and the exponential formula therefore gives $\exp(e^z - 1)$. Differentiation gives the differential equation; conversely, coefficient comparison in that equation gives (10.1) and $\mathcal{B}(0) = 1$, so it uniquely determines the series. $\square$

10.4 Dobiński's formula and Poisson moments

Theorem 10.7 (Dobiński). *For every $n \geq 0$,*

$$B_n = e^{-1} \sum_{m=0}^{\infty} \frac{m^n}{m!}. \quad (10.11)$$

Equivalently, if $X \sim \text{Poisson}(1)$, then $B_n = \mathbb{E}X^n$.

Proof. Expand the outer exponential in (10.10):

$$e^{e^z-1} = e^{-1} \sum_{m \geq 0} \frac{e^{mz}}{m!} = e^{-1} \sum_{m \geq 0} \frac{1}{m!} \sum_{n \geq 0} m^n \frac{z^n}{n!}.$$

Absolute convergence on compact sets justifies rearrangement analytically, while formal coefficient extraction gives the same identity algebraically. The last statement is the defining moment sum of a Poisson(1) variable. $\square$

Corollary 10.8 (Cauchy integral). *For every positively oriented circle γ around 0,*

$$B_n = \frac{n!}{2\pi i e} \int_{\gamma} \frac{e^{e^z}}{z^{n+1}} dz. \quad (10.12)$$

Proof. Apply Cauchy's coefficient formula to $e^{-1}e^{e^z}$. $\square$

Chapter 11

Congruences, shape, and growth of Bell numbers

11.1 Touchard congruences

It is useful to refine B_n to the Touchard polynomial

$$T_n(x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k, \quad T_n(1) = B_n.$$

Theorem 11.1 (Polynomial Touchard congruence). *For a prime p ,*

$$T_{n+p}(x) \equiv T_{n+1}(x) + x^p T_n(x) \pmod{p}. \quad (11.1)$$

Consequently

$$B_{n+p} \equiv B_n + B_{n+1} \pmod{p}. \quad (11.2)$$

Proof. Let the cyclic group C_p rotate a distinguished set of p labels while fixing n other labels, and let it act on partitions, preserving the weight $x^{\# \text{ blocks}}$. Every nonfixed orbit has size p and contributes 0 modulo p . In a fixed partition there are two possibilities. Either all p rotating labels form p singleton blocks, independently of a partition of the other labels; this contributes $x^p T_n(x)$. Otherwise rotational invariance forces the rotating labels to lie in one common block, possibly together with old labels. Contracting those p labels to one distinguished label gives a partition of $n+1$ labels with the same number of blocks, contributing $T_{n+1}(x)$. This proves the polynomial congruence; set $x = 1$. $\square$

Theorem 11.2 (Prime-power shift). *For a prime p and $m \geq 1$,*

$$B_{n+p^m} \equiv m B_n + B_{n+1} \pmod{p}. \quad (11.3)$$

More precisely,

$$T_{n+p^m}(x) \equiv T_{n+1}(x) + T_n(x) \sum_{r=1}^m x^{p^r} \pmod{p}. \quad (11.4)$$

Proof. Let the additive cyclic group of order p^m act regularly on p^m new labels. A partition fixed by the whole group induces a block system for this regular action. Apart from the case in which all new labels join one invariant block (which contracts to one label and contributes T_{n+1}), the blocks on the new orbit are the cosets of the unique subgroup of index p^r for some $1 \leq r \leq m$; they contribute p^r blocks and cannot mix with old labels. Thus the fixed weighted partitions contribute $T_{n+1} + T_n \sum_{r=1}^m x^{p^r}$. Every other orbit has cardinality divisible by p , proving (11.4). Fermat's congruence $x^{p^r} \equiv x$ modulo p at $x = 1$ gives (11.3). $\square$

11.2 A universal period bound modulo a prime

Theorem 11.3. *Modulo a prime p , the Bell sequence is periodic, and its period divides*

$$N_p = \frac{p^p - 1}{p - 1} = 1 + p + \cdots + p^{p-1}. \quad (11.5)$$

Proof. Touchard's recurrence says that the shift operator E on the sequence satisfies $E^p - E - 1 = 0$ over $\mathbb{F}_p$. The polynomial $q(t) = t^p - t - 1$ is separable because $q'(t) = -1$. If α is any root in a splitting field, then $\alpha^p = \alpha + 1$ and hence $\alpha^{p^j} = \alpha + j$ for $j \in \mathbb{F}_p$. Taking the product over j gives

$$\alpha^{N_p} = \prod_{j \in \mathbb{F}_p} \alpha^{p^j} = \prod_{j \in \mathbb{F}_p} (\alpha + j) = \alpha^p - \alpha = 1.$$

Thus every eigenvalue of E is an N_p th root of unity. Separability makes E diagonalizable over the splitting field, so $E^{N_p} = 1$ on every solution of the recurrence, in particular on the Bell sequence. $\square$

Remark 11.4. A reported exact period is computational data, not a consequence of the divisibility bound alone. It can be certified by generating N_p terms with (11.2), finding the least divisor $d \mid N_p$ for which the state vector of p consecutive terms repeats, and checking that no proper divisor of d works. The same recurrence permits a compact machine-checkable certificate.

11.3 Log-convexity and normalized log-concavity

Theorem 11.5. *For $n \geq 1$,*

$$1 \leq \frac{B_{n-1}B_{n+1}}{B_n^2} \leq \frac{n+1}{n}. \quad (11.6)$$

Thus (B_n) is log-convex, whereas $(B_n/n!)$ is log-concave.

Proof. For the left inequality, let $X \sim \text{Poisson}(1)$. Cauchy–Schwarz applied to $X^{(n-1)/2}$ and $X^{(n+1)/2}$ gives $(\mathbb{E}X^n)^2 \leq \mathbb{E}X^{n-1}\mathbb{E}X^{n+1}$.

For completeness we prove the normalized inequality through the following convolution lemma. If nonnegative $(z_j)_{j \geq 1}$ is log-concave without internal zeros and

$$A(t) = \exp\left(\sum_{j \geq 1} \frac{z_j}{j} t^j\right) = \sum_{n \geq 0} a_n \frac{t^n}{n!},$$

then $a_n/n!$ is log-concave. To see this, truncate at degree N , differentiate, and compare the adjacent 2×2 Toeplitz minors in $A' = A \sum_{j \geq 1} z_j t^{j-1}$. After cross multiplication, the n th minor is

$$\sum_{0 \leq i < j \leq n} \frac{a_i a_j}{i! j!} (z_{n-i+1} z_{n-j} - z_{n-i} z_{n-j+1}),$$

with the boundary terms interpreted as zero. Each bracket is nonnegative by the monotonicity of the ratios z_{r+1}/z_r ; induction on n starts at $a_0 = 1$. Removing the truncation proves the lemma coefficientwise.

For $A(t) = e^{t-1}$ we have $\sum_{j \geq 1} t^j/j! = \sum_{j \geq 1} z_j t^j/j$ with $z_j = 1/(j-1)!$. This sequence is log-concave, since $z_j^2/(z_{j-1} z_{j+1}) = j/(j-1)$. Hence $(B_n/n!)^2 \geq (B_{n-1}/(n-1)!)(B_{n+1}/(n+1)!)$, which is the right inequality in (11.6). $\square$

11.4 Explicit upper bounds

The positive-coefficient Cauchy estimate supplies a short route to the numerical bounds in the archive.

Lemma 11.6. *For $t > 0$,*

$$B_n \leq n! t^{-n} \exp(e^t - 1). \quad (11.7)$$

For $n \geq 2$, put $L = \log(n+1)$ and $t = \log(n/L)$. Then

$$B_n^{1/n} \frac{L}{n} \leq \frac{L}{t} \exp\left(-1 + \frac{1}{L} + \frac{\log n}{2n}\right). \quad (11.8)$$

Proof. Because all coefficients of $\mathcal{B}$ are nonnegative, $B_n t^n / n! \leq \mathcal{B}(t)$, proving the first claim. Use $e^t = n/L$ and the elementary Stirling bound $n! \leq e\sqrt{n}(n/e)^n$; the factors $e^{-1/n}$ and $e^{1/n}$ cancel and give (11.8). $\square$

Theorem 11.7 (Uniform numerical bounds). *For every $n \geq 1$,*

$$B_n < \left(\frac{0.792 n}{\log(n+1)} \right)^n. \quad (11.9)$$

Define, for $x > 1$,

$$d(x) = \log \log(x+1) - \log \log x + \frac{1 + e^{-1}}{\log x}. \quad (11.10)$$

For every $\varepsilon > 0$ there is $n_0(\varepsilon) = \max\{e^4, d^{-1}(\varepsilon)\}$ such that, for $n > n_0(\varepsilon)$,

$$B_n < \left(\frac{e^{-0.6+\varepsilon} n}{\log(n+1)} \right)^n. \quad (11.11)$$

Proof. Set

$$R(x) = \frac{\log(x+1)}{\log(x/\log(x+1))} \exp\left(-1 + \frac{1}{\log(x+1)} + \frac{\log x}{2x}\right).$$

Direct differentiation, after multiplying by the positive denominator $2x \log x \log(x+1) \log(x/\log(x+1))^2$, shows that $R'(x) < 0$ for $x \geq 39$; substitution gives $R(39) < 0.792$. Thus (11.8) proves (11.9) for $n \geq 39$. The remaining $1 \leq n \leq 38$ are obtained successively from (10.1); substituting them gives strict inequalities. This is a finite exact integer check.

A second differentiation gives, for $x \geq e^4$,

$$\log R(x) \leq -0.6 + d(x), \quad d'(x) < 0, \quad \lim_{x \rightarrow \infty} d(x) = 0.$$

(The numerator after clearing positive denominators is a sum of $-(\log x - 4)^2$, $-(x - 1 - \log x)$, and negative multiples of these two standard nonnegative quantities.) Hence $d(n) < \varepsilon$ when $n > d^{-1}(\varepsilon)$, and (11.11) follows from (11.8). $\square$

11.5 Saddle-point asymptotics and Lambert W

Let $r = W(n)$, so that $re^r = n$, and put $A = n(1+r)$.

Theorem 11.8 (Leading Bell asymptotic). *As $n \rightarrow \infty$,*

$$B_n \sim \frac{n! \exp(e^r - 1)}{r^n \sqrt{2\pi n(1+r)}} \quad (11.12)$$

$$\sim \frac{1}{\sqrt{n}} \left(\frac{n}{W(n)} \right)^{n+1/2} \exp\left(\frac{n}{W(n)} - n - 1\right). \quad (11.13)$$

Proof. In (10.12) take the circle $z = re^{i\theta}$. The coefficient integral becomes

$$\frac{B_n}{n!} = \frac{e^{e^r} - 1}{2\pi r^n} \int_{-\pi}^{\pi} \exp(e^{re^{i\theta}} - e^r - in\theta) d\theta.$$

The linear term vanishes because $re^r = n$; the quadratic term is $-A\theta^2/2$. On $|\theta| \leq A^{-2/5}$, Taylor expansion is uniform and the scaled integral tends to $\int_{\mathbb{R}} e^{-u^2/2} du/\sqrt{A}$. Outside that arc, $\Re(e^{re^{i\theta}} - e^r) \leq -cA\theta^2$ on the intermediate range and is $\leq -c'e^r$ on the remaining compact range, so the contribution is negligible. This proves (11.12). Stirling's formula and $e^r = n/r$ simplify it to (11.13). $\square$

The archived refined expansion used undefined symbols. The following construction makes every coefficient unambiguous. Let $T_j(r) = \sum_{q=0}^j \{j\}_q r^q$ be the Touchard polynomial and let $Z \sim N(0, 1)$. For an integer $h = O(r)$ define

$$\mathcal{R}_{r,h}(Z) = \exp\left(-\frac{ihZ}{\sqrt{A}} + \sum_{j \geq 3} \frac{i^j e^r T_j(r)}{j! A^{j/2}} Z^j\right). \quad (11.14)$$

Expand this formal exponential and take Gaussian moments termwise; retaining all monomials of total order at most e^{-Mr} defines $C_0(r, h), \dots, C_M(r, h)$, with $C_0 = 1$.

Theorem 11.9 (Uniform Moser–Wyman scheme). *For every fixed M and $|h| \leq Cr$,*

$$B_{n+h} = \frac{(n+h)! \exp(e^r - 1)}{r^{n+h} \sqrt{2\pi A}} \left(\sum_{q=0}^{M-1} C_q(r, h) + O_C(e^{-Mr} P_M(r)) \right), \quad (11.15)$$

where P_M is an explicitly computable polynomial. The first correction is

$$C_1(r, h) = -\frac{h^2}{2A} - \frac{he^r T_3(r)}{2A^2} + \frac{e^r T_4(r)}{8A^2} - \frac{5e^{2r} T_3(r)^2}{24A^3}. \quad (11.16)$$

In particular the coefficient of e^{-qr} is a polynomial in h of degree at most $2q$, with rational functions of r as coefficients.

Proof. Use the same circle as in the preceding proof but replace n by $n + h$. After $\theta = Z/\sqrt{A}$, the ratio of the integrand to its Gaussian quadratic part is exactly the formal series (11.14). Watson's lemma with a Gaussian majorant permits termwise integration on the central arc; the complementary arcs have the same exponential bound uniformly for $|h| \leq Cr$. Odd Gaussian moments vanish, so only integral powers of e^{-r} remain. Since $\mathbb{E}Z^2 = 1$, $\mathbb{E}Z^4 = 3$, and $\mathbb{E}Z^6 = 15$, the terms from the square of the h -term, its product with the cubic term, the quartic term, and the square of the cubic term give (11.16). The same finite bookkeeping defines every subsequent C_q and bounds the omitted Taylor remainder. $\square$

Corollary 11.10 (de Bruijn logarithmic expansion). *As $n \rightarrow \infty$,*

$$\begin{aligned} \frac{\log B_n}{n} &= \log n - \log \log n - 1 + \frac{\log \log n}{\log n} + \frac{1}{\log n} \\ &\quad + \frac{1}{2} \left(\frac{\log \log n}{\log n} \right)^2 + O\left(\frac{\log \log n}{(\log n)^2} \right). \end{aligned} \quad (11.17)$$

Proof. Take logarithms in (11.13). The standard reversion of $r + \log r = \log n$ gives

$$r = \log n - \log \log n + r \log \log n \log n + O\left(\frac{(\log \log n)^2}{(\log n)^2} \right).$$

Substitution and one further Taylor expansion of $\log r$ produce the displayed terms; the prefactor contributes only $O(\log n/n)$. $\square$

11.6 Bell primes and verifiable computation

A *Bell prime* is a prime value of B_n . The archive records the early indices found by computation. Such a list is not a symbolic theorem, but each entry has a rigorous certificate: compute B_n by the integer recurrence (10.1), apply a probable-prime filter, and attach an ECPP or Pratt-style primality certificate. Composite entries are certified by one nontrivial factor. This distinction between an identity and a finite certified computation will also be used for tabulated values later.

Part IV

Bell Polynomials and Composition

Chapter 12

Exponential and ordinary Bell polynomials

12.1 Definitions

Definition 12.1 (Partial exponential Bell polynomial). For $0 \leq k \leq n$, define

$$\mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) = n! \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+3j_3+\dots=n}} \prod_{i \geq 1} \frac{x_i^{j_i}}{(i!)^{j_i} j_i!}. \quad (12.1)$$

Only $i \leq n - k + 1$ can occur. The complete exponential Bell polynomial is

$$\mathcal{B}_n(x_1, \dots, x_n) = \sum_{k=0}^n \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}), \quad \mathcal{B}_0 = 1. \quad (12.2)$$

Definition 12.2 (Partial ordinary Bell polynomial).

$$\hat{\mathcal{B}}_{n,k}(x_1, \dots, x_{n-k+1}) = \sum_{\substack{j_1+j_2+\dots=k \\ j_1+2j_2+\dots=n}} \frac{k!}{j_1! j_2! \dots} \prod_{i \geq 1} x_i^{j_i}. \quad (12.3)$$

Equivalently,

$$\hat{\mathcal{B}}_{n,k}(x_1, x_2, \dots) = \frac{k!}{n!} \mathcal{B}_{n,k}(1!x_1, 2!x_2, \dots). \quad (12.4)$$

Proof of (12.4). Substitute $i!x_i$ for x_i in (12.1); the factorials $(i!)^{j_i}$ cancel, and multiplication by $k!/n!$ leaves exactly (12.3). $\square$

12.2 Combinatorial interpretation

Give each block of size i a weight x_i , and define the weight of a partition as the product of its block weights.

Theorem 12.3 (Weighted-partition interpretation). $\mathcal{B}_{n,k}$ is the total weight of partitions of an n -element labelled set into exactly k blocks, and $\mathcal{B}_n$ is the total weight of all its partitions.

Proof. Fix multiplicities j_i , where j_i blocks have size i . The number of such partitions is

$$\frac{n!}{\prod_i (i!)^{j_i} j_i!}.$$

The factors $i!$ forget order inside each block and $j_i!$ forget order among blocks of equal size. Multiplication by $\prod_i x_i^{j_i}$ and summation gives (12.1); summing over k gives the complete polynomial. $\square$

Thus

$$\begin{aligned}\mathcal{B}_{3,2}(x_1, x_2) &= 3x_1x_2, \\ \mathcal{B}_{6,2}(x_1, \dots, x_5) &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\ \mathcal{B}_{6,3}(x_1, \dots, x_4) &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3.\end{aligned}$$

The monomials correspond to integer partitions of n into k parts, while their coefficients count the labelled set partitions having the prescribed block sizes.

12.3 The master generating functions

Put

$$X(t) = \sum_{j \geq 1} x_j \frac{t^j}{j!}.$$

Theorem 12.4 (Bell-polynomial generating functions). *One has*

$$\frac{X(t)^k}{k!} = \sum_{n \geq k} \mathcal{B}_{n,k}(x_1, \dots) \frac{t^n}{n!}, \quad (12.5)$$

$$\exp(uX(t)) = \sum_{n \geq k \geq 0} \mathcal{B}_{n,k}(x_1, \dots) u^k \frac{t^n}{n!}, \quad (12.6)$$

$$\exp(X(t)) = \sum_{n \geq 0} \mathcal{B}_n(x_1, \dots, x_n) \frac{t^n}{n!}. \quad (12.7)$$

For $\hat{X}(t) = \sum_{j \geq 1} x_j t^j$,

$$\hat{X}(t)^k = \sum_{n \geq k} \hat{\mathcal{B}}_{n,k}(x_1, \dots) t^n, \quad (12.8)$$

$$e^{u\hat{X}(t)} = \sum_{n \geq k \geq 0} \hat{\mathcal{B}}_{n,k}(x_1, \dots) t^n \frac{u^k}{k!}. \quad (12.9)$$

Proof. Expand $X(t)^k$ multinomially. Choosing j_i copies of the term $x_i t^i / i!$ gives the constraints $\sum j_i = k$ and $\sum i j_i = n$, and its coefficient is $k! \prod_i x_i^{j_i} / ((i!)^{j_i} j_i!)$. Division by $k!$ gives (12.1). Summing in k proves the exponential identities. The ordinary identities follow by the same multinomial argument or by (12.4). $\square$

Master principle

Nearly every algebraic identity for Bell polynomials follows by performing a simple operation—multiplication, differentiation, substitution, or logarithm—on (1.10) or (1.12). The combinatorial proofs are the same operations interpreted as selecting, pointing, colouring, or gluing blocks.

12.4 Tables and low-degree polynomials

The nonzero partial polynomials through $n = 6$ are

$$\begin{aligned}
\mathcal{B}_{1,1} &= x_1, \\
\mathcal{B}_{2,1} &= x_2, & \mathcal{B}_{2,2} &= x_1^2, \\
\mathcal{B}_{3,1} &= x_3, & \mathcal{B}_{3,2} &= 3x_1x_2, & \mathcal{B}_{3,3} &= x_1^3, \\
\mathcal{B}_{4,1} &= x_4, & \mathcal{B}_{4,2} &= 4x_1x_3 + 3x_2^2, \\
\mathcal{B}_{4,3} &= 6x_1^2x_2, & \mathcal{B}_{4,4} &= x_1^4, \\
\mathcal{B}_{5,1} &= x_5, & \mathcal{B}_{5,2} &= 5x_1x_4 + 10x_2x_3, \\
\mathcal{B}_{5,3} &= 10x_1^2x_3 + 15x_1x_2^2, \\
\mathcal{B}_{5,4} &= 10x_1^3x_2, & \mathcal{B}_{5,5} &= x_1^5, \\
\mathcal{B}_{6,1} &= x_6, & \mathcal{B}_{6,2} &= 6x_1x_5 + 15x_2x_4 + 10x_3^2, \\
\mathcal{B}_{6,3} &= 15x_1^2x_4 + 60x_1x_2x_3 + 15x_2^3, \\
\mathcal{B}_{6,4} &= 20x_1^3x_3 + 45x_1^2x_2^2, \\
\mathcal{B}_{6,5} &= 15x_1^4x_2, & \mathcal{B}_{6,6} &= x_1^6.
\end{aligned}$$

Every coefficient is supplied by (12.1), so the table is a computation from the theorem rather than an independent assertion.

The first complete polynomials are

$$\begin{aligned}
\mathcal{B}_0 &= 1, \\
\mathcal{B}_1 &= x_1, \\
\mathcal{B}_2 &= x_1^2 + x_2, \\
\mathcal{B}_3 &= x_1^3 + 3x_1x_2 + x_3, \\
\mathcal{B}_4 &= x_1^4 + 6x_1^2x_2 + 3x_2^2 + 4x_1x_3 + x_4, \\
\mathcal{B}_5 &= x_1^5 + 10x_1^3x_2 + 15x_1x_2^2 + 10x_1^2x_3 + 10x_2x_3 + 5x_1x_4 + x_5, \\
\mathcal{B}_6 &= x_1^6 + 15x_1^4x_2 + 45x_1^2x_2^2 + 15x_2^3 + 20x_1^3x_3 + 60x_1x_2x_3 + 10x_3^2 \\
&\quad + 15x_1^2x_4 + 15x_2x_4 + 6x_1x_5 + x_6, \\
\mathcal{B}_7 &= x_1^7 + 21x_1^5x_2 + 105x_1^3x_2^2 + 105x_1x_2^3 + 35x_1^4x_3 + 210x_1^2x_2x_3 \\
&\quad + 70x_1x_3^2 + 105x_2^2x_3 + 35x_1^3x_4 + 105x_1x_2x_4 + 35x_3x_4 \\
&\quad + 21x_1^2x_5 + 21x_2x_5 + 7x_1x_6 + x_7.
\end{aligned}$$

Chapter 13

Recurrences, derivatives, and algebraic identities

13.1 Pointing recurrences

Theorem 13.1 (Complete and partial recurrences). For $n \geq 0$,

$$\mathcal{B}_{n+1}(x_1, \dots, x_{n+1}) = \sum_{i=0}^n \binom{n}{i} x_{i+1} \mathcal{B}_{n-i}(x_1, \dots, x_{n-i}), \quad (13.1)$$

$$\mathcal{B}_{n+1,k+1} = \sum_{i=0}^{n-k} \binom{n}{i} x_{i+1} \mathcal{B}_{n-i,k}, \quad (13.2)$$

with $\mathcal{B}_{0,0} = 1$, $\mathcal{B}_{n,0} = 0$ for $n > 0$, and $\mathcal{B}_{0,k} = 0$ for $k > 0$.

Proof. In a weighted partition of $[n+1]$, inspect the block containing the distinguished label $n+1$. If that block contains i further labels, choose them in $\binom{n}{i}$ ways, give the block weight x_{i+1} , and partition the remaining labels. Keeping or forgetting the total block count yields the two formulas. $\square$

Theorem 13.2 (Block-colour convolution). For $k_1, k_2 \geq 0$,

$$\mathcal{B}_{n,k_1+k_2}(x) = \frac{k_1!k_2!}{(k_1+k_2)!} \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{i,k_1}(x) \mathcal{B}_{n-i,k_2}(x). \quad (13.3)$$

Proof. Multiply (1.10) for k_1 and k_2 . Their product is $X^{k_1+k_2}/(k_1!k_2!)$; comparison with the same formula for k_1+k_2 gives the factor $(k_1!k_2!)/(k_1+k_2)!$ and the binomial EGF convolution. $\square$

13.2 Near-diagonal reduction

Theorem 13.3. For $1 \leq a < n$,

$$\begin{aligned} \mathcal{B}_{n,n-a}(x_1, \dots, x_{a+1}) \\ = \sum_{j=a+1}^{2a} \frac{j!}{a!} \binom{n}{j} x_1^{n-j} \mathcal{B}_{a,j-a}\left(\frac{x_2}{2}, \frac{x_3}{3}, \dots, \frac{x_{2a-j+2}}{2a-j+2}\right). \end{aligned} \quad (13.4)$$

Proof. A partition into $n-a$ blocks has total excess $\sum_B (|B|-1) = a$. Let j be the number of elements in nonsingleton blocks. Then the number of those blocks is $j-a$, so $a+1 \leq j \leq 2a$. Choose the j elements, leave the other $n-j$ as singleton blocks, and count the nonsingleton partition. Replacing a

block of size $r + 1$ by an atom of size r removes one distinguished element per block; the factor $j!/a!$ and weights $x_{r+1}/(r+1)$ are exactly the labelled pointing factors. Equivalently, make the substitution in (12.1); the coefficient of every multiplicity vector on both sides is $n!x_1^{n-j} \prod_{r \geq 2} x_r^{j_r} / ((r!)^{j_r} j_r!)$. $\square$

The first cases, obtained by listing integer partitions of the excess, are

$$\mathcal{B}_{n,1} = x_n, \quad \mathcal{B}_{n,2} = \frac{1}{2} \sum_{j=1}^{n-1} \binom{n}{j} x_j x_{n-j}, \quad (13.5)$$

$$\mathcal{B}_{n,n} = x_1^n, \quad \mathcal{B}_{n,n-1} = \binom{n}{2} x_1^{n-2} x_2, \quad (13.6)$$

$$\mathcal{B}_{n,n-2} = \binom{n}{3} x_1^{n-3} x_3 + 3 \binom{n}{4} x_1^{n-4} x_2^2, \quad (13.7)$$

$$\mathcal{B}_{n,n-3} = \binom{n}{4} x_1^{n-4} x_4 + 10 \binom{n}{5} x_1^{n-5} x_2 x_3 + 15 \binom{n}{6} x_1^{n-6} x_2^3, \quad (13.8)$$

$$\begin{aligned} \mathcal{B}_{n,n-4} = & \binom{n}{5} x_1^{n-5} x_5 + 5 \binom{n}{6} x_1^{n-6} (3x_2 x_4 + 2x_3^2) \\ & + 105 \binom{n}{7} x_1^{n-7} x_2^2 x_3 + 105 \binom{n}{8} x_1^{n-8} x_2^4. \end{aligned} \quad (13.9)$$

Their coefficients are the corresponding block-size counts from Theorem 12.3.

13.3 Partial derivatives

Theorem 13.4 (Derivative identities). *For $i \geq 1$,*

$$\frac{\partial \mathcal{B}_n}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i}, \quad (13.10)$$

$$\frac{\partial \mathcal{B}_{n,k}}{\partial x_i} = \binom{n}{i} \mathcal{B}_{n-i,k-1}. \quad (13.11)$$

Consequently, for differentiable $a_i(x)$,

$$\frac{d}{dx} \mathcal{B}_{n,k}(a_1(x), \dots) = \sum_{i=1}^{n-k+1} \binom{n}{i} a'_i(x) \mathcal{B}_{n-i,k-1}(a_1(x), \dots). \quad (13.12)$$

Proof. Differentiate (1.10) with respect to x_i :

$$\frac{\partial}{\partial x_i} \frac{X(t)^k}{k!} = \frac{t^i X(t)^{k-1}}{i! (k-1)!}.$$

Coefficient comparison gives (13.11); summing over k gives (13.10). The ordinary chain rule then gives (13.12). $\square$

13.4 A quadratic differential recurrence

The less familiar differential recurrence in the archive is also a disguised coefficient identity.

Theorem 13.5. For $n \geq 2$,

$$\begin{aligned} \mathcal{B}_n = \frac{1}{n-1} & \left[\sum_{i=2}^n \sum_{j=1}^{i-1} (i-1) \binom{i-2}{j-1} x_j x_{i-j} \frac{\partial \mathcal{B}_{n-1}}{\partial x_{i-1}} \right. \\ & \left. + \sum_{i=2}^n \sum_{j=1}^{i-1} \frac{x_{i+1}}{\binom{i}{j}} \frac{\partial^2 \mathcal{B}_{n-1}}{\partial x_j \partial x_{i-j}} + \sum_{i=2}^n x_i \frac{\partial \mathcal{B}_{n-1}}{\partial x_{i-1}} \right]. \end{aligned} \quad (13.13)$$

Terms containing unavailable variables or negative-index Bell polynomials are zero.

Proof. Let $F(t) = e^{X(t)} = \sum_m \mathcal{B}_m t^m / m!$. After using (13.10), the first double sum is

$$(n-1)! [t^{n-1}] (tX'(t))^2 F(t).$$

The second is $(n-1)! [t^{n-1}] ((tX'' - X' + x_1)F)$, and the last is $(n-1)! [t^{n-1}] ((X' - x_1)F)$. Their sum is therefore

$$(n-1)! [t^{n-1}] t(X'' + X'^2)F = (n-1)! [t^{n-1}] tF'' = (n-1)\mathcal{B}_n.$$

Division by $n-1$ proves the formula. □

Chapter 14

Specializations, inversions, determinants, and convolutions

14.1 Stirling, Bell, Lah, and Touchard specializations

Theorem 14.1. *For admissible n, k ,*

$$\mathcal{B}_{n,k}(0!, 1!, \dots, (n-k)!) = \begin{bmatrix} n \\ k \end{bmatrix}, \quad (14.1)$$

$$\mathcal{B}_n(0!, 1!, \dots, (n-1)!) = n!, \quad (14.2)$$

$$\mathcal{B}_{n,k}(1, 1, \dots, 1) = \left\{ \begin{matrix} n \\ k \end{matrix} \right\}, \quad (14.3)$$

$$\mathcal{B}_n(1, 1, \dots, 1) = \mathcal{B}_n, \quad (14.4)$$

$$\mathcal{B}_{n,k}(1!, 2!, \dots, (n-k+1)!) = \binom{n-1}{k-1} \frac{n!}{k!} = L(n, k), \quad (14.5)$$

$$T_n(x) = \mathcal{B}_n(x, x, \dots, x) = \sum_{k=0}^n \left\{ \begin{matrix} n \\ k \end{matrix} \right\} x^k. \quad (14.6)$$

Proof. In (1.10), the choices $x_j = (j-1)!$, $x_j = 1$, and $x_j = j!$ make $X(t)$ respectively

$$-\log(1-t), \quad e^t - 1, \quad \frac{t}{1-t}.$$

The resulting coefficient formulae are exactly the column EGFs for unsigned first-kind Stirling numbers, second-kind Stirling numbers, and Lah numbers proved earlier. Summing over k yields (14.2) and (14.4). Finally $x_j = x$ gives $\exp(x(e^t - 1))$, the EGF of $T_n(x)$. $\square$

14.2 Homogeneity and a useful evaluation

Theorem 14.2 (Bigraded homogeneity). *For scalars α, β ,*

$$\mathcal{B}_{n,k}(\alpha\beta x_1, \alpha\beta^2 x_2, \dots) = \alpha^k \beta^n \mathcal{B}_{n,k}(x_1, x_2, \dots). \quad (14.7)$$

Moreover,

$$\mathcal{B}_{n,k}(1, 2, 3, \dots, n-k+1) = \binom{n}{k} k^{n-k}. \quad (14.8)$$

Proof. Every monomial in (12.1) has $\sum j_i = k$ and $\sum i j_i = n$, proving the first formula. For the second, $x_j = j$ gives $X(t) = te^t$; hence

$$\frac{X(t)^k}{k!} = \frac{t^k e^{kt}}{k!}.$$

The coefficient of $t^n/n!$ is $n!k^{n-k}/(k!(n-k)!) = \binom{n}{k}k^{n-k}$. $\square$

14.3 Addition and shifted-zero identities

Theorem 14.3 (Complete Bell addition formula).

$$\mathcal{B}_n(x_1 + y_1, \dots, x_n + y_n) = \sum_{i=0}^n \binom{n}{i} \mathcal{B}_{n-i}(x) \mathcal{B}_i(y). \quad (14.9)$$

Proof. The complete EGF for the left side is $\exp(X(t) + Y(t)) = \exp X(t) \exp Y(t)$. Coefficient extraction gives the binomial convolution. $\square$

Theorem 14.4 (Insertion of q leading zeros). For $q \geq 0$,

$$\begin{aligned} \mathcal{B}_{n,k} \left(\frac{x_{q+1}}{\binom{q+1}{q}}, \frac{x_{q+2}}{\binom{q+2}{q}}, \dots \right) \\ = \frac{n!(q!)^k}{(n+qk)!} \mathcal{B}_{n+qk,k}(\underbrace{0, \dots, 0}_{q \text{ entries}}, x_{q+1}, x_{q+2}, \dots). \end{aligned} \quad (14.10)$$

Proof. Since

$$\frac{x_{q+i}}{\binom{q+i}{q}} \frac{t^i}{i!} = q! \frac{x_{q+i} t^i}{(q+i)!},$$

the k th power of the left component series is $(q!)^k t^{-qk}$ times the k th power of the component series on the right. Comparing the coefficient of t^n in (1.10) gives the factorial factor displayed. $\square$

14.4 Exponential–logarithmic inversion

Theorem 14.5 (Bell transform and its inverse). Suppose

$$y_n = \mathcal{B}_n(x_1, \dots, x_n) = \sum_{k=1}^n \mathcal{B}_{n,k}(x_1, \dots). \quad (14.11)$$

Then

$$x_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(y_1, \dots, y_{n-k+1}). \quad (14.12)$$

Proof. Let $X(t) = \sum_{n \geq 1} x_n t^n / n!$ and $Y(t) = \sum_{n \geq 1} y_n t^n / n!$. Equation (1.12) says $1 + Y = e^X$, so

$$X = \log(1 + Y) = \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} Y^k.$$

Use (1.10) to extract the coefficient of $t^n/n!$. $\square$

More generally, if f and $g = f^{-1}$ are locally inverse and

$$y_n = \sum_{k=0}^n f^{(k)}(a) \mathcal{B}_{n,k}(x_1, \dots), \quad (14.13)$$

then

$$x_n = \sum_{k=0}^n g^{(k)}(f(a)) \mathcal{B}_{n,k}(y_1, \dots). \quad (14.14)$$

Indeed, these are the derivative coefficients of $f \circ u$ and $g \circ f \circ u = u$; a complete proof is given with Faà di Bruno's formula in Equation (19.3).

14.5 Two Hessenberg determinants

For $n \geq 1$, define $H_n = (h_{ij})$ and $K_n = (k_{ij})$ by

$$h_{ij} = \begin{cases} \binom{n-i}{j-i} x_{j-i+1}, & j \geq i, \\ -1, & j = i-1, \\ 0, & j < i-1, \end{cases} \quad k_{ij} = \begin{cases} x_{j-i+1}/(j-i)!, & j \geq i, \\ -(i-1), & j = i-1, \\ 0, & j < i-1. \end{cases}$$

Thus H_n begins with

$$\begin{bmatrix} x_1 & \binom{n-1}{1} x_2 & \binom{n-1}{2} x_3 & \cdots & x_n \\ -1 & x_1 & \binom{n-2}{1} x_2 & \cdots & x_{n-1} \\ 0 & -1 & x_1 & \cdots & x_{n-2} \\ \vdots & & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & -1 & x_1 \end{bmatrix},$$

while K_n has first row $x_1, x_2/1!, \dots, x_n/(n-1)!$ and subdiagonal $-1, -2, \dots, -(n-1)$.

Theorem 14.6 (Bell determinants).

$$\mathcal{B}_n(x_1, \dots, x_n) = \det H_n = \det K_n. \quad (14.15)$$

Proof. Expansion of the upper Hessenberg determinant $\det H_n$ along its first row gives

$$D_n = \sum_{j=1}^n \binom{n-1}{j-1} x_j D_{n-j}, \quad D_0 = 1;$$

the cofactor sign is cancelled by the product of the $j-1$ entries -1 on the subdiagonal. This is exactly (13.1), so $D_n = \mathcal{B}_n$.

Let J reverse the coordinate order and let $D = \text{diag}(0!, 1!, \dots, (n-1)!)$. A direct entrywise calculation gives

$$K_n = D(JH_n^T J)D^{-1}.$$

Indeed, $\frac{(i-1)!}{(j-1)!} \binom{j-1}{i-1} = 1/(j-i)!$ above the diagonal, and the subdiagonal becomes $-(i-1)$. Similarity, transposition, and reversal preserve the determinant, proving the second equality. $\square$

14.6 The diamond convolution

For sequences $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$ define

$$(x \diamond y)_n = \sum_{j=1}^{n-1} \binom{n}{j} x_j y_{n-j}. \quad (14.16)$$

This operation is commutative and associative because it corresponds to multiplication of the EGFs $X(t)$ and $Y(t)$.

Theorem 14.7. *If $x^{k\Diamond}$ denotes the k -fold diamond power, then*

$$\mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}) = \frac{(x^{k\Diamond})_n}{k!}. \quad (14.17)$$

Proof. The EGF of $x^{k\Diamond}$ is $X(t)^k$. Compare with (1.10). □

For example,

$$x\Diamond x = (0, 2x_1^2, 6x_1x_2, 8x_1x_3 + 6x_2^2, \dots)$$

and $(x^{3\Diamond})_4 = 36x_1^2x_2$, whence $\mathcal{B}_{4,3} = 6x_1^2x_2$.

Chapter 15

Applications of Bell polynomials

15.1 Symmetric functions and Newton identities

Let e_n be the elementary symmetric function and $p_n = \sum_i u_i^n$ the power sum in an alphabet $u_1, u_2, \dots$

Theorem 15.1.

$$e_n = \frac{1}{n!} \mathcal{B}_n(p_1, -1!p_2, 2!p_3, \dots, (-1)^{n-1}(n-1)!p_n) \quad (15.1)$$

$$\begin{aligned} &= \frac{(-1)^n}{n!} \mathcal{B}_n(-p_1, -1!p_2, -2!p_3, \dots, -(n-1)!p_n), \\ p_n &= \frac{(-1)^{n-1}}{(n-1)!} \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(1!e_1, 2!e_2, \dots) \\ &= (-1)^n \sum_{k=1}^n \frac{1}{k} \widehat{\mathcal{B}}_{n,k}(-e_1, -e_2, \dots). \end{aligned} \quad (15.2)$$

Proof. The classical product identity

$$E(t) = \sum_{n \geq 0} e_n t^n = \prod_i (1 + u_i t) = \exp \left(\sum_{r \geq 1} (-1)^{r-1} p_r \frac{t^r}{r} \right)$$

and (1.12) give (15.1). Taking logarithms, $\sum_{r \geq 1} (-1)^{r-1} p_r t^r / r = \log(1 + \sum_{j \geq 1} e_j t^j)$, and expanding the logarithm through ordinary Bell polynomials yields the second pair of forms. The exponential form follows from (12.4). $\square$

Corollary 15.2 (Determinant from traces). *For an $n \times n$ matrix A , set $s_k = -(k-1)! \operatorname{tr}(A^k)$. Then*

$$\det A = \frac{(-1)^n}{n!} \mathcal{B}_n(s_1, s_2, \dots, s_n). \quad (15.3)$$

Proof. Apply (15.1) to the eigenvalues of A , using $e_n = \det A$ and $p_k = \operatorname{tr}(A^k)$. Both sides are polynomial in the matrix entries, so the argument remains valid for nondiagonalizable matrices. $\square$

15.2 Cycle indices

Let a_j mark cycles of length j in a permutation.

Theorem 15.3. *The cycle index of the symmetric group is*

$$Z(S_n) = \frac{1}{n!} \mathcal{B}_n(0!a_1, 1!a_2, \dots, (n-1)!a_n). \quad (15.4)$$

Proof. A permutation is a set of cycles. A labelled cycle of size j has $(j-1)!$ linear realizations, so its component EGF is $(j-1)!a_j t^j/j! = a_j t^j/j$. The exponential formula gives $\exp(\sum_{j \geq 1} a_j t^j/j)$; comparison with (1.12) proves the identity after division by $n!$. $\square$

15.3 Moments and cumulants

Let $M(t) = \mathbb{E}e^{tX} = \sum_{n \geq 0} \mu'_n t^n/n!$ and $K(t) = \log M(t) = \sum_{n \geq 1} \kappa_n t^n/n!$.

Theorem 15.4 (Moment–cumulant relations).

$$\mu'_n = \mathcal{B}_n(\kappa_1, \dots, \kappa_n) = \sum_{k=1}^n \mathcal{B}_{n,k}(\kappa_1, \dots), \quad (15.5)$$

$$\kappa_n = \sum_{k=1}^n (-1)^{k-1} (k-1)! \mathcal{B}_{n,k}(\mu'_1, \dots, \mu'_{n-k+1}). \quad (15.6)$$

Proof. The identity $M = e^K$ is (1.12); its logarithmic inverse is Theorem 14.5. $\square$

15.4 Hermite polynomials

The probabilists' Hermite polynomials are defined by

$$e^{xt-t^2/2} = \sum_{n \geq 0} \text{He}_n(x) \frac{t^n}{n!}.$$

Comparison with (1.12) gives and proves

$$\text{He}_n(x) = \mathcal{B}_n(x, -1, 0, \dots, 0). \quad (15.7)$$

In particular, the usual recurrence and differential equation follow by differentiating this generating function.

15.5 Polynomial sequences of binomial type

Let $h(t) = \sum_{j \geq 1} a_j t^j/j!$ with $a_1 \neq 0$, and define

$$p_n(x) = \mathcal{B}_n(a_1 x, a_2 x, \dots, a_n x) = \sum_{k=0}^n \mathcal{B}_{n,k}(a_1, \dots, a_{n-k+1}) x^k. \quad (15.8)$$

Theorem 15.5. *The polynomials p_n satisfy*

$$\sum_{n \geq 0} p_n(x) \frac{t^n}{n!} = e^{xh(t)}, \quad (15.9)$$

$$p_n(x+y) = \sum_{k=0}^n \binom{n}{k} p_k(x) p_{n-k}(y). \quad (15.10)$$

If $Q = h^{-1}(D_x)$ is the associated delta operator, then

$$Qp_n(x) = np_{n-1}(x). \quad (15.11)$$

Proof. Substitute xa_j in (1.12). The product $e^{(x+y)h} = e^{xh}e^{yh}$ gives the binomial identity. Finally $h^{-1}(D_x)e^{xh(t)} = h^{-1}(h(t))e^{xh(t)} = te^{xh(t)}$; coefficient comparison gives (15.11). $\square$

Editorial normalization

In some sources the symbol $B_n(x)$ denotes the Touchard polynomial, in others a complete Bell polynomial, a Bernoulli polynomial, or a binary-tree sequence. The notations T_n , $\mathcal{B}_n$, $\beta_n(x)$, and C_n are kept separate here; this is essential when the objects occur in the same inversion formula.

Part V

Eulerian Numbers and Descent Calculus

Chapter 16

Ordinary Eulerian numbers

16.1 Descents, symmetry, and the basic recurrence

For a permutation $\pi = \pi_1 \cdots \pi_n$, a descent is an index $i \in \{1, \dots, n-1\}$ with $\pi_i > \pi_{i+1}$.

Definition 16.1. The Eulerian number $\langle n \rangle_k = A(n, k)$ counts permutations of $[n]$ with exactly k descents. Put

$$A_0(t) = 1, \quad A_n(t) = \sum_{k=0}^{n-1} \langle n \rangle_k t^k \quad (n \geq 1).$$

Outside $0 \leq k < n$ the number is zero.

Theorem 16.2 (Eulerian recurrence).

$$\langle n \rangle_k = (k+1) \langle n-1 \rangle_k + (n-k) \langle n-1 \rangle_{k-1}. \quad (16.1)$$

Equivalently,

$$A_n(t) = (1 + (n-1)t)A_{n-1}(t) + t(1-t)A'_{n-1}(t). \quad (16.2)$$

Proof. Insert n into a permutation of $[n-1]$. If the old permutation has k descents, insertion after one of those k descents or at the end preserves the number of descents, giving $k+1$ choices. If it has $k-1$ descents, insertion in any of the remaining $n-k$ slots creates one new descent. This proves (16.1). Multiply by t^k , sum over k , and collect the terms containing k to obtain (16.2). $\square$

Complementation $\pi_i \mapsto n+1-\pi_i$ interchanges descents and ascents. Therefore

$$\langle n \rangle_k = \langle n \rangle_{n-1-k}, \quad A_n(t) = t^{n-1}A_n(t^{-1}). \quad (16.3)$$

In particular $\langle n \rangle_0 = \langle n \rangle_{n-1} = 1$ and $\sum_k \langle n \rangle_k = n!$. The next diagonal is

$$\langle n \rangle_1 = 2^n - (n+1), \quad (16.4)$$

which follows either from the recurrence or from the explicit formula below.

16.2 Exponential generating function and explicit formula

Theorem 16.3 (Eulerian EGF).

$$\sum_{n \geq 0} A_n(t) \frac{x^n}{n!} = \frac{t-1}{t - e^{(t-1)x}} = \frac{(1-t)e^{(1-t)x}}{1 - te^{(1-t)x}}. \quad (16.5)$$

Proof. Let $F(x, t)$ denote the left side. Multiplying (16.2) by $x^{n-1}/(n-1)!$ and summing gives the first-order equation

$$(1 - tx)F_x = t(1 - t)F_t + F, \quad F(0, t) = 1.$$

Direct substitution verifies that the displayed rational-exponential function solves this equation and initial condition. Formal uniqueness follows by recursively comparing coefficients of x . $\square$

Expanding the denominator geometrically gives the central power-series identity.

Theorem 16.4. For $n \geq 0$,

$$\sum_{m=0}^{\infty} (m+1)^n t^m = \frac{A_n(t)}{(1-t)^{n+1}}. \quad (16.6)$$

Consequently,

$$\left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n. \quad (16.7)$$

Proof. From (16.5),

$$F(x, t) = (1-t) \sum_{m \geq 0} t^m e^{(m+1)(1-t)x}.$$

Extracting $x^n/n!$ proves (16.6). Multiply by $(1-t)^{n+1}$ and extract t^k to obtain (16.7). Formula (16.4) is its case $k = 1$. $\square$

Theorem 16.5 (A second polynomial recurrence). For $n > 1$,

$$A_n(t) = \sum_{k=0}^{n-1} \binom{n}{k} A_k(t) (t-1)^{n-1-k}. \quad (16.8)$$

Proof. The EGF of the right sides, summed over $n \geq 1$, is

$$F(x, t) \frac{e^{(t-1)x} - 1}{t-1}.$$

Using (16.5), this equals $F(x, t) - 1$, which is the EGF of the left sides. $\square$

16.3 Worpitzky's identity and sums of powers

Theorem 16.6 (Worpitzky). For every indeterminate x and $n \geq 1$,

$$x^n = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{x+k}{n}. \quad (16.9)$$

Proof. The coefficient of t^{x-1} in $t^{-k}(1-t)^{-n-1}$ is $\binom{x+k}{n}$ for positive integer x . Equivalently, taking the coefficient of t^{x-1} in (16.6) after multiplying by the numerator gives (16.9). Since both sides are polynomials of degree n in x and agree for all positive integers, they agree identically.

A direct combinatorial proof standardizes a word in $[x]^n$: its stable sorting permutation records the relative order, and its k descents specify k mandatory separations among the n sorted positions. Stars and bars gives $\binom{x+k}{n}$ words for each permutation with k descents. $\square$

Corollary 16.7 (Power sums). For $m, n \geq 1$,

$$\sum_{r=1}^m r^n = \sum_{k=0}^{n-1} \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle \binom{m+k+1}{n+1}. \quad (16.10)$$

Proof. Sum (16.9) over $x = 1, \dots, m$ and use the hockey-stick identity $\sum_{x=1}^m \binom{x+k}{n} = \binom{m+k+1}{n+1}$; the omitted lower boundary terms vanish for $0 \leq k < n$. $\square$

16.4 Negative polylogarithms

For $|z| < 1$, $\text{Li}_{-n}(z) = \sum_{m \geq 1} m^n z^m$. Shifting m in (16.6) gives

$$\text{Li}_{-n}(z) = \frac{z A_n(z)}{(1-z)^{n+1}} = \frac{1}{(1-z)^{n+1}} \sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle z^{n-k}, \quad (16.11)$$

where the second equality uses symmetry (16.3). Since the right side is rational, it also supplies the meromorphic continuation of the negative polylogarithm.

16.5 Eulerian and Stirling bases

Theorem 16.8.

$$A_n(t) = \sum_{k=0}^n \left\{ n \atop k \right\} k! (t-1)^{n-k}. \quad (16.12)$$

Proof. Use $m^n = \sum_k \left\{ n \atop k \right\} m^k$ in $\sum_{m \geq 0} m^n t^m$. Since $\sum_{m \geq 0} m^k t^m = k! t^k / (1-t)^{k+1}$, multiplication by $(1-t)^{n+1}/t$ yields $A_n(t) = \sum_k \left\{ n \atop k \right\} k! t^{k-1} (1-t)^{n-k}$. Replacing t by $1/t$ and using (16.3), or simply expanding around $t = 1$, gives the equivalent displayed form. $\square$

16.6 An Irwin–Hall integral and hypersimplex volumes

Theorem 16.9. For every function $f : \mathbb{R} \rightarrow \mathbb{C}$ for which the following finite sum is defined,

$$\sum_{k=0}^{n-1} \left\langle n \atop k \right\rangle f(k) = n! \int_{[0,1]^n} f(\lfloor x_1 + \cdots + x_n \rfloor) dx_1 \cdots dx_n. \quad (16.13)$$

In particular, the slab

$$\Delta_{n,k} = \{x \in [0, 1]^n : k \leq x_1 + \cdots + x_n < k+1\}$$

has volume $\left\langle n \atop k \right\rangle / n!$.

Proof. The volume of $\{x \in [0, 1]^n : \sum x_i < y\}$ is obtained from the simplex $\{x_i \geq 0 : \sum x_i < y\}$ by inclusion–exclusion over coordinates exceeding 1:

$$V_n(y) = \frac{1}{n!} \sum_{j=0}^{\lfloor y \rfloor} (-1)^j \binom{n}{j} (y-j)^n.$$

Thus the volume of the k th slab is $V_n(k+1) - V_n(k)$. Combining adjacent terms by Pascal’s identity gives

$$\frac{1}{n!} \sum_{j=0}^k (-1)^j \binom{n+1}{j} (k+1-j)^n = \frac{1}{n!} \left\langle n \atop k \right\rangle$$

by (16.7). Summing $f(k)$ times these volumes proves the integral identity. $\square$

16.7 Alternating sums and Bernoulli numbers

Theorem 16.10. For $n > 0$,

$$\sum_{k=0}^{n-1} (-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle = 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1}. \quad (16.14)$$

For $n > 1$,

$$\sum_{k=0}^{n-1} \frac{(-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle}{\binom{n-1}{k}} = 0, \quad (16.15)$$

$$\sum_{k=0}^{n-1} \frac{(-1)^k \left\langle \begin{matrix} n \\ k \end{matrix} \right\rangle}{\binom{n}{k}} = (n+1) \beta_n. \quad (16.16)$$

Proof. At $t = -1$, (16.5) becomes $2/(1 + e^{-2x}) = 1 + \tanh x$. The Bernoulli generating function gives

$$\tanh x = \sum_{n \geq 1} 2^{n+1} (2^{n+1} - 1) \frac{\beta_{n+1}}{n+1} \frac{x^n}{n!},$$

which proves (16.14).

For the remaining identities use

$$\binom{n-1}{k}^{-1} = n \int_0^1 u^k (1-u)^{n-k-1} du, \quad \binom{n}{k}^{-1} = (n+1) \int_0^1 u^k (1-u)^{n-k} du.$$

Substitute $t = -u/(1-u)$ in (16.12); then

$$(1-u)^n A_n \left(-\frac{u}{1-u} \right) = \sum_{j=0}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j! (1-u)^j. \quad (16.17)$$

For the first integral divide by $1-u$ and integrate. The result is $n \sum_{j=1}^n (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} (j-1)! = 0$ for $n > 1$, because it is the coefficient of $x^n/n!$ in $\log(1 + (e^x - 1)) = x$. For the second integral one obtains $(n+1) \sum_j (-1)^{n-j} \left\{ \begin{matrix} n \\ j \end{matrix} \right\} j!/(j+1)$. The standard Stirling formula $\beta_n = \sum_j (-1)^j j! \left\{ \begin{matrix} n \\ j \end{matrix} \right\} / (j+1)$ proves the claim; for odd $n > 1$ both sides vanish, and for even n the signs coincide. $\square$

Chapter 17

Geometric interpretations

17.1 The cube and its Ehrhart numerator

For a lattice polytope P of dimension d , its Ehrhart series has the form $\sum_{m \geq 0} |mP \cap \mathbb{Z}^d| t^m = h^*(t)/(1-t)^{d+1}$. For the unit cube $C = [0, 1]^n$ one has $|mC \cap \mathbb{Z}^n| = (m+1)^n$; hence (16.6) proves:

Theorem 17.1. *The h^* -polynomial of the standard n -cube is $A_n(t)$, so its h^* -vector is the n th Eulerian row.*

Proof. The dilate $m[0, 1]^n = [0, m]^n$ contains exactly $(m+1)^n$ lattice points. Therefore its Ehrhart series is

$$\sum_{m \geq 0} (m+1)^n t^m.$$

By (16.6) this equals $A_n(t)/(1-t)^{n+1}$. The numerator in the Ehrhart-series normal form is unique, hence $h^*_{[0,1]^n}(t) = A_n(t)$; coefficient comparison gives the stated h^* -vector. $\square$

The regions $\Delta_{n,k}$ in Theorem 16.9 are half-open hypersimplices. Their normalized volumes are $n! \operatorname{vol} \Delta_{n,k} = \langle \begin{smallmatrix} n \\ k \end{smallmatrix} \rangle$.

17.2 The permutohedron

The type- A permutohedron is the convex hull of the permutations of $(1, 2, \dots, n)$; its dimension is $n-1$. Faces of its braid fan are indexed by ordered set partitions. There are $k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\}$ ordered partitions into k blocks.

Theorem 17.2. *The h -polynomial of the simple polytope dual to the type- A permutohedron is $A_n(t)$.*

Proof. For a simple $(n-1)$ -polytope the h -polynomial is the transform $h(t) = \sum_F (t-1)^{\dim F}$ over faces of the dual simplicial complex. Grouping braid-fan faces by their k ordered blocks gives $\sum_k k! \left\{ \begin{smallmatrix} n \\ k \end{smallmatrix} \right\} (t-1)^{n-k}$, which is $A_n(t)$ by (16.12). $\square$

Chapter 18

Type-B and second-order Eulerian numbers

18.1 Signed permutations and the corrected type-B indexing

A signed permutation is a word $\pi_1 \cdots \pi_n$ whose absolute values form a permutation of $[n]$. Put $\pi_0 = 0$ and call $i \in \{0, \dots, n-1\}$ a type- B descent when $\pi_i > \pi_{i+1}$. Let $E_B(n, k)$ count signed permutations with k such descents, and set $M_n(t) = \sum_{k=0}^n E_B(n, k)t^k$.

Editorial normalization

The archived display reverses the two entries in the descent inequality and pairs that convention with an incompatible explicit formula. The definition above is the standard Coxeter descent convention. It gives the archived table 1, 6, 1 for $n = 2$ and the midpoint-Eulerian generating function below.

Theorem 18.1 (Type- B recurrence and Worpitzky series).

$$E_B(n, k) = (2k + 1)E_B(n - 1, k) + (2n - 2k + 1)E_B(n - 1, k - 1), \quad (18.1)$$

$$\sum_{m \geq 0} (2m + 1)^n t^m = \frac{M_n(t)}{(1 - t)^{n+1}}, \quad (18.2)$$

$$E_B(n, k) = \sum_{j=0}^k (-1)^j \binom{n+1}{j} (2(k-j) + 1)^n. \quad (18.3)$$

The first rows are

$$1; \quad 1, 1; \quad 1, 6, 1; \quad 1, 23, 23, 1; \quad 1, 76, 230, 76, 1.$$

Proof. Insert either n or $-n$ into a signed permutation of size $n - 1$. Marking the slots according to whether insertion preserves or creates a descent gives $2k + 1$ preserving slots for an object with k descents and $2n - 2k + 1$ creating slots for one with $k - 1$ descents; this proves the recurrence.

A type- B standardization partitions the $(2m + 1)^n$ words in the ordered alphabet $-m < \cdots < -1 < 0 < 1 < \cdots < m$ by their signed sorting permutation. If that permutation has k type- B descents, stars and bars gives $\binom{m+n-k}{n}$ compatible weakly increasing absolute levels. Hence

$$(2m + 1)^n = \sum_{k=0}^n E_B(n, k) \binom{m+n-k}{n}.$$

Multiply by t^m , sum in m , and use $\sum_{m \geq 0} \binom{m+n-k}{n} t^m = t^k / (1 - t)^{n+1}$ to obtain (18.2). Multiplication by $(1 - t)^{n+1}$ and coefficient extraction gives (18.3). $\square$

As in type A , these coefficients form the h -vector of the simple polytope dual to the type- B permutohedron; the proof repeats the braid-fan argument with signed ordered partitions.

18.2 Stirling permutations and second-order Eulerian numbers

A *Stirling permutation of order n* is a permutation of the multiset $\{1, 1, 2, 2, \dots, n, n\}$ such that all entries between the two copies of i are larger than i . Let $\langle\langle n \rangle\rangle_k$ count those with k ordinary descents and put

$$P_n(t) = \sum_{k=0}^{n-1} \langle\langle n \rangle\rangle_k t^k, \quad P_0(t) = 1.$$

Theorem 18.2 (Second-order recurrence).

$$\langle\langle n \rangle\rangle_k = (2n - k - 1) \langle\langle n-1 \rangle\rangle_{k-1} + (k+1) \langle\langle n-1 \rangle\rangle_k, \quad (18.4)$$

$$P_{n+1}(t) = (2nt + 1)P_n(t) - t(t-1)P'_n(t), \quad (18.5)$$

$$P_n(1) = (2n-1)!! \quad (18.6)$$

Proof. The two copies of the largest label n must be adjacent. Insert the pair nn into one of the $2n-1$ gaps of a Stirling permutation of order $n-1$. A descent gap or the terminal gap preserves the number of descents, giving $k+1$ choices from an object with k descents. Each of the remaining $2n-k-1$ gaps creates one descent from an object with $k-1$ descents. This is the first recurrence; summing against t^k gives the second. At each stage there are $2n-1$ insertion gaps, so the total count is $(2n-1)!!$. $\square$

The differential recurrence has two useful equivalent forms:

$$(t-1)^{-2n-2}P_{n+1}(t) = (t(1-t)^{-2n-1}P_n(t))', \quad (18.7)$$

$$u_{n+1}(t) = \left(\frac{t}{1-t} u_n(t) \right)', \quad u_n(t) = (t-1)^{-2n}P_n(t), \quad u_0 = 1. \quad (18.8)$$

They follow by the product rule from (18.5).

Theorem 18.3 (Diagonal Stirling generating function). For $n \geq 1$,

$$\sum_{m=0}^{\infty} \left\{ \begin{matrix} n+m \\ m \end{matrix} \right\} t^m = \frac{tP_n(t)}{(1-t)^{2n+1}}. \quad (18.9)$$

Proof. Let $F_n(t)$ denote the left side. The second-kind recurrence gives $(1-t)F_{n+1} = tF'_n$, with $F_0 = (1-t)^{-1}$. Hence $F_1 = t/(1-t)^3$. If the displayed formula holds for n , differentiating $tP_n/(1-t)^{2n+1}$ and using (18.5) gives the formula for $n+1$. $\square$

18.3 Second-order Eulerian interpolation of first-kind diagonals

The same second-order Eulerian row that controls (18.9) also gives a binomial-basis expansion for the near-diagonals of the unsigned first-kind triangle.

Theorem 18.4 (Second-order Eulerian diagonal formula). For $p, m \geq 0$,

$$\left[\begin{matrix} m \\ m-p \end{matrix} \right] = \sum_{j=0}^p \langle\langle p \rangle\rangle_j \binom{m+j}{2p}, \quad (18.10)$$

where an out-of-range first-kind number is zero. Equivalently, the polynomial continuation in a variable x satisfies

$$\begin{bmatrix} x \\ x-p \end{bmatrix} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle \binom{x+j}{2p}.$$

The first instances are

$$\begin{aligned} \begin{bmatrix} x \\ x-1 \end{bmatrix} &= \binom{x}{2}, \\ \begin{bmatrix} x \\ x-2 \end{bmatrix} &= \binom{x}{4} + 2 \binom{x+1}{4}, \\ \begin{bmatrix} x \\ x-3 \end{bmatrix} &= \binom{x}{6} + 8 \binom{x+1}{6} + 6 \binom{x+2}{6}. \end{aligned}$$

Proof. For fixed p define

$$D_p(t) = \sum_{m \geq 0} \begin{bmatrix} m \\ m-p \end{bmatrix} t^m.$$

Writing $a_{m,p} = \begin{bmatrix} m \\ m-p \end{bmatrix}$, the first-kind recurrence becomes $a_{m,p} = a_{m-1,p} + (m-1)a_{m-1,p-1}$. Therefore

$$(1-t)D_p(t) = t^2 D'_{p-1}(t), \quad D_0(t) = \frac{1}{1-t}. \quad (18.11)$$

Let

$$Q_p(t) = t^{2p} P_p(t^{-1}).$$

The polynomial recurrence (18.5) is equivalent, after the substitution $t \mapsto t^{-1}$, to

$$Q_p(t) = t^2 ((1-t)Q'_{p-1}(t) + (2p-1)Q_{p-1}(t)).$$

It follows inductively from (18.11) that

$$D_p(t) = \frac{Q_p(t)}{(1-t)^{2p+1}} = \frac{t^{2p} P_p(t^{-1})}{(1-t)^{2p+1}}. \quad (18.12)$$

Since $P_p(u) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle u^j$, coefficient extraction gives

$$[t^m] D_p(t) = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle [t^{m-2p+j}] (1-t)^{-2p-1} = \sum_{j=0}^p \left\langle \left\langle p \right\rangle \right\rangle \left\langle \left\langle j \right\rangle \right\rangle \binom{m+j}{2p},$$

which is (18.10). Both sides are polynomials in m of degree at most $2p$, so the same identity gives the stated polynomial continuation. $\square$

Part VI

Differentiation, Composition, and Reversion

Chapter 19

Faà di Bruno's formula

19.1 The set-partition form

For a block $B \subseteq [n]$, write $|B|$ for its size.

Theorem 19.1 (Faà di Bruno, partition form). *For $n \geq 0$ and sufficiently differentiable f, g ,*

$$D^n(f \circ g)(x) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(g(x)) \prod_{B \in \pi} g^{(|B|)}(x). \quad (19.1)$$

For $n = 0$, the empty partition contributes $f(g(x))$.

Proof. Induct on n . Differentiate a term indexed by a partition π of $[n]$. If the derivative hits $f^{(|\pi|)}(g)$, the chain rule contributes g' and creates the new singleton block $\{n+1\}$. If it hits the factor associated with a block B , it changes $g^{(|B|)}$ to $g^{(|B|+1)}$, which inserts $n+1$ into B . Every partition of $[n+1]$ arises uniquely in exactly one of these ways: remove $n+1$, either deleting its singleton block or removing it from its non-singleton block. Hence all terms occur once. $\square$

19.2 Multiplicity and Bell-polynomial forms

Grouping partitions by the number m_j of blocks of size j gives the classical formula.

Theorem 19.2 (Faà di Bruno, multiplicity form).

$$D^n f(g(x)) = \sum_{\substack{m_1, \dots, m_n \geq 0 \\ m_1 + 2m_2 + \dots + nm_n = n}} \frac{n!}{m_1! m_2! \dots m_n!} \times f^{(m_1 + \dots + m_n)}(g(x)) \prod_{j=1}^n \left(\frac{g^{(j)}(x)}{j!} \right)^{m_j}. \quad (19.2)$$

Equivalently,

$$D^n f(g(x)) = \sum_{k=0}^n f^{(k)}(g(x)) \mathcal{B}_{n,k}(g'(x), g''(x), \dots, g^{(n-k+1)}(x)). \quad (19.3)$$

Proof. For fixed multiplicities (m_j) , the number of set partitions of $[n]$ with m_j blocks of size j is $n! / \prod_j ((j!)^{m_j} m_j!)$. Substitution in (19.1) gives (19.2). Grouping further by $k = \sum m_j$ and invoking (12.1) gives (19.3). $\square$

For $n = 4$ this reads

$$\begin{aligned} (f \circ g)^{(4)} &= f^{(4)}(g)(g')^4 + 6f^{(3)}(g)(g')^2 g'' + 3f''(g)(g'')^2 \\ &\quad + 4f''(g)g'g^{(3)} + f'(g)g^{(4)}. \end{aligned} \quad (19.4)$$

The coefficients 1, 6, 3, 4, 1 count partitions of types $1+1+1+1$, $2+1+1$, $2+2$, $3+1$, and 4 respectively. This also proves the expanded example in the archive without four separate differentiations.

19.3 The exponential special case

Because every derivative of e^u equals e^u , summing the partial Bell polynomials in (19.3) yields

$$D^n e^{g(x)} = e^{g(x)} \mathcal{B}_n(g'(x), g''(x), \dots, g^{(n)}(x)). \quad (19.5)$$

When g is a cumulant generating function, this is precisely the moment–cumulant relation of Theorem 15.4.

19.4 Mixed partial derivatives

For $B \subseteq [n]$, write $\partial_B = \prod_{j \in B} \partial / \partial x_j$.

Theorem 19.3 (Multivariate Faà di Bruno for distinct variables). *If $y = g(x_1, \dots, x_n)$, then*

$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} f(y) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(y) \prod_{B \in \pi} \partial_B y. \quad (19.6)$$

Proof. The proof of Theorem 19.1 applies verbatim, except that the new derivative is $\partial / \partial x_{n+1}$. It either creates a singleton block or joins the new index to one existing block. $\square$

For three variables, (19.6) becomes

$$\begin{aligned} \partial_1 \partial_2 \partial_3 f(y) &= f'(y) \partial_{123} y \\ &\quad + f''(y) ((\partial_1 y)(\partial_{23} y) + (\partial_2 y)(\partial_{13} y) + (\partial_3 y)(\partial_{12} y)) \\ &\quad + f^{(3)}(y) (\partial_1 y)(\partial_2 y)(\partial_3 y). \end{aligned}$$

For repeated variables or vector-valued inner maps, one labels derivative slots first and then contracts tensor indices; the same partition lattice remains the combinatorial skeleton.

Chapter 20

Composition of ordinary and exponential series

20.1 Ordinary formal power series

Let

$$F(u) = \sum_{k \geq 0} a_k u^k, \quad G(x) = \sum_{j \geq 1} b_j x^j, \quad F(G(x)) = \sum_{n \geq 0} c_n x^n.$$

Theorem 20.1 (Ordinary composition). *For $n \geq 1$,*

$$c_n = \sum_{k=1}^n a_k \sum_{\substack{i_1 + \dots + i_k = n \\ i_r \geq 1}} b_{i_1} \dots b_{i_k}, \quad (20.1)$$

$$= \sum_{k=1}^n a_k \sum_{\substack{\pi_1 + \dots + \pi_n = k \\ \pi_1 + 2\pi_2 + \dots + n\pi_n = n}} \binom{k}{\pi_1, \dots, \pi_n} b_1^{\pi_1} \dots b_n^{\pi_n} \quad (20.2)$$

$$= \sum_{k=1}^n a_k \widehat{\mathcal{B}}_{n,k}(b_1, \dots, b_{n-k+1}), \quad (20.3)$$

and $c_0 = a_0$.

Proof. The coefficient of x^n in $G(x)^k$ is obtained by choosing an ordered composition $i_1 + \dots + i_k = n$, proving the first formula. Collect equal part sizes; the number of orderings with multiplicities π_j is the multinomial coefficient, proving the second. The third is the definition of the ordinary Bell polynomial. $\square$

Two useful specializations are immediate. If $F(u) = e^u$ and the series are exponentially normalized, one obtains the exponential formula. If $A(x) = 1 + \sum_{n \geq 1} a_n x^n$, then

$$[x^n] \frac{1}{A(x)} = \sum_{k=1}^n (-1)^k \widehat{\mathcal{B}}_{n,k}(a_1, \dots, a_{n-k+1}) \quad (n \geq 1), \quad (20.4)$$

by taking $F(u) = 1/(1-u)$ and $G(x) = -\sum_{n \geq 1} a_n x^n$.

20.2 Exponential formal power series

Let

$$F(x) = \sum_{j \geq 1} a_j \frac{x^j}{j!}, \quad G(u) = \sum_{k \geq 0} b_k \frac{u^k}{k!}, \quad G(F(x)) = \sum_{n \geq 0} c_n \frac{x^n}{n!}.$$

Theorem 20.2 (Exponential composition).

$$c_n = \sum_{k=0}^n b_k \mathcal{B}_{n,k}(a_1, \dots, a_{n-k+1}). \quad (20.5)$$

Equivalently,

$$c_n = \sum_{\pi \in \text{Part}([n])} b_{|\pi|} \prod_{B \in \pi} a_{|B|}. \quad (20.6)$$

Proof. Substitute $F(x)$ into G and apply (1.10). The set-partition form is Theorem 12.3 with an additional weight b_k for the number of blocks. $\square$

Taking $b_k = 1$ gives the complete Bell-polynomial exponential identity

$$\exp \left(\sum_{j \geq 1} a_j \frac{x^j}{j!} \right) = \sum_{n \geq 0} \mathcal{B}_n(a_1, \dots, a_n) \frac{x^n}{n!}. \quad (20.7)$$

If the series converge, (20.5) is exactly (19.3) evaluated at the expansion point; formally, no convergence notion is needed.

Chapter 21

Lagrange inversion and Lagrange–Bürmann

21.1 Formal theorem

Let R be a commutative $\mathbb{Q}$ -algebra and let $\phi(w) \in R[[w]]$ have invertible constant term. There is a unique $g(z) \in zR[[z]]$ satisfying

$$g(z) = z\phi(g(z)). \quad (21.1)$$

Uniqueness follows recursively because the coefficient of z^n on the right depends only on coefficients of g below degree n .

Theorem 21.1 (Lagrange–Bürmann). *For $n \geq 1$ and every formal series H ,*

$$[z^n]H(g(z)) = \frac{1}{n}[w^{n-1}]H'(w)\phi(w)^n, \quad (21.2)$$

$$= [w^n]H(w)\phi(w)^{n-1}(\phi(w) - w\phi'(w)). \quad (21.3)$$

In particular,

$$[z^n]g(z) = \frac{1}{n}[w^{n-1}]\phi(w)^n. \quad (21.4)$$

Proof. Put $f(w) = w/\phi(w)$, so that $f(g(z)) = z$. Formal residues and the change of variables theorem Theorem 1.2 give

$$\begin{aligned} [z^n]H(g(z)) &= \operatorname{Res}_z H(g(z))z^{-n-1} dz \\ &= \operatorname{Res}_w H(w)f(w)^{-n-1}f'(w) dw \\ &= -\frac{1}{n} \operatorname{Res}_w H(w)(f(w)^{-n})' dw \\ &= \frac{1}{n} \operatorname{Res}_w H'(w)f(w)^{-n} dw, \end{aligned}$$

where the residue of a derivative is zero. Since $f(w)^{-n} = w^{-n}\phi(w)^n$, this is (21.2). Applying the same zero-residue rule to $H(w)\phi(w)^n w^{-n}$ gives

$$\frac{1}{n} \operatorname{Res} H' \phi^n w^{-n} = \operatorname{Res} H \phi^n w^{-n-1} - \operatorname{Res} H \phi^{n-1} \phi' w^{-n},$$

which is (21.3). Take $H(w) = w$ for (21.4). □

21.2 Analytic inverse functions

Theorem 21.2 (Analytic Lagrange inversion). *Let f be analytic near a , let $b = f(a)$, and suppose $f'(a) \neq 0$. Its local inverse g is analytic near b and has*

$$g(z) = a + \sum_{n \geq 1} g_n \frac{(z - b)^n}{n!}, \quad (21.5)$$

where

$$g_n = \frac{d^{n-1}}{dw^{n-1}} \left(\frac{w - a}{f(w) - f(a)} \right)^n \Big|_{w=a}. \quad (21.6)$$

The series has a positive radius of convergence.

Proof. The analytic inverse-function theorem gives an analytic local inverse and therefore a convergent Taylor series. Put $u = w - a$, $\zeta = z - b$, and

$$\phi(u) = \frac{u}{f(a + u) - f(a)};$$

its constant term is $1/f'(a)$. The inverse displacement satisfies $u = \zeta \phi(u)$. Formula (21.4), multiplied by $n!$, is exactly (21.6). $\square$

If $f'(a) = 0$ but $f(w) - f(a) = c(w - a)^q + O((w - a)^{q+1})$ with $c \neq 0$, factor $f(w) - f(a) = (w - a)^q U(w)$ with $U(a) = c$. Choosing one of the q values of $c^{-1/q}$ reduces the equation to ordinary inversion in the variable $(z - f(a))^{1/q}$. Thus the inverse has q local Puiseux branches. This is the precise sense in which Lagrange inversion extends to a critical point; a single-valued Taylor inverse cannot exist there.

21.3 Inverse coefficients in Bell-polynomial form

Suppose

$$f(w) = \sum_{j \geq 1} f_j \frac{w^j}{j!}, \quad f_1 \neq 0, \quad g(z) = f^{(-1)}(z) = \sum_{n \geq 1} g_n \frac{z^n}{n!}.$$

Define

$$\hat{f}_j = \frac{f_{j+1}}{(j+1)f_1}. \quad (21.7)$$

Theorem 21.3 (Explicit inverse coefficients). *One has $g_1 = f_1^{-1}$ and, for $n \geq 2$,*

$$g_n = \frac{1}{f_1^n} \sum_{k=1}^{n-1} (-1)^k n^{\bar{k}} \mathcal{B}_{n-1,k}(\hat{f}_1, \dots, \hat{f}_{n-k}). \quad (21.8)$$

Proof. Write

$$\frac{f(w)}{f_1 w} = 1 + U(w), \quad U(w) = \sum_{j \geq 1} \hat{f}_j \frac{w^j}{j!}.$$

By (21.4), $g_n = (n-1)! [w^{n-1}] (w/f(w))^n$. Now

$$(1 + U)^{-n} = \sum_{k \geq 0} (-1)^k \binom{n+k-1}{k} U^k = \sum_{k \geq 0} (-1)^k \frac{n^{\bar{k}}}{k!} U^k.$$

Apply (1.10) and extract degree $n-1$; the $k=0$ term vanishes when $n \geq 2$. $\square$

21.4 The associahedron interpretation, with normalized notation

The formula in the archive mixes exponential coefficients with an ordinary-series face formula. The exact geometric statement is the following.

Let

$$A(x) = x + \sum_{n \geq 1} a_n x^{n+1}, \quad B(x) = A^{\langle -1 \rangle}(x) = x + \sum_{n \geq 1} b_n x^{n+1}. \quad (21.9)$$

Use the indexing in which the associahedron K_n has dimension $n - 1$ and its faces are plane rooted trees with $n + 1$ leaves and internal arities at least 2. If an internal vertex has arity $i + 1$, give it weight a_i ; let a_F be the product of all vertex weights. Equivalently, if $F \cong K_{i_1} \times \cdots \times K_{i_m}$, then $a_F = a_{i_1} \cdots a_{i_m}$.

Theorem 21.4 (Associahedral inversion).

$$b_n = \sum_{F \text{ a face of } K_n} (-1)^{n - \dim F} a_F. \quad (21.10)$$

Proof. The equation $A(B) = x$ is

$$B = x - \sum_{i \geq 1} a_i B^{i+1}.$$

Iterating it recursively expands b_n as a sum over plane rooted trees with $n + 1$ leaves: every internal vertex of arity $i + 1$ contributes $-a_i$. A face F of K_n is exactly such a tree after contracting a chosen set of internal edges. If it has m internal vertices, then $\dim F = n - m$, so its sign is $(-1)^m = (-1)^{n - \dim F}$ and its weight is a_F . Thus the tree expansion is the alternating face sum. $\square$

For example,

$$\begin{aligned} b_1 &= -a_1, \\ b_2 &= -a_2 + 2a_1^2, \\ b_3 &= -a_3 + 5a_2a_1 - 5a_1^3, \\ b_4 &= -a_4 + 6a_3a_1 + 3a_2^2 - 21a_2a_1^2 + 14a_1^4. \end{aligned}$$

For K_4 , the coefficients 1, 6, 3, 21, 14 count respectively the whole 3-polytope, its six pentagonal and three square facets, its edges, and its vertices.

Chapter 22

Algebraic and transcendental examples

22.1 The Fuss–Catalan algebraic equation

Let $p \geq 2$ be an integer and consider

$$x^p - x + z = 0, \quad x(0) = 0. \quad (22.1)$$

It is equivalent to $x = z\phi(x)$ with $\phi(x) = (1 - x^{p-1})^{-1}$.

Theorem 22.1. *The distinguished solution is*

$$x(z) = \sum_{k=0}^{\infty} \binom{pk}{k} \frac{z^{(p-1)k+1}}{(p-1)k+1}. \quad (22.2)$$

Its radius of convergence is

$$R_p = (p-1)p^{-p/(p-1)}, \quad (22.3)$$

and the series converges absolutely also on $|z| = R_p$.

Proof. By (21.4),

$$[z^n]x(z) = \frac{1}{n}[w^{n-1}](1 - w^{p-1})^{-n}.$$

This vanishes unless $n-1 = (p-1)k$. In that case the coefficient is $\frac{1}{n} \binom{n+k-1}{k} = \binom{pk}{k} / ((p-1)k+1)$, proving the series.

The inverse map is $z = f(x) = x - x^p$. Its nearest critical points satisfy $f'(x) = 1 - px^{p-1} = 0$ and all have modulus $p^{-1/(p-1)}$; their critical values have modulus $(p-1)p^{-p/(p-1)}$. No local inverse can pass analytically through a critical value, while the inverse-function theorem continues the branch up to one, so this is the radius. Stirling's formula gives coefficients of order $Ck^{-3/2}R_p^{-((p-1)k+1)}$, which proves absolute convergence on the boundary. $\square$

22.2 Lambert's W function at the origin

The Lambert function satisfies $W(z)e^{W(z)} = z$.

Theorem 22.2. *On its principal branch near 0,*

$$W(z) = \sum_{n=1}^{\infty} (-n)^{n-1} \frac{z^n}{n!} = z - z^2 + \frac{3}{2}z^3 - \frac{8}{3}z^4 + O(z^5), \quad (22.4)$$

with radius of convergence e^{-1} .

Proof. Here $f(w) = we^w$, so $w/f(w) = e^{-w}$. Formula (21.6) gives $g_n = D^{n-1}e^{-nw}|_{w=0} = (-n)^{n-1}$. The inverse map has its nearest critical point at $w = -1$, where $f(-1) = -e^{-1}$; this square-root branch point determines the radius. $\square$

22.3 A rapidly useful expansion around $W(e) = 1$

Put $\xi = \log z - 1$ and $u = W(z) - 1$. Then

$$u + \log(1 + u) = \xi. \quad (22.5)$$

Reversion gives

$$W(e^{1+\xi}) = 1 + \frac{\xi}{2} + \frac{\xi^2}{16} - \frac{\xi^3}{192} - \frac{\xi^4}{3072} + \frac{13\xi^5}{61440} + O(\xi^6). \quad (22.6)$$

Proof. The derivative at $u = 0$ of the left side of (22.5) is 2, so it has an ordinary inverse. Substitute $u = c_1\xi + \cdots + c_5\xi^5$ into $u + \log(1 + u) = \xi$ and compare successive powers. The triangular equations give $c_1 = 1/2$, $c_2 = 1/16$, $c_3 = -1/192$, $c_4 = -1/3072$, $c_5 = 13/61440$. $\square$

The closest critical points of $u + \log(1 + u)$ satisfy $1 + (1 + u)^{-1} = 0$, hence $u = -2$; using the two adjacent logarithm values gives images $-2 \pm i\pi$. Therefore the radius in the ξ -plane is $\sqrt{4 + \pi^2}$. For positive real z , the expansion converges whenever $|\log z - 1| < \sqrt{4 + \pi^2}$, approximately $0.0655 < z < 112.63$. Taking $\xi = -1$ yields a convergent evaluation of $W(1) = 0.567143\dots$

22.4 Binary trees and Catalan numbers

Let C_n count rooted ordered full binary trees with n internal vertices, and let $B(z) = \sum_{n \geq 0} C_n z^n$. Removing the root gives

$$B(z) = 1 + zB(z)^2. \quad (22.7)$$

Put $C(z) = B(z) - 1$; then $C = z(1 + C)^2$. Lagrange inversion gives

$$C_n = [z^n]C(z) = \frac{1}{n}[w^{n-1}](1 + w)^{2n} = \frac{1}{n} \binom{2n}{n-1} = \frac{1}{n+1} \binom{2n}{n}. \quad (22.8)$$

This simultaneously proves the functional equation, its unique formal solution, and the Catalan closed form. The symbol C_n is used here to avoid confusing these binary-tree numbers with Bell numbers.

Chapter 23

Lagrange's reversion theorem in shifted form

23.1 The two-parameter implicit equation

Let $v = v(x, y)$ be the unique formal solution of

$$v = x + yf(v), \quad v(x, 0) = x. \quad (23.1)$$

Theorem 23.1 (Lagrange reversion). *For every formal or analytic g ,*

$$g(v) = g(x) + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} (f(x)^k g'(x)). \quad (23.2)$$

In particular,

$$v = x + \sum_{k=1}^{\infty} \frac{y^k}{k!} \left(\frac{\partial}{\partial x} \right)^{k-1} f(x)^k. \quad (23.3)$$

Proof. Put $u = v - x$. Then $u = y\phi(u)$ with $\phi(u) = f(x + u)$, treating x as a coefficient parameter. Apply (21.2) to $H(u) = g(x + u)$:

$$[y^k]g(v) = \frac{1}{k} [u^{k-1}]g'(x + u)f(x + u)^k = \frac{1}{k!} D_x^{k-1}(g'(x)f(x)^k).$$

Summation proves the first formula; take $g(z) = z$ for the second. $\square$

23.2 The delta-function proof decoded

The archive also gives a distributional derivation. Assume temporarily that $x, y \in \mathbb{R}$, the equation $z - x - yf(z) = 0$ has one simple root v , and $1 - yf'(v) > 0$. The elementary change-of-variables rule for the Dirac delta gives

$$g(v) = \int_{\mathbb{R}} \delta(yf(z) - z + x)g(z)(1 - yf'(z)) \, dz. \quad (23.4)$$

Using $\delta(s) = \int_{\mathbb{R}} e^{iks} \, dk / (2\pi)$ and expanding only in the formal parameter y ,

$$\begin{aligned} g(v) &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} (1 - yf'(x)) \right] \\ &= \sum_{n \geq 0} D_x^n \left[\frac{y^n f(x)^n g(x)}{n!} - \frac{y^{n+1}}{(n+1)!} ((gf^{n+1})' - g'f^{n+1}) \right]. \end{aligned} \quad (23.5)$$

The first term of level $n + 1$ cancels the total derivative in the second term of level n ; what remains is exactly (23.2). Thus the Fourier–delta calculation is a compact analytic avatar of the formal residue proof. Formula (23.4) requires an absolute Jacobian in general; the displayed sign is valid in the stated orientation-preserving neighbourhood.

Chapter 24

Laplace-type asymptotic integrals

24.1 Endpoint expansion

Consider

$$I(\lambda) = \int_a^b e^{-\lambda f(x)} g(x) \, dx, \quad \lambda \rightarrow +\infty, \quad (24.1)$$

where the endpoint a is the unique relevant minimum and, with $t = x - a$,

$$f(a+t) \sim f(a) + \sum_{k \geq 0} a_k t^{k+\alpha}, \quad a_0 > 0, \quad \alpha > 0, \quad (24.2)$$

$$g(a+t) \sim \sum_{k \geq 0} b_k t^{k+\beta-1}, \quad \beta > 0. \quad (24.3)$$

Let $\widehat{\mathcal{B}}_{k,j}$ denote the ordinary Bell polynomial.

Theorem 24.1 (Laplace–Erdélyi coefficient formula). *Under the standard differentiability and remainder hypotheses that permit Watson’s lemma,*

$$I(\lambda) \sim e^{-\lambda f(a)} \sum_{n=0}^{\infty} \Gamma\left(\frac{n+\beta}{\alpha}\right) \frac{c_n}{\lambda^{(n+\beta)/\alpha}}, \quad (24.4)$$

where

$$c_n = \frac{1}{\alpha a_0^{(n+\beta)/\alpha}} \sum_{k=0}^n b_{n-k} \sum_{j=0}^k \binom{-\frac{n+\beta}{\alpha}}{j} \frac{\widehat{\mathcal{B}}_{k,j}(a_1, \dots, a_{k-j+1})}{a_0^j}. \quad (24.5)$$

Proof. Write $f(a+t) - f(a) = t^\alpha A(t)$ with $A(t) = a_0 + a_1 t + \dots$, and set $s = (f(a+t) - f(a))^{1/\alpha} = tA(t)^{1/\alpha}$. This has an inverse $t = t(s)$. Expand

$$g(a+t(s))t'(s) = \sum_{n \geq 0} d_n s^{n+\beta-1}.$$

A generalized Lagrange–Bürmann calculation, first for integral β and then by formal continuation in the exponent, gives

$$d_n = \sum_{k=0}^n b_{n-k} [t^k] A(t)^{-(n+\beta)/\alpha}.$$

Indeed, apply (21.2) to an antiderivative of $t^{m+\beta-1}$ and differentiate back after substitution.

Put $\rho = (n + \beta)/\alpha$. The ordinary Bell expansion yields

$$\begin{aligned} [t^k]A(t)^{-\rho} &= a_0^{-\rho} [t^k] \sum_{j \geq 0} \binom{-\rho}{j} \left(\sum_{r \geq 1} \frac{a_r}{a_0} t^r \right)^j \\ &= a_0^{-\rho} \sum_{j=0}^k \binom{-\rho}{j} a_0^{-j} \widehat{\mathcal{B}}_{k,j}(a_1, \dots, a_{k-j+1}). \end{aligned}$$

Thus $c_n = d_n/\alpha$ is (24.5). Finally

$$\int_0^\infty e^{-\lambda s^\alpha} s^{n+\beta-1} ds = \frac{1}{\alpha} \Gamma\left(\frac{n+\beta}{\alpha}\right) \lambda^{-(n+\beta)/\alpha},$$

and Watson's lemma controls the replacement of the finite endpoint interval by the asymptotic expansion. $\square$

Remark 24.2. The appearance of ordinary rather than exponential Bell polynomials is forced by the ordinary Taylor expansion of $A(t)^{-\rho}$. Lagrange inversion enters in the change from t to the natural Laplace coordinate s ; the gamma factor enters only in the final monomial integral.

Part VII

Factorial, Umbral, and Special-Function Extensions

Chapter 25

Pochhammer symbols, binomial transforms, and factorial calculus

The five drafts use several mutually compatible factorial conventions. This chapter places them in one calculus and makes explicit the binomial inversion machinery that is repeatedly used later. The notation and analytic continuation conventions agree with the standard special-function references [13, 25]. We retain $x^{\underline{n}}$ and $x^{\overline{n}}$ for falling and rising factorials; when a formula is standardly written with the Pochhammer symbol, we set

$$(x)_n := x^{\overline{n}} = x(x+1) \cdots (x+n-1), \quad (x)_0 = 1. \quad (25.1)$$

Thus $x^{\underline{n}} = (-1)^n (-x)_n$.

25.1 Elementary identities and analytic continuation

Proposition 25.1 (Shift, splitting, and gamma quotient). *For nonnegative integers m, n ,*

$$(x)_{m+n} = (x)_m (x+m)_n, \quad (25.2)$$

$$x^{\overline{m+n}} = x^{\overline{m}} x^{\overline{n}}, \quad (25.3)$$

$$(x)_n = \frac{\Gamma(x+n)}{\Gamma(x)} \quad (25.4)$$

whenever neither side of (25.4) has an uncanceled pole. The quotient on the right gives the meromorphic continuation of $(x)_n$ in x .

Proof. Split the defining product for $(x)_{m+n}$ after its first m factors; the remaining factors are $x+m, x+m+1, \dots, x+m+n-1$. The falling-factorial identity is identical after reversing the direction of the shifts. Repeatedly applying $\Gamma(z+1) = z\Gamma(z)$ gives $\Gamma(x+n) = (x+n-1) \cdots x \Gamma(x)$. Division proves (25.4) away from poles, and meromorphic continuation extends the identity wherever the quotient is defined. $\square$

Proposition 25.2 (Vandermonde convolution for rising factorials). *For every $n \geq 0$,*

$$(x+y)_n = \sum_{k=0}^n \binom{n}{k} (x)_k (y)_{n-k}. \quad (25.5)$$

Equivalently, for generalized binomial coefficients,

$$\binom{x+y}{n} = \sum_{k=0}^n \binom{x}{k} \binom{y}{n-k}. \quad (25.6)$$

Proof. The generalized binomial theorem, proved below, gives

$$(1 - z)^{-x}(1 - z)^{-y} = (1 - z)^{-(x+y)}.$$

The coefficient of z^n on the left is $\sum_{k=0}^n (x)_k (y)_{n-k} / (k!(n-k)!)$, whereas on the right it is $(x+y)_n / n!$. Multiplication by $n!$ proves (25.5). Replacing x, y by $-x, -y$, using $\binom{x}{n} = x^n / n! = (-1)^n (-x)_n / n!$, gives (25.6). $\square$

25.2 The generalized binomial theorem

Theorem 25.3 (Formal generalized binomial theorem). *Let R be a commutative $\mathbb{Q}$ -algebra and let $\alpha \in R$. In $R[[z]]$,*

$$(1 + z)^\alpha = \sum_{n \geq 0} \binom{\alpha}{n} z^n = \sum_{n \geq 0} \frac{\alpha^n}{n!} z^n, \quad (25.7)$$

$$(1 - z)^{-\alpha} = \sum_{n \geq 0} \frac{(\alpha)_n}{n!} z^n. \quad (25.8)$$

Analytically, the same series converge for $|z| < 1$, with any consistent choice of the logarithm used to define the powers.

Proof. Let $F(z) = \sum_{n \geq 0} \alpha^n z^n / n!$. Since $\alpha^{n+1} = (\alpha - n)\alpha^n$, coefficient comparison gives

$$(1 + z)F'(z) = \alpha F(z), \quad F(0) = 1.$$

The formal differential equation has a unique solution: once the constant term is fixed, its coefficient of degree $n + 1$ is determined by coefficients of lower degree. The formal exponential $\exp(\alpha \log(1 + z))$ satisfies the same equation and initial condition, so it equals F . This proves (25.7); replacing z by $-z$ and α by $-\alpha$ proves (25.8). In the analytic setting, the ratio test gives radius one unless the series terminates, and differentiation identifies the sum on every simply connected subdomain of the unit disk. $\square$

25.3 Binomial inversion and finite differences

Theorem 25.4 (Binomial inversion). *For sequences (a_n) and (b_n) in a commutative ring, the two systems*

$$b_n = \sum_{k=0}^n \binom{n}{k} a_k, \quad (25.9)$$

$$a_n = \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} b_k \quad (25.10)$$

are equivalent.

Proof. Assume (25.9) and substitute it into the right side of (25.10). Interchanging the two finite sums gives

$$\begin{aligned} \sum_{k=0}^n (-1)^{n-k} \binom{n}{k} b_k &= \sum_{j=0}^n a_j \sum_{k=j}^n (-1)^{n-k} \binom{n}{k} \binom{k}{j} \\ &= \sum_{j=0}^n a_j \binom{n}{j} \sum_{r=0}^{n-j} (-1)^{n-j-r} \binom{n-j}{r}. \end{aligned}$$

The inner sum is $(1 - 1)^{n-j}$, hence is one for $j = n$ and zero otherwise. The result is a_n . The same calculation in the opposite order proves the converse, because the inverse matrix is its own inverse after the alternating signs are inserted. $\square$

Corollary 25.5 (Generating-function forms). *Let $A(z) = \sum_{n \geq 0} a_n z^n$ and $B(z) = \sum_{n \geq 0} b_n z^n$. Then (25.9) is equivalent to*

$$B(z) = \frac{1}{1-z} A\left(\frac{z}{1-z}\right), \quad A(z) = \frac{1}{1+z} B\left(\frac{z}{1+z}\right). \quad (25.11)$$

For exponential generating functions $\hat{A}(z) = \sum a_n z^n / n!$ and $\hat{B}(z) = \sum b_n z^n / n!$, the equivalent relation is

$$\hat{B}(z) = e^z \hat{A}(z), \quad \hat{A}(z) = e^{-z} \hat{B}(z). \quad (25.12)$$

Proof. For ordinary generating functions, sum (25.9) over n and use $\sum_{n \geq k} \binom{n}{k} z^n = z^k / (1-z)^{k+1}$. Solving the fractional-linear substitution for its inverse gives the second formula. For exponential generating functions,

$$\sum_{n \geq 0} \sum_{k=0}^n \binom{n}{k} a_k \frac{z^n}{n!} = \left(\sum_{k \geq 0} a_k \frac{z^k}{k!} \right) \left(\sum_{j \geq 0} \frac{z^j}{j!} \right),$$

which is (25.12). $\square$

Proposition 25.6 (Newton series and binomial inversion). *For every polynomial p of degree at most d ,*

$$p(x) = \sum_{k=0}^d \binom{x}{k} \Delta^k p(0), \quad \Delta^k p(0) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} p(j). \quad (25.13)$$

Proof. The first statement is the Newton expansion already proved in Theorem 1.3. Evaluating it at integers $x = n$ gives $p(n) = \sum_{k=0}^n \binom{n}{k} \Delta^k p(0)$. Applying Theorem 25.4 yields the second identity. Conversely, the second identity and binomial inversion recover the first at all nonnegative integers; two polynomials of degree at most d agreeing at $d+1$ points are equal. $\square$

25.4 Factorial differential and difference rules

Proposition 25.7. *For $n \geq 1$,*

$$\Delta x^n = nx^{\overline{n-1}}, \quad (25.14)$$

$$\nabla x^{\overline{n}} = nx^{\overline{n-1}}, \quad (25.15)$$

$$\frac{d}{dx}(x)_n = (x)_n \sum_{j=0}^{n-1} \frac{1}{x+j} = (x)_n (\psi(x+n) - \psi(x)), \quad (25.16)$$

where $\nabla f(x) = f(x) - f(x-1)$, and the last identity holds away from poles.

Proof. Factor the two neighboring products:

$$x + 1^{\overline{n}} - x^{\overline{n}} = x^{\overline{n-1}}((x+1) - (x-n+1)) = nx^{\overline{n-1}}.$$

The rising-factorial statement is the reflected identity. Logarithmic differentiation of $(x)_n = \prod_{j=0}^{n-1} (x+j)$ gives the finite sum. Logarithmic differentiation of (25.4) gives the digamma difference. $\square$

Remark 25.8. The formulas in this chapter explain why factorial bases are the natural coordinates for finite-difference operators, just as monomials are natural for differentiation. Stirling numbers are precisely the transition matrices between these two coordinate systems.

Chapter 26

Bernoulli, Euler, Genocchi, and Cauchy polynomial families

To avoid collision with Bell numbers, Bernoulli numbers remain β_n , and we write $B_n(x)$ for Bernoulli polynomials, $E_n(x)$ for Euler polynomials, G_n for Genocchi numbers, and $C_n(x)$ for the Bernoulli–Cauchy polynomials of the second kind, following the classical difference-calculus tradition [23].

26.1 Bernoulli polynomials as an Appell sequence

Definition 26.1. The Bernoulli polynomials are defined by

$$\frac{te^{xt}}{e^t - 1} = \sum_{n \geq 0} B_n(x) \frac{t^n}{n!}. \quad (26.1)$$

Thus $B_n(0) = \beta_n$.

Theorem 26.2 (Appell, translation, difference, and reflection identities). *For $n \geq 1$,*

$$B'_n(x) = nB_{n-1}(x), \quad (26.2)$$

$$B_n(x+y) = \sum_{k=0}^n \binom{n}{k} B_k(x) y^{n-k}, \quad (26.3)$$

$$B_n(x+1) - B_n(x) = nx^{n-1}, \quad (26.4)$$

$$B_n(1-x) = (-1)^n B_n(x). \quad (26.5)$$

Moreover,

$$B_n(x) = \sum_{k=0}^n \binom{n}{k} \beta_k x^{n-k}. \quad (26.6)$$

Proof. Differentiate (26.1) with respect to x ; the result is t times the same generating function, and coefficient comparison gives (26.2). Multiplying the generating function by e^{yt} proves (26.3); taking $x = 0$ gives (26.6). For the finite difference,

$$\frac{te^{(x+1)t}}{e^t - 1} - \frac{te^{xt}}{e^t - 1} = te^{xt},$$

whose coefficient of $t^n/n!$ is nx^{n-1} . Finally,

$$\frac{te^{(1-x)t}}{e^t - 1} = \frac{(-t)e^{x(-t)}}{e^{-t} - 1};$$

replace t by $-t$ in the defining series to obtain (26.5). $\square$

Corollary 26.3 (Faulhaber's formula). *For integers $N \geq 1$ and $p \geq 0$,*

$$\sum_{m=0}^{N-1} m^p = \frac{B_{p+1}(N) - B_{p+1}(0)}{p+1} = \frac{1}{p+1} \sum_{k=0}^p \binom{p+1}{k} \beta_k N^{p+1-k}. \quad (26.7)$$

Proof. Set $n = p + 1$ in (26.4), substitute $x = m$, and sum from $m = 0$ to $N - 1$. The left side telescopes. Formula (26.6) gives the second equality; the $k = p + 1$ term cancels in the difference. $\square$

26.2 Euler polynomials and alternating sums

Definition 26.4. The Euler polynomials are defined by

$$\frac{2e^{xt}}{e^t + 1} = \sum_{n \geq 0} E_n(x) \frac{t^n}{n!}. \quad (26.8)$$

Theorem 26.5. *For $n \geq 1$,*

$$E'_n(x) = nE_{n-1}(x), \quad (26.9)$$

$$E_n(x+1) + E_n(x) = 2x^n, \quad (26.10)$$

$$E_n(1-x) = (-1)^n E_n(x), \quad (26.11)$$

$$E_n(x+y) = \sum_{k=0}^n \binom{n}{k} E_k(x) y^{n-k}. \quad (26.12)$$

Consequently, for $N \geq 1$,

$$\sum_{m=0}^{N-1} (-1)^m m^n = \frac{E_n(0) - (-1)^N E_n(N)}{2}. \quad (26.13)$$

Proof. The derivative and translation identities follow exactly as for Bernoulli polynomials. Multiplying (26.8) by $e^t + 1$ gives

$$\sum_{n \geq 0} (E_n(x+1) + E_n(x)) \frac{t^n}{n!} = 2e^{xt},$$

which proves (26.10). Replacing t by $-t$ proves reflection. Multiply (26.10) at $x = m$ by $(-1)^m$ and sum. Adjacent terms cancel, leaving $E_n(0) - (-1)^N E_n(N)$, and division by two gives (26.13). $\square$

26.3 Genocchi numbers

Definition 26.6. The Genocchi numbers G_n are defined by

$$\frac{2t}{e^t + 1} = \sum_{n \geq 0} G_n \frac{t^n}{n!}. \quad (26.14)$$

Proposition 26.7 (Reduction to Bernoulli numbers). *For every $n \geq 0$,*

$$G_n = 2(1 - 2^n) \beta_n. \quad (26.15)$$

In particular, $G_0 = 0$, $G_1 = 1$, and $G_{2m+1} = 0$ for $m \geq 1$.

Proof. The algebraic identity

$$\frac{2t}{e^t + 1} = \frac{2t}{e^t - 1} - \frac{4t}{e^{2t} - 1}$$

expresses the left side as

$$2 \sum_{n \geq 0} \beta_n \frac{t^n}{n!} - 2 \sum_{n \geq 0} \beta_n \frac{(2t)^n}{n!}.$$

Coefficient comparison proves (26.15). The stated vanishing follows from $\beta_{2m+1} = 0$ for $m \geq 1$, which itself follows from Bernoulli reflection at $x = 0, 1$ together with (26.4). $\square$

26.4 Bernoulli–Cauchy polynomials of the second kind

Definition 26.8. Define $C_n(x)$ by

$$\frac{t(1+t)^x}{\log(1+t)} = \sum_{n \geq 0} C_n(x) \frac{t^n}{n!}. \quad (26.16)$$

These are also called Bernoulli polynomials of the second kind. We use a distinct letter to prevent confusion with $B_n(x)$.

Theorem 26.9 (Stirling expansion, integral representation, and reflection). *For $n \geq 1$,*

$$C_n(x) = C_n(0) + \sum_{k=1}^n \frac{n}{k} s(n-1, k-1) x^k, \quad (26.17)$$

$$C_n(x) = \int_0^1 x + u^n \, du. \quad (26.18)$$

Moreover,

$$C_n(n-2-x) = (-1)^n C_n(x). \quad (26.19)$$

Equivalently, the graph is even or odd about $(n-2)/2$ according as n is even or odd.

Proof. Differentiate (26.16) with respect to x :

$$\sum_{n \geq 0} C'_n(x) \frac{t^n}{n!} = t(1+t)^x.$$

Using $(1+t)^x = \sum_{m \geq 0} x^m t^m / m!$, comparison gives

$$C'_n(x) = nx^{n-1}. \quad (26.20)$$

Expand $x^{n-1} = \sum_{j=0}^{n-1} s(n-1, j) x^j$ and integrate from zero to x . This proves (26.17) after putting $k = j+1$.

For the integral representation, observe directly that

$$\int_0^1 (1+t)^{x+u} \, du = (1+t)^x \frac{(1+t) - 1}{\log(1+t)} = \frac{t(1+t)^x}{\log(1+t)}.$$

Expanding $(1+t)^{x+u}$ by falling factorials and comparing coefficients proves (26.18). Finally, use the elementary reflection $n-1-y^n = (-1)^n y^n$:

$$\begin{aligned} C_n(n-2-x) &= \int_0^1 n-2-x+u^n \, du \\ &= \int_0^1 n-1-(x+1-u)^n \, du \\ &= (-1)^n \int_0^1 x+1-u^n \, du = (-1)^n C_n(x), \end{aligned}$$

where the last equality uses the substitution $v = 1 - u$. This proves (26.19) without an undetermined integration constant. $\square$

26.5 Complementary Bell numbers

Definition 26.10. The complementary Bell numbers are the signed partition sums

$$\widetilde{B}_n := \sum_{k=0}^n (-1)^k \left\{ \begin{matrix} n \\ k \end{matrix} \right\}. \quad (26.21)$$

Theorem 26.11 (Generating function and signed Dobinski formula). *One has*

$$\sum_{n \geq 0} \widetilde{B}_n \frac{z^n}{n!} = \exp(1 - e^z), \quad (26.22)$$

$$\widetilde{B}_n = e \sum_{m \geq 0} \frac{(-1)^m m^n}{m!}. \quad (26.23)$$

The second series converges absolutely for each fixed n .

Proof. Insert the Stirling EGF (4.3) into (26.21) and interchange the locally finite formal sums:

$$\sum_{n \geq 0} \widetilde{B}_n \frac{z^n}{n!} = \sum_{k \geq 0} \frac{(-1)^k (e^z - 1)^k}{k!} = e^{-(e^z - 1)}.$$

This is (26.22). Expanding the outer exponential in the equivalent form $e e^{-e^z}$ gives

$$e \sum_{m \geq 0} \frac{(-1)^m}{m!} e^{mz}.$$

The coefficient of $z^n/n!$ is (26.23). Absolute convergence follows from the ratio test applied to $m^n/m!$. $\square$

26.6 Modified Bernoulli numbers from a logarithmic hyperbolic series

Proposition 26.12. Define b_{2m} by

$$\frac{1}{2} \log \left(\frac{e^{x/2} - e^{-x/2}}{x/2} \right) = \frac{1}{2} \log 2 + \sum_{m \geq 1} b_{2m} x^{2m}. \quad (26.24)$$

Then

$$b_{2m} = \frac{\beta_{2m}}{4m(2m)!}, \quad b_{2m+1} = 0. \quad (26.25)$$

In particular,

$$b_2 = \frac{1}{48}, \quad b_4 = -\frac{1}{5760}, \quad b_6 = \frac{1}{362880}.$$

Proof. After removing the constant $\frac{1}{2} \log 2$, differentiate the left side:

$$\frac{1}{4} \coth(x/2) - \frac{1}{2x}.$$

From the Bernoulli generating function one obtains

$$\coth(x/2) = \frac{2}{x} + 2 \sum_{m \geq 1} \beta_{2m} \frac{x^{2m-1}}{(2m)!}.$$

Substitution cancels the pole and leaves $\frac{1}{2} \sum_{m \geq 1} \beta_{2m} x^{2m-1} / (2m)!$. Termwise integration, with zero constant after the explicit $\frac{1}{2} \log 2$, gives (26.25). Odd powers are absent because the logarithm after removal of the constant is an even function. $\square$

Chapter 27

Sheffer sequences and exponential Riordan arrays

Polynomial sequences of binomial type already arose from complete Bell polynomials in Theorem 15.5. The natural closure of that class under multiplication by an invertible exponential series is the class of Sheffer sequences. Exponential Riordan arrays package the same operations as lower-triangular matrix multiplication [15].

27.1 Characterization of sequences of binomial type

Theorem 27.1 (Generating-function characterization). *Let $p_n(x)$ be polynomials over a field of characteristic zero, with $p_0(x) = 1$ and $\deg p_n \leq n$. The following are equivalent.*

1. For all x, y ,

$$p_n(x + y) = \sum_{k=0}^n \binom{n}{k} p_k(x) p_{n-k}(y). \quad (27.1)$$

2. There is a unique series $\beta(z) = \beta_1 z + \beta_2 z^2/2! + \cdots$ such that

$$\sum_{n \geq 0} p_n(x) \frac{z^n}{n!} = \exp(x\beta(z)). \quad (27.2)$$

Moreover, $\deg p_n = n$ for every n if and only if $\beta_1 = \beta'(0) \neq 0$; in that case β is a delta series in the usual umbral-calculus terminology.

Proof. Assume (27.2). Replacing x by $x + y$ and factoring the exponential yields a product of two EGFs; the EGF product rule (1.7) gives (27.1).

Conversely, put $P(x, z) = \sum_{n \geq 0} p_n(x) z^n / n!$. The binomial identity is exactly

$$P(x + y, z) = P(x, z)P(y, z). \quad (27.3)$$

Setting $x = y = 0$ gives $P(0, z) = P(0, z)^2$. Because $P(0, z)$ has constant term one, it is invertible, and hence $P(0, z) = 1$. The formal logarithm is therefore defined. Let $L(x, z) = \log P(x, z)$. Equation (27.3) becomes $L(x + y, z) = L(x, z) + L(y, z)$. Write $L(x, z) = \sum_{m \geq 0} \ell_m(x) z^m$. Each $\ell_m(x)$ is a polynomial satisfying the additive Cauchy equation; comparing coefficients in $\ell_m(x + y)$ shows that every term of degree other than one vanishes. Hence $\ell_m(x) = c_m x$, so $L(x, z) = x\beta(z)$. Since the constant term of $P(x, z)$ in z is $p_0(x) = 1$, one has $\beta(0) = 0$. Uniqueness follows from $\beta(z) = \partial_x P(0, z)$. If $\beta_1 \neq 0$, the contribution $(x\beta_1 z)^n / n!$ shows that the coefficient of x^n in $p_n(x)$ is β_1^n , so $\deg p_n = n$. Conversely, exact degree for p_1 already forces $\beta_1 \neq 0$. $\square$

27.2 Sheffer sequences

Definition 27.2. A polynomial sequence $(s_n(x))_{n \geq 0}$ is Sheffer for the pair (A, β) if

$$\sum_{n \geq 0} s_n(x) \frac{z^n}{n!} = A(z) e^{x\beta(z)}, \quad A(0) \neq 0, \quad \beta(0) = 0, \quad \beta'(0) \neq 0. \quad (27.4)$$

The associated sequence $p_n(x)$ is defined by $\sum p_n(x) z^n / n! = e^{x\beta(z)}$.

Theorem 27.3 (Sheffer identity and convolution). Write $A(z) = \sum_{j \geq 0} a_j z^j / j!$. Then

$$s_n(x+y) = \sum_{j=0}^n \binom{n}{j} s_j(x) p_{n-j}(y), \quad (27.5)$$

$$s_n(x) = \sum_{j=0}^n \binom{n}{j} a_j p_{n-j}(x). \quad (27.6)$$

Conversely, either identity together with the EGF of p_n reconstructs (27.4).

Proof. For the first identity,

$$A(z) e^{(x+y)\beta(z)} = (A(z) e^{x\beta(z)}) e^{y\beta(z)}.$$

Apply the EGF product rule. The second identity follows from $A(z) e^{x\beta(z)} = A(z) \sum p_n(x) z^n / n!$. Conversely, summing either convolution against $z^n / n!$ recovers the corresponding product of EGFs. $\square$

Theorem 27.4 (Delta operator and lowering law). Let $\bar{\beta}$ be the compositional inverse of β , and define the shift-invariant differential operator

$$Q = \bar{\beta}(D_x). \quad (27.7)$$

Then the associated sequence satisfies

$$Qp_n(x) = np_{n-1}(x), \quad (27.8)$$

and a Sheffer sequence for (A, β) satisfies the same lowering law $Qs_n = ns_{n-1}$.

Proof. For any formal series q , $q(D_x) e^{x\beta(z)} = q(\beta(z)) e^{x\beta(z)}$. With $q = \bar{\beta}$, this becomes

$$Q e^{x\beta(z)} = z e^{x\beta(z)}.$$

Comparing the coefficient of $z^n / n!$ gives (27.8). Since $A(z)$ is independent of x , the same calculation applies to $A(z) e^{x\beta(z)}$. $\square$

Example 27.5 (Appell families). When $\beta(z) = z$, one has $Q = D_x$, and the Sheffer sequence is an Appell sequence. Bernoulli polynomials correspond to $A(z) = z/(e^z - 1)$; Euler polynomials correspond to $A(z) = 2/(e^z + 1)$. Their derivative laws (26.2) and (26.9) are therefore instances of the general lowering law.

Example 27.6 (Bernoulli–Cauchy family). Equation (26.16) has the Sheffer form

$$A(t) e^{x\beta(t)}, \quad A(t) = \frac{t}{\log(1+t)}, \quad \beta(t) = \log(1+t).$$

Its delta operator is $Q = e^{D_x} - 1 = \Delta$, and hence

$$C_n(x+1) - C_n(x) = n C_{n-1}(x). \quad (27.9)$$

This identity follows directly by multiplying the generating function by $(1+t) - 1 = t$.

27.3 Abel polynomials from Lagrange inversion

Theorem 27.7 (Abel's binomial-type sequence). *Fix a scalar a and put*

$$p_0(x) = 1, \quad p_n(x) = x(x - an)^{n-1} \quad (n \geq 1). \quad (27.10)$$

Then (p_n) is of binomial type and therefore obeys

$$(x + y)(x + y - an)^{n-1} = \sum_{k=0}^n \binom{n}{k} x(x - ak)^{k-1} y(y - a(n - k))^{n-k-1}, \quad (27.11)$$

with the endpoint factors interpreted through $p_0 = 1$.

Proof. Let $T(z)$ be the unique series satisfying $T = ze^{-aT}$. Lagrange–Bürmann gives, for $n \geq 1$,

$$\begin{aligned} [z^n]e^{xT(z)} &= \frac{1}{n}[u^{n-1}]xe^{xu}e^{-anu} \\ &= \frac{x(x - an)^{n-1}}{n!}. \end{aligned}$$

Thus

$$e^{xT(z)} = \sum_{n \geq 0} p_n(x) \frac{z^n}{n!}.$$

This is of the form $e^{x\beta(z)}$, so Theorem 27.1 proves the binomial law. Writing that law explicitly gives (27.11). $\square$

27.4 Exponential Riordan arrays

Definition 27.8. Let $g(0) \neq 0$ and let $f(z) = f_1z + O(z^2)$ with $f_1 \neq 0$. The exponential Riordan array (g, f) is the lower-triangular matrix

$$R_{n,k} = \frac{n!}{k!} [z^n]g(z)f(z)^k. \quad (27.12)$$

Theorem 27.9 (Action, group law, and inverse). *If $a = (a_k)$ has EGF $A(z) = \sum a_k z^k / k!$, then $b = Ra$ has EGF*

$$B(z) = g(z)A(f(z)). \quad (27.13)$$

Consequently,

$$(g, f)(h, \ell) = (g(h \circ f), \ell \circ f), \quad (27.14)$$

$$(g, f)^{-1} = \left(\frac{1}{g \circ \bar{f}}, \bar{f} \right), \quad (27.15)$$

where $\bar{f} = f^{\langle -1 \rangle}$.

Proof. By (27.12),

$$\begin{aligned} \sum_{n \geq 0} \left(\sum_{k=0}^n R_{n,k} a_k \right) \frac{z^n}{n!} &= \sum_{k \geq 0} \frac{a_k}{k!} g(z) f(z)^k \\ &= g(z) A(f(z)), \end{aligned}$$

which proves the action. Apply first (h, ℓ) and then (g, f) :

$$A(z) \mapsto h(z)A(\ell(z)) \mapsto g(z)h(f(z))A(\ell(f(z))).$$

This proves (27.14). Substituting $(1/(g \circ \bar{f}), \bar{f})$ on either side of (g, f) reduces the second component to z and the first to one, proving (27.15). $\square$

Corollary 27.10 (Stirling and Bell arrays). *The second-kind and signed first-kind Stirling matrices are the inverse exponential Riordan arrays*

$$(1, e^z - 1), \quad (1, \log(1 + z)). \quad (27.16)$$

The Bell-polynomial matrix associated with $F(z) = \sum_{j \geq 1} x_j z^j / j!$ is $(1, F)$, since

$$\frac{n!}{k!} [z^n] F(z)^k = \mathcal{B}_{n,k}(x_1, \dots, x_{n-k+1}). \quad (27.17)$$

Proof. The first statement is precisely the column EGF formulas (4.1) and (4.3), together with the fact that $e^z - 1$ and $\log(1 + z)$ are compositional inverses. The second is the defining Bell-polynomial EGF (1.10). $\square$

Remark 27.11. Faà di Bruno composition is multiplication in the exponential Riordan group, while Lagrange inversion computes the compositional inverse needed by the group inverse. Thus three topics that are often presented separately are different faces of one triangular-matrix calculus.

Chapter 28

Higher products, multivariate composition, and Good inversion

28.1 Products, reciprocals, and quotients

Theorem 28.1 (Multinomial Leibniz rule). *If $f_1, \dots, f_q$ possess n derivatives, then*

$$D^n \prod_{r=1}^q f_r(x) = \sum_{j_1 + \dots + j_q = n} \binom{n}{j_1, \dots, j_q} \prod_{r=1}^q f_r^{(j_r)}(x). \quad (28.1)$$

Proof. Induct on the number q of factors. The case $q = 1$ is immediate. If the formula holds for $q - 1$ factors, apply the ordinary two-factor Leibniz rule to $F = \prod_{r=1}^{q-1} f_r$ and f_q :

$$D^n (F f_q) = \sum_{j_q=0}^n \binom{n}{j_q} D^{n-j_q} F f_q^{(j_q)}.$$

Insert the induction hypothesis in $D^{n-j_q} F$. The coefficient of the term indexed by $j_1 + \dots + j_q = n$ is

$$\binom{n}{j_q} \binom{n-j_q}{j_1, \dots, j_{q-1}} = \frac{n!}{j_1! \dots j_q!},$$

which is the multinomial coefficient in (28.1). This proof requires only the stated finite differentiability. $\square$

Theorem 28.2 (Higher reciprocal and quotient rules). *Let $g(x) \neq 0$. For $r \geq 0$,*

$$D^r \frac{1}{g(x)} = \sum_{k=0}^r (-1)^k k! \frac{\mathcal{B}_{r,k}(g'(x), g''(x), \dots, g^{(r-k+1)}(x))}{g(x)^{k+1}}, \quad (28.2)$$

where $\mathcal{B}_{0,0} = 1$. Hence

$$D^n \frac{f}{g} = \sum_{r=0}^n \binom{n}{r} f^{(n-r)} \sum_{k=0}^r (-1)^k k! \frac{\mathcal{B}_{r,k}(g', g'', \dots)}{g^{k+1}}. \quad (28.3)$$

Proof. Apply the Bell-polynomial form of Faà di Bruno, (19.3), to $h(u) = u^{-1}$ and $u = g(x)$. Since $h^{(k)}(u) = (-1)^k k! u^{-k-1}$, coefficient substitution gives (28.2). The ordinary two-factor Leibniz rule applied to $f \cdot g^{-1}$, followed by the reciprocal formula, gives (28.3). $\square$

28.2 Mixed partial derivatives

For a smooth scalar field $y = g(x_1, \dots, x_n)$, write $\partial_B y = \prod_{j \in B} \partial_{x_j} y$, with the derivatives applied jointly rather than multiplied; explicitly, $\partial_B y = \partial^{|B|} y / \prod_{j \in B} \partial_{x_j}$.

Theorem 28.3 (Set-partition formula for mixed partials). *For $y = g(x_1, \dots, x_n)$,*

$$\frac{\partial^n}{\partial x_1 \cdots \partial x_n} f(y) = \sum_{\pi \in \text{Part}([n])} f^{(|\pi|)}(y) \prod_{B \in \pi} \partial_B y. \quad (28.4)$$

Proof. Induct on n . The statement for $n = 1$ is the chain rule. Suppose it holds for n and differentiate with respect to x_{n+1} . In each term, the derivative either strikes $f^{(|\pi|)}(y)$, producing $f^{(|\pi|+1)}(y) \partial_{n+1} y$; combinatorially this adjoins the singleton block $\{n+1\}$. Or it strikes one factor $\partial_B y$, replacing it by $\partial_{B \cup \{n+1\}} y$; combinatorially this inserts $n+1$ into the chosen existing block B . Every partition of $[n+1]$ is obtained exactly once by one of these alternatives, proving the induction step. $\square$

For example, the five partitions of $[3]$ give

$$\begin{aligned} \partial_{123} f(y) &= f'(y) \partial_{123} y + f''(y) ((\partial_1 y)(\partial_{23} y) + (\partial_2 y)(\partial_{13} y) + (\partial_3 y)(\partial_{12} y)) \\ &\quad + f^{(3)}(y) (\partial_1 y)(\partial_2 y)(\partial_3 y). \end{aligned} \quad (28.5)$$

28.3 The Fréchet Faà di Bruno formula

Theorem 28.4 (Fréchet form). *Let X, Y, Z be Banach spaces, let $g : X \rightarrow Y$, $f : Y \rightarrow Z$, and assume that all Fréchet derivatives through order n exist and are continuous near x . Then*

$$D^n(f \circ g)(x)[h_1, \dots, h_n] = \sum_{\pi \in \text{Part}([n])} D^{|\pi|} f(g(x)) \left[D^{|B|} g(x)[h_j : j \in B] \right]_{B \in \pi}. \quad (28.6)$$

The order in which the blocks are supplied is immaterial because higher Fréchet derivatives are symmetric multilinear maps.

Proof. The proof is the same partition induction as in Theorem 28.3, but we spell out the differential step. Assume the formula for n and differentiate its right side at x in direction h_{n+1} . Differentiating the outer map $D^{|\pi|} f(g(x))$ gives

$$D^{|\pi|+1} f(g(x)) [Dg(x)h_{n+1}, D^{|B|} g(x)[h_j : j \in B]_{B \in \pi}],$$

which corresponds to adjoining the singleton $\{n+1\}$. Differentiating the argument attached to a block B gives

$$D^{|B|+1} g(x)[h_j : j \in B, h_{n+1}],$$

which corresponds to inserting $n+1$ into B . The product rule for a multilinear map says that these are all terms. As before, every partition of $[n+1]$ has a unique predecessor obtained either by deleting its singleton $\{n+1\}$ or by deleting $n+1$ from the unique larger block containing it. Thus every term appears exactly once. $\square$

28.4 Compositions of several functions

For three sufficiently differentiable scalar functions, two applications of Faà di Bruno give the explicit nested formula

$$D^n f(g(h(x))) = \sum_{k=0}^n f^{(k)}(g(h(x))) \mathcal{B}_{n,k}((g \circ h)', (g \circ h)'', \dots), \quad (28.7)$$

$$(g \circ h)^{(j)} = \sum_{\ell=0}^j g^{(\ell)}(h(x)) \mathcal{B}_{j,\ell}(h', h'', \dots). \quad (28.8)$$

Substitution of the second line into the first is a finite closed formula. Its combinatorial meaning is a hierarchy of partitions: inner blocks record collections of derivatives acting on the same occurrence of h , and outer blocks collect those inner blocks according to occurrences of g . Flattening this hierarchy is the partition-theoretic shadow of associativity of formal series composition.

28.5 Good's multivariate Lagrange inversion formula

The determinant form below is due to Good; combinatorial and further formal proofs were developed by Gessel and others [20, 21].

Let $\mathbf{w} = (w_1, \dots, w_d)$ be the unique vector of formal series satisfying

$$w_i = t_i \phi_i(\mathbf{w}), \quad \phi_i(\mathbf{0}) \neq 0, \quad 1 \leq i \leq d. \quad (28.9)$$

For multi-indices $\mathbf{n} \in \mathbb{N}^d$, write $\mathbf{t}^{\mathbf{n}} = \prod_i t_i^{n_i}$ and $\phi^{\mathbf{n}} = \prod_i \phi_i^{n_i}$.

Lemma 28.5 (Multivariate residue substitution). *Suppose $\mathbf{t} = \mathbf{T}(\mathbf{x})$ is a formal change of variables of the form $T_i = x_i u_i(\mathbf{x})$ with each $u_i(\mathbf{0})$ invertible. Then*

$$\operatorname{Res}_{\mathbf{t}} A(\mathbf{t}) \, dt_1 \cdots dt_d = \operatorname{Res}_{\mathbf{x}} A(\mathbf{T}(\mathbf{x})) \det \left(\frac{\partial T_i}{\partial x_j} \right) dx_1 \cdots dx_d. \quad (28.10)$$

Proof. By linearity it is enough to treat a Laurent monomial $A(\mathbf{t}) = \prod_i t_i^{m_i}$. Regard the integrand as the top differential form

$$\Omega = \prod_i t_i^{m_i} dt_1 \wedge \cdots \wedge dt_d.$$

If $m_i \neq -1$ for some i , then, over characteristic zero,

$$\Omega = d \left((-1)^{i-1} \frac{t_i^{m_i+1}}{m_i+1} \prod_{j \neq i} t_j^{m_j} dt_1 \wedge \cdots \wedge \widehat{dt_i} \cdots \wedge dt_d \right).$$

Pullback commutes with the exterior derivative. The multivariate residue of an exact top form is zero: in coordinates it is a sum of derivatives $\partial Q_i / \partial x_i$, and no such derivative has an x_i^{-1} coefficient. Thus both sides of (28.10) vanish whenever some $m_i \neq -1$.

It remains to take every $m_i = -1$. Since $T_i = x_i u_i$,

$$\bigwedge_{i=1}^d \frac{dT_i}{T_i} = \det \left(\delta_{ij} + \frac{x_j}{u_i} \frac{\partial u_i}{\partial x_j} \right)_{i,j=1}^d \bigwedge_{j=1}^d \frac{dx_j}{x_j}.$$

Every entry after the Kronecker delta has zero constant term, so the determinant has constant term one and no other term can contribute to the coefficient of $x_1^{-1} \cdots x_d^{-1}$. The transformed residue is therefore one, which equals the residue of $\prod_i t_i^{-1}$. Linearity now proves the rule for every Laurent series for which the indicated coefficient extractions are locally finite. $\square$

Theorem 28.6 (Lagrange–Good inversion). *For every formal series F ,*

$$[\mathbf{t}^{\mathbf{n}}] F(\mathbf{w}(\mathbf{t})) = [\mathbf{x}^{\mathbf{n}}] F(\mathbf{x}) \phi(\mathbf{x})^{\mathbf{n}} \det \left(\delta_{ij} - \frac{x_j}{\phi_i(\mathbf{x})} \frac{\partial \phi_i}{\partial x_j}(\mathbf{x}) \right)_{i,j=1}^d. \quad (28.11)$$

Proof. Represent the desired coefficient by an iterated residue and make the formal substitution

$$t_i = \frac{x_i}{\phi_i(\mathbf{x})}. \quad (28.12)$$

The inverse substitution is exactly $x_i = w_i(\mathbf{t})$. The Jacobian has entries

$$\frac{\partial t_i}{\partial x_j} = \frac{1}{\phi_i} \left(\delta_{ij} - \frac{x_i}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right).$$

Therefore

$$\begin{aligned} [t^n]F(\mathbf{w}(\mathbf{t})) &= \operatorname{Res}_{\mathbf{t}} F(\mathbf{w}(\mathbf{t})) \prod_i t_i^{-n_i-1} dt_1 \cdots dt_d \\ &= \operatorname{Res}_{\mathbf{x}} F(\mathbf{x}) \prod_i x_i^{-n_i-1} \phi_i(\mathbf{x})^{n_i} \det \left(\delta_{ij} - \frac{x_i}{\phi_i} \frac{\partial \phi_i}{\partial x_j} \right) d\mathbf{x}. \end{aligned}$$

It remains only to put the determinant in the displayed form. Let $X = \operatorname{diag}(x_1, \dots, x_d)$, $\Phi = \operatorname{diag}(\phi_1, \dots, \phi_d)$, and let $J = (\partial \phi_i / \partial x_j)$. The determinant just obtained is $\det(I - X\Phi^{-1}J)$. Since X and Φ commute and $\det(I - AB) = \det(I - BA)$,

$$\det(I - X\Phi^{-1}J) = \det(I - \Phi^{-1}JX),$$

whose (i, j) -entry is $\delta_{ij} - x_j \phi_i^{-1} \partial \phi_i / \partial x_j$. The remaining residue is exactly the coefficient on the right of (28.11). $\square$

Corollary 28.7 (Reduction to one variable). *For $d = 1$, Good's formula reads*

$$[t^n]F(w(t)) = [x^n]F(x)\phi(x)^n \left(1 - \frac{x\phi'(x)}{\phi(x)} \right). \quad (28.13)$$

Formal integration by parts transforms this into the usual Lagrange–Bürmann formula (21.2).

Proof. The residue of the derivative of $F(x)\phi(x)^n x^{-n}$ is zero. Expanding it gives

$$0 = \operatorname{Res} F'(x)\phi(x)^n x^{-n} dx + n \operatorname{Res} F(x)\phi(x)^{n-1} \phi'(x) x^{-n} dx - n \operatorname{Res} F(x)\phi(x)^n x^{-n-1} dx.$$

Consequently the *sum* of the two residue terms represented by the right-hand side of (28.13) is

$$\operatorname{Res} F(x)\phi(x)^n x^{-n-1} dx - \operatorname{Res} F(x)\phi(x)^{n-1} \phi'(x) x^{-n} dx = \frac{1}{n} \operatorname{Res} F'(x)\phi(x)^n x^{-n} dx.$$

The last member is precisely the residue form of (21.2). $\square$

Chapter 29

Catalan refinements, polygon dissections, and associahedra

The base chapters derive Catalan and Fuss–Catalan coefficients from Lagrange inversion and explain the signed face expansion underlying compositional inversion. Here we add two refinements supplied by the supporting dossiers: Narayana peak enumeration and a complete count of faces of every dimension of the associahedron [24, 22].

29.1 Narayana numbers from a bivariate Lagrange equation

Let $C(z, u)$ count Dyck paths, where z marks semilength and u marks peaks; the empty path contributes one. Every nonempty Dyck path has the unique first-return decomposition

$$U A D B.$$

If A is empty, the initial UD is a peak; otherwise it is not. Hence

$$C(z, u) = 1 + zC(z, u)(C(z, u) + u - 1). \quad (29.1)$$

Theorem 29.1 (Narayana formula). *For $1 \leq k \leq n$, the number of Dyck paths of semilength n with exactly k peaks is*

$$N(n, k) = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}. \quad (29.2)$$

Consequently,

$$\sum_{k=1}^n N(n, k) = \frac{1}{n+1} \binom{2n}{n}. \quad (29.3)$$

Proof. Put $Y = C - 1$. Equation (29.1) becomes

$$Y = z(1 + Y)(u + Y).$$

Lagrange inversion gives

$$\begin{aligned} [z^n u^k] Y &= \frac{1}{n} [t^{n-1} u^k] (1+t)^n (u+t)^n \\ &= \frac{1}{n} \binom{n}{k} [t^{k-1}] (1+t)^n = \frac{1}{n} \binom{n}{k} \binom{n}{k-1}, \end{aligned}$$

proving (29.2). Summing over k is equivalent to setting $u = 1$, where the functional equation reduces to $C = 1 + zC^2$. Formula (22.8) then proves (29.3). Alternatively, Vandermonde convolution sums the closed form directly. $\square$

Corollary 29.2 (Symmetry).

$$N(n, k) = N(n, n + 1 - k). \quad (29.4)$$

Proof. Apply $\binom{n}{k} = \binom{n}{n-k}$ to both binomial factors in (29.2). $\square$

29.2 Polygon dissections and Schröder trees

Fix a convex polygon with $N \geq 3$ sides and distinguish one boundary edge as the root. A dissection using j noncrossing diagonals has $j + 1$ regions. Its rooted planar dual is a plane rooted tree: each region is an internal node, each non-root boundary edge is a leaf, and the cyclic order around regions orders the children. Every internal node has at least two children. Conversely, gluing polygons according to such a tree reconstructs the dissection. Thus dissections of an N -gon with j diagonals correspond to plane rooted trees with

$$L = N - 1 \text{ leaves,} \quad r = j + 1 \text{ internal nodes,} \quad (29.5)$$

all internal outdegrees at least two.

Let $T(z, u)$ count these trees, with z marking leaves and u internal nodes. A tree is either a leaf or an internal node carrying an ordered sequence of at least two subtrees, so

$$T = z + u \frac{T^2}{1 - T}. \quad (29.6)$$

Theorem 29.3 (Number of polygon dissections). *The number $D(N, j)$ of dissections of a convex N -gon by exactly j noncrossing diagonals is*

$$D(N, j) = \frac{1}{N - 1} \binom{N - 3}{j} \binom{N + j - 1}{j + 1}, \quad 0 \leq j \leq N - 3. \quad (29.7)$$

Proof. Solve (29.6) for the Lagrange form:

$$T = z \frac{1 - T}{1 - (1 + u)T} = z \left(1 - \frac{uT}{1 - T} \right)^{-1}.$$

For $L \geq 1$, Lagrange inversion yields

$$\begin{aligned} [z^L u^r] T &= \frac{1}{L} [t^{L-1} u^r] \left(1 - \frac{ut}{1-t} \right)^{-L} \\ &= \frac{1}{L} \binom{L + r - 1}{r} [t^{L-1-r}] (1-t)^{-r} \\ &= \frac{1}{L} \binom{L + r - 1}{r} \binom{L - 2}{r - 1}. \end{aligned}$$

Substitute $L = N - 1$ and $r = j + 1$ from (29.5); this is exactly (29.7). The rooted edge causes no additional factor because every polygon in the count is rooted in the same prescribed way. $\square$

At the two extremes, $D(N, 0) = 1$, while

$$D(N, N - 3) = \frac{1}{N - 1} \binom{2N - 4}{N - 2}$$

is the Catalan number counting triangulations of the N -gon.

29.3 The complete face vector of the associahedron

We now use the common convention that K_n is the $(n - 2)$ -dimensional associahedron whose vertices are parenthesizations of n letters, equivalently triangulations of an $(n + 1)$ -gon. A set of j pairwise noncrossing diagonals determines a face of codimension j : its vertices are the triangulations containing those diagonals. Hence a d -dimensional face corresponds to a dissection with $j = n - 2 - d$ diagonals.

Theorem 29.4 (Face numbers). *For $0 \leq d \leq n - 2$, the number of d -faces of K_n is*

$$f_d(K_n) = \frac{1}{n} \binom{n-2}{d} \binom{2n-d-2}{n-1}. \quad (29.8)$$

In particular,

$$f_0(K_n) = \frac{1}{n} \binom{2n-2}{n-1} = C_{n-1}, \quad (29.9)$$

$$f_{n-3}(K_n) = \frac{(n+1)(n-2)}{2}, \quad (29.10)$$

$$f_{n-2}(K_n) = 1. \quad (29.11)$$

Proof. Apply Theorem 29.3 with $N = n + 1$ and $j = n - 2 - d$:

$$\begin{aligned} f_d(K_n) &= \frac{1}{n} \binom{n-2}{n-2-d} \binom{2n-d-2}{n-1-d} \\ &= \frac{1}{n} \binom{n-2}{d} \binom{2n-d-2}{n-1}, \end{aligned}$$

where the last equality uses symmetry of the second binomial coefficient. This proves (29.8). The three special cases follow by direct simplification. For facets, the result is also the number of diagonals in an $(n + 1)$ -gon. $\square$

Example 29.5. For K_4 , the formula gives $(f_0, f_1, f_2) = (5, 5, 1)$, the face vector of a pentagon. For K_5 , it gives $(14, 21, 9, 1)$: fourteen vertices, twenty-one edges, nine polygonal facets, and the polytope itself.

29.4 The Narayana h -vector

For a simple D -polytope P , define its h -polynomial by

$$h_P(u) = \sum_{d=0}^D f_d(P)(u-1)^d. \quad (29.12)$$

Theorem 29.6 (Associahedral h -polynomial). *For K_n ,*

$$h_{K_n}(u) = \sum_{r=0}^{n-2} N(n-1, r+1)u^r = \sum_{r=0}^{n-2} \frac{1}{n-1} \binom{n-1}{r} \binom{n-1}{r+1} u^r. \quad (29.13)$$

Proof. The coefficient of u^r in (29.12), using (29.8), is

$$\begin{aligned} &\sum_{d=r}^{n-2} (-1)^{d-r} \binom{d}{r} \frac{1}{n} \binom{n-2}{d} \binom{2n-d-2}{n-1} \\ &= \frac{1}{n} \binom{n-2}{r} \sum_{j=0}^{n-2-r} (-1)^j \binom{n-2-r}{j} \binom{2n-r-j-2}{n-1}. \end{aligned}$$

The alternating Vandermonde identity

$$\sum_{j=0}^M (-1)^j \binom{M}{j} \binom{R-j}{S} = \binom{R-M}{S-M} \quad (29.14)$$

follows by extracting $[x^S]$ from $(1+x)^R(1-(1+x)^{-1})^M = x^M(1+x)^{R-M}$. Apply it with $M = n-2-r$, $R = 2n-r-2$, and $S = n-1$. The coefficient becomes

$$\frac{1}{n} \binom{n-2}{r} \binom{n}{r+1} = \frac{1}{n-1} \binom{n-1}{r} \binom{n-1}{r+1},$$

which is $N(n-1, r+1)$ by (29.2). □

Corollary 29.7 (Dehn–Sommerville symmetry in this case). *The coefficients of $h_{K_n}(u)$ are palindromic.*

Proof. This is the Narayana symmetry (29.4) with n replaced by $n-1$. □

29.5 Tamari adjacency and the geometric meaning of associativity

Two vertices of K_n are adjacent exactly when their triangulations differ by a single diagonal flip, or equivalently when their parenthesizations differ by one local use of $(ab)c \leftrightarrow a(bc)$. Orienting each edge by the forward rotation $(ab)c \rightarrow a(bc)$ produces the Hasse diagram of the Tamari lattice. The pentagonal K_4 is therefore not merely a picture of five parenthesizations: its boundary records the coherence relation asserting that all reassociation paths between four factors agree. Higher-dimensional faces encode simultaneous compatible reassociations, while (29.8) counts them exactly.

Chapter 30

Gamma, digamma, polygamma, and Barnes G

The first-kind Stirling chapters already derive convergent series for $\log \Gamma$, ψ , and the Stieltjes constants. We complement that coefficient viewpoint with the classical products and with the next function in the gamma hierarchy, Barnes's G -function [13, 25].

30.1 Euler's integral and product for the gamma function

For $\Re z > 0$, define

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt. \quad (30.1)$$

Integration by parts gives

$$\Gamma(z+1) = z\Gamma(z), \quad \Gamma(1) = 1. \quad (30.2)$$

The boundary term vanishes because $t^z e^{-t} \rightarrow 0$ at infinity and $t^z \rightarrow 0$ at zero when $\Re z > 0$.

Theorem 30.1 (Euler limit and Weierstrass product). *For $\Re z > 0$,*

$$\Gamma(z) = \lim_{N \rightarrow \infty} \frac{N! N^z}{z(z+1) \cdots (z+N)}. \quad (30.3)$$

Consequently, for all $z \in \mathbb{C}$,

$$\frac{1}{\Gamma(z+1)} = e^{\gamma z} \prod_{k=1}^{\infty} \left(1 + \frac{z}{k}\right) e^{-z/k}, \quad (30.4)$$

where $\gamma = \lim_{N \rightarrow \infty} (H_N - \log N)$. The product converges locally uniformly.

Proof. The beta integral and N integrations by parts give

$$\int_0^1 t^{z-1} (1-t)^N dt = \frac{N!}{z(z+1) \cdots (z+N)}.$$

Multiply by N^z and substitute $u = Nt$:

$$\frac{N! N^z}{z(z+1) \cdots (z+N)} = \int_0^N u^{z-1} \left(1 - \frac{u}{N}\right)^N du.$$

For fixed u , the factor in parentheses tends to e^{-u} ; moreover $(1 - u/N)^N \leq e^{-u}$ on $0 \leq u \leq N$. Dominated convergence on a large fixed interval, followed by an exponentially small tail estimate, proves (30.3).

Apply the limit to $\Gamma(z+1)$ and use the recurrence to write

$$\frac{1}{\Gamma(z+1)} = \lim_{N \rightarrow \infty} N^{-z} \prod_{k=1}^N \left(1 + \frac{z}{k}\right).$$

Since $\log N = H_N - \gamma + o(1)$, this becomes

$$\lim_{N \rightarrow \infty} e^{\gamma z + o(1)} \prod_{k=1}^N \left(1 + \frac{z}{k}\right) e^{-z/k},$$

which is (30.4). For every compact set, the estimate $\log(1 + z/k) - z/k = O(k^{-2})$ is uniform once the finitely many small values of k are removed. Hence the product converges locally uniformly and defines an entire function. It agrees with $1/\Gamma(z+1)$ on the half-plane where Euler's limit was derived, and therefore everywhere by analytic continuation. $\square$

30.2 Digamma and polygamma series

Define

$$\psi(z) = \frac{\Gamma'(z)}{\Gamma(z)}, \quad \psi^{(m)}(z) = \frac{d^m}{dz^m} \psi(z). \quad (30.5)$$

Theorem 30.2 (Series and Hurwitz-zeta representation). *For $z \notin \{-1, -2, \dots\}$,*

$$\psi(z+1) = -\gamma + \sum_{k=1}^{\infty} \frac{z}{k(k+z)}. \quad (30.6)$$

For $m \geq 1$ and $z \notin \{0, -1, -2, \dots\}$,

$$\psi^{(m)}(z) = (-1)^{m+1} m! \sum_{k=0}^{\infty} \frac{1}{(z+k)^{m+1}} = (-1)^{m+1} m! \zeta(m+1, z). \quad (30.7)$$

Also,

$$\psi(z+1) = \psi(z) + \frac{1}{z}, \quad \psi^{(m)}(z+1) = \psi^{(m)}(z) + \frac{(-1)^m m!}{z^{m+1}}. \quad (30.8)$$

Proof. Take logarithms in (30.4) and differentiate. Local uniform convergence permits termwise differentiation and gives

$$-\psi(z+1) = \gamma + \sum_{k \geq 1} \left(\frac{1}{k+z} - \frac{1}{k} \right),$$

which is (30.6). Replacing z by $z-1$ and differentiating m times gives the absolutely and locally uniformly convergent series (30.7). Finally, logarithmic differentiation of $\Gamma(z+1) = z\Gamma(z)$ proves the first recurrence; further differentiation proves the second. $\square$

Corollary 30.3 (Pochhammer logarithmic derivatives). *For $n \geq 1$,*

$$\frac{d}{dx} \log(x)_n = \psi(x+n) - \psi(x) = \sum_{j=0}^{n-1} \frac{1}{x+j}, \quad (30.9)$$

$$\frac{d^m}{dx^m} \log(x)_n = \psi^{(m-1)}(x+n) - \psi^{(m-1)}(x) = (-1)^{m-1} (m-1)! \sum_{j=0}^{n-1} \frac{1}{(x+j)^m} \quad (30.10)$$

for $m \geq 1$.

Proof. Differentiate the gamma quotient (25.4) and use the recurrences in (30.8); telescoping gives the finite sums. $\square$

30.3 Barnes's canonical product

Definition 30.4. Barnes's G -function is defined by the locally uniformly convergent product

$$G(z+1) = (2\pi)^{z/2} \exp\left[-\frac{z+(1+\gamma)z^2}{2}\right] \prod_{k=1}^{\infty} \left(1 + \frac{z}{k}\right)^k \exp\left(\frac{z^2}{2k} - z\right). \quad (30.11)$$

Proposition 30.5 (Convergence and functional equation). *The product (30.11) converges locally uniformly and defines an entire function satisfying*

$$G(z+1) = \Gamma(z)G(z), \quad G(1) = 1. \quad (30.12)$$

Consequently, for integers $n \geq 2$,

$$G(n) = 1! 2! \cdots (n-2)! = \prod_{k=1}^{n-2} k^{n-1-k}. \quad (30.13)$$

Proof. For z in a compact set,

$$k \log(1 + z/k) + \frac{z^2}{2k} - z = O(k^{-2})$$

uniformly, so the logarithm of the product converges locally uniformly. To verify the recurrence, let G_N denote the product truncated at $k = N$. The weighted factors telescope:

$$\prod_{k=1}^N \left(\frac{1 + (z+1)/k}{1 + z/k} \right)^k = \frac{(N+z+1)^N}{\prod_{k=1}^N (z+k)}.$$

A direct division of the remaining exponential and $(2\pi)^{z/2}$ factors gives

$$\begin{aligned} \frac{G_N(z+2)}{G_N(z+1)} &= \sqrt{2\pi} \frac{(N+z+1)^N}{\prod_{k=1}^N (z+k)} \\ &\quad \times \exp\left(- (1+\gamma)z - \frac{2+\gamma}{2} + \frac{2z+1}{2} H_N - N\right). \end{aligned}$$

Use Stirling's formula $N! \sim \sqrt{2\pi N} (N/e)^N$, $H_N = \log N + \gamma + O(N^{-1})$, and $(1 + (z+1)/N)^N \rightarrow e^{z+1}$. The displayed expression differs by a factor tending to one from Euler's limit

$$\Gamma(z+1) = \lim_{N \rightarrow \infty} \frac{N^z N!}{\prod_{k=1}^N (z+k)}.$$

Therefore $G(z+2)/G(z+1) = \Gamma(z+1)$, equivalent to (30.12). Setting $z = 0$ in the product gives $G(1) = 1$. Iterating the recurrence at positive integers proves the first product in (30.13); collecting equal factors proves the second. $\square$

Proposition 30.6 (Alexeiewsky's integral). *For real $x \geq 0$,*

$$\int_0^x \log \Gamma(1+t) dt = \frac{x}{2} \log(2\pi) - \frac{x(x+1)}{2} + x \log \Gamma(x+1) - \log G(x+1). \quad (30.14)$$

The identity extends holomorphically to any simply connected domain avoiding the poles and zeros after compatible logarithm branches are chosen.

Proof. Termwise differentiation of the locally uniformly convergent Barnes product, followed by the digamma series (30.6), gives

$$\begin{aligned}\frac{d}{dx} \log G(x+1) &= \frac{1}{2} \log(2\pi) - \frac{1}{2} - (1+\gamma)x + \sum_{k \geq 1} \frac{x^2}{k(k+x)} \\ &= \frac{1}{2} \log(2\pi) - \frac{1}{2} - x + x\psi(x+1).\end{aligned}$$

Differentiate the right-hand side of (30.14). The terms other than $\log \Gamma(x+1)$ cancel by the displayed logarithmic derivative, so its derivative is $\log \Gamma(x+1)$, exactly the derivative of the left side. Both sides vanish at $x = 0$, proving the identity. $\square$

30.4 Euler–Maclaurin asymptotics for Barnes G

Theorem 30.7 (Complete algebraic asymptotic expansion). *For every fixed $p \geq 1$, as $x \rightarrow +\infty$,*

$$\begin{aligned}\log G(x+1) &= \frac{x^2}{2} \log x - \frac{3x^2}{4} + \frac{x}{2} \log(2\pi) - \frac{1}{12} \log x + \zeta'(-1) \\ &\quad + \sum_{r=1}^{p-1} \frac{\beta_{2r+2}}{4r(r+1)x^{2r}} + O(x^{-2p}).\end{aligned}\tag{30.15}$$

The proof first determines every coefficient and the constant on positive integers, then uses Alexeiewsky's integral to establish the same expansion for all real $x \rightarrow +\infty$.

Proof. We give the real integer derivation, which determines every displayed coefficient and the constant. From (30.13),

$$\log G(n+1) = n \log \Gamma(n+1) - \sum_{k=1}^n k \log k.\tag{30.16}$$

Apply Euler–Maclaurin to $f(x) = x \log x$. Since

$$f'(x) = \log x + 1, \quad f^{(m)}(x) = (-1)^m (m-2)! x^{1-m} \quad (m \geq 2),$$

one obtains, to arbitrary order,

$$\begin{aligned}\sum_{k=1}^n k \log k &= \left(\frac{n^2}{2} + \frac{n}{2} + \frac{1}{12} \right) \log n - \frac{n^2}{4} + \frac{1}{12} - \zeta'(-1) \\ &\quad - \sum_{r=1}^{p-1} \frac{\beta_{2r+2}}{2r(2r+1)(2r+2)n^{2r}} + O(n^{-2p}).\end{aligned}\tag{30.17}$$

Here the constant is identified by applying Euler–Maclaurin to $\zeta(s) = \sum_{k \geq 1} k^{-s}$, differentiating the continued formula at $s = -1$, and comparing finite parts. Explicitly, the first three tail terms at $s = -1$ are

$$\frac{n^2}{2} \log n - \frac{n^2}{4} + \frac{n}{2} \log n + \frac{1}{12} (\log n + 1),$$

while the differentiated Bernoulli tails give exactly the inverse even powers in (30.17); moving $-\sum_{k \leq n} k \log k$ to the other side leaves $\zeta'(-1)$.

The ordinary Stirling expansion is

$$\log \Gamma(n+1) = \left(n + \frac{1}{2} \right) \log n - n + \frac{1}{2} \log(2\pi) + \sum_{r=1}^p \frac{\beta_{2r}}{2r(2r-1)n^{2r-1}} + O(n^{-2p-1}).\tag{30.18}$$

Substitute (30.17) and (30.18) into (30.16). The polynomial and logarithmic terms collect to the first line of (30.15). At each inverse even power, the contribution from $n \log \Gamma(n+1)$ and that from the hyperfactorial sum combine as

$$\frac{\beta_{2r+2}}{(2r+2)(2r+1)} + \frac{\beta_{2r+2}}{2r(2r+1)(2r+2)} = \frac{\beta_{2r+2}}{4r(r+1)},$$

which proves every coefficient in the second line. The Euler–Maclaurin remainder estimates give $O(n^{-2p})$.

It remains to justify the asserted expansion between the integers. Solve Alexeiewsky's identity (30.14) for $\log G(x+1)$, split the integral at 1, and insert the Stirling expansion for $\log \Gamma(1+t)$ uniformly for $t \geq 1$. If that expansion is truncated after β_{2p} , its remainder is $O(t^{-2p-1})$; multiplication by x at the endpoint contributes $O(x^{-2p})$, while integration of the remainder contributes an x -independent constant plus $O(x^{-2p})$. The elementary terms give

$$\frac{x^2}{2} \log x - \frac{3x^2}{4} + \frac{x}{2} \log(2\pi) - \frac{1}{12} \log x,$$

and, for $r \geq 1$, the term β_{2r+2} contributes $\beta_{2r+2}/(4r(r+1)x^{2r})$. Thus (30.15) holds for all real $x \rightarrow +\infty$ with one constant independent of x . The integer calculation above identifies that constant as $\zeta'(-1)$, completing the proof. $\square$

Remark 30.8. The constant $A = \exp(1/12 - \zeta'(-1))$ is the Glaisher–Kinkelin constant. Equation (30.17) is equivalently the asymptotic law for the hyperfactorial $\prod_{k=1}^n k^k$. Thus the constant appearing in Barnes G is forced by the same finite-part calculation that continues the zeta function to -1 .

Chapter 31

Euler–Maclaurin summation and Darboux methods

Bernoulli polynomials have two complementary analytic roles. Periodizing them turns sums into integrals with controlled remainders; using an auxiliary polynomial in repeated integration by parts turns endpoint derivatives into an exact Taylor-type identity. Both mechanisms are developed here from first principles; the normalization agrees with the modern DLMF convention [13].

31.1 Periodic Bernoulli functions

For $m \geq 1$, write

$$\tilde{B}_m(x) = B_m(\{x\}), \quad \{x\} = x - \lfloor x \rfloor. \quad (31.1)$$

For $m \geq 2$ this is continuous at the integers because $B_m(1) = B_m(0)$. Away from integers,

$$\frac{d}{dx} \frac{\tilde{B}_m(x)}{m!} = \frac{\tilde{B}_{m-1}(x)}{(m-1)!}. \quad (31.2)$$

Proposition 31.1 (Fourier series). *For $m \geq 2$,*

$$\tilde{B}_m(x) = -\frac{m!}{(2\pi i)^m} \sum_{k \in \mathbb{Z} \setminus \{0\}} \frac{e^{2\pi i k x}}{k^m}, \quad (31.3)$$

with pointwise convergence everywhere and absolute convergence for $m \geq 2$ except that the $m = 2$ derivative series is only conditionally convergent. In particular,

$$\tilde{B}_{2p}(x) = (-1)^{p+1} \frac{2(2p)!}{(2\pi)^{2p}} \sum_{k=1}^{\infty} \frac{\cos(2\pi k x)}{k^{2p}}, \quad (31.4)$$

and therefore

$$\left| \tilde{B}_{2p}(x) \right| \leq \frac{2(2p)! \zeta(2p)}{(2\pi)^{2p}}. \quad (31.5)$$

Proof. For nonzero k , let $c_k^{(m)} = \int_0^1 B_m(x) e^{-2\pi i k x} dx$. Integration by parts, using $B_m(1) = B_m(0)$ for $m \neq 1$, gives

$$c_k^{(m)} = \frac{m}{2\pi i k} c_k^{(m-1)}.$$

At the last step, direct integration of $B_1(x) = x - 1/2$ gives $c_k^{(1)} = -1/(2\pi i k)$. Hence $c_k^{(m)} = -m!/(2\pi i k)^m$. The mean coefficient is zero because $\int_0^1 B_m(x) dx = 0$ for $m \geq 1$, by (26.2). This proves (31.3); pairing k with $-k$ gives the even form. Taking absolute values and summing proves (31.5). $\square$

31.2 Euler–Maclaurin with an explicit remainder

Theorem 31.2 (Euler–Maclaurin summation). *Let $a < b$ be integers and $f \in C^{2p}([a, b])$. Then*

$$\begin{aligned} \sum_{k=a}^b f(k) &= \int_a^b f(x) dx + \frac{f(a) + f(b)}{2} + \sum_{r=1}^p \frac{\beta_{2r}}{(2r)!} (f^{(2r-1)}(b) - f^{(2r-1)}(a)) \\ &\quad - \frac{1}{(2p)!} \int_a^b \tilde{B}_{2p}(x) f^{(2p)}(x) dx. \end{aligned} \quad (31.6)$$

The remainder satisfies

$$|R_p| \leq \frac{2\zeta(2p)}{(2\pi)^{2p}} \int_a^b |f^{(2p)}(x)| dx. \quad (31.7)$$

Proof. On each interval $(k, k+1)$, $\tilde{B}_1(x) = x - k - 1/2$. Integration by parts gives

$$\int_k^{k+1} \tilde{B}_1(x) f'(x) dx = \frac{f(k) + f(k+1)}{2} - \int_k^{k+1} f(x) dx.$$

Summing from $k = a$ to $b - 1$ yields the exact starting identity

$$\sum_{k=a}^b f(k) = \int_a^b f(x) dx + \frac{f(a) + f(b)}{2} + \int_a^b \tilde{B}_1(x) f'(x) dx. \quad (31.8)$$

Now use (31.2) repeatedly. The first integration gives

$$\int_a^b \tilde{B}_1 f' = \frac{\beta_2}{2!} (f'(b) - f'(a)) - \frac{1}{2!} \int_a^b \tilde{B}_2 f''.$$

The next integration introduces $\tilde{B}_3$, but its boundary term vanishes because $\beta_3 = 0$; the following integration produces the β_4 endpoint term. Continuing in pairs gives exactly (31.6). The odd Bernoulli numbers beyond β_1 vanish by reflection, so no omitted endpoint terms remain. Finally apply (31.5) inside the remainder integral to obtain (31.7). $\square$

Corollary 31.3 (Infinite-tail form). *Suppose $f^{(j)}(x) \rightarrow 0$ as $x \rightarrow \infty$ for $0 \leq j \leq 2p - 1$, and the remainder integral converges. Then*

$$\begin{aligned} \sum_{k=a}^{\infty} f(k) &= \int_a^{\infty} f(x) dx + \frac{f(a)}{2} - \sum_{r=1}^p \frac{\beta_{2r}}{(2r)!} f^{(2r-1)}(a) \\ &\quad - \frac{1}{(2p)!} \int_a^{\infty} \tilde{B}_{2p}(x) f^{(2p)}(x) dx. \end{aligned} \quad (31.9)$$

Proof. Apply Theorem 31.2 on $[a, b]$ and let the integer $b \rightarrow \infty$. The endpoint hypotheses remove all terms at b ; the assumed convergence of the improper remainder integral gives the limit of its truncations. $\square$

Example 31.4 (Harmonic numbers). Apply the finite formula (31.6) to $f(x) = 1/x$ on $[1, n]$, and absorb all lower-end terms and the limiting remainder into the constant $\gamma = \lim_{n \rightarrow \infty} (H_n - \log n)$. Since $f^{(m)}(x) = (-1)^m m! x^{-m-1}$, one obtains, for fixed p ,

$$H_n = \log n + \gamma + \frac{1}{2n} - \sum_{r=1}^{p-1} \frac{\beta_{2r}}{2r n^{2r}} + O(n^{-2p}). \quad (31.10)$$

Indeed, the tail of the remainder integral is $O(\int_n^{\infty} x^{-2p-1} dx) = O(n^{-2p})$ by (31.5), which gives the displayed error order.

31.3 Darboux's exact integration-by-parts identity

Theorem 31.5 (Darboux formula). *Let f be analytic on a neighborhood of the line segment joining a to z , and let ϕ be a polynomial of degree n . Then*

$$\begin{aligned} \phi^{(n)}(0)(f(z) - f(a)) &= \sum_{m=1}^n (-1)^{m-1} (z-a)^m \left(\phi^{(n-m)}(1) f^{(m)}(z) - \phi^{(n-m)}(0) f^{(m)}(a) \right) \\ &\quad + (-1)^n (z-a)^{n+1} \int_0^1 \phi(t) f^{(n+1)}(a + t(z-a)) dt. \end{aligned} \quad (31.11)$$

Proof. Put $d = z - a$ and define

$$S(t) = \sum_{m=1}^n (-1)^m d^m \phi^{(n-m)}(t) f^{(m)}(a + td).$$

Differentiation produces two sums. After shifting the index in the terms where the derivative hits $f^{(m)}$, all intermediate terms cancel, leaving

$$S'(t) = -d \phi^{(n)}(t) f'(a + td) + (-1)^n d^{n+1} \phi(t) f^{(n+1)}(a + td). \quad (31.12)$$

Because $\phi^{(n)}$ is constant, integration from zero to one turns the first term into $-\phi^{(n)}(0)(f(z) - f(a))$. Move it to the other side and substitute the endpoint values of S ; the result is exactly (31.11). $\square$

Corollary 31.6 (Taylor formula with integral remainder). *Taking $\phi(t) = (t-1)^n$ gives*

$$f(z) = \sum_{m=0}^n \frac{f^{(m)}(a)}{m!} (z-a)^m + \frac{(z-a)^{n+1}}{n!} \int_0^1 (1-t)^n f^{(n+1)}(a + t(z-a)) dt. \quad (31.13)$$

Proof. Here $\phi^{(n)} = n!$, all derivatives of order below n vanish at $t = 1$, and $\phi^{(n-m)}(0) = n!(-1)^m/m!$. Substitute these values into (31.11) and divide by $n!$. $\square$

31.4 Darboux transfer for algebraic singularities

Theorem 31.7 (Elementary Darboux transfer). *Let $0 < |\rho| < R$, let $\alpha_j \notin \{0, -1, -2, \dots\}$, and suppose*

$$F(z) = \sum_{j=1}^J c_j (1 - z/\rho)^{-\alpha_j} + H(z), \quad (31.14)$$

where H is analytic in $|z| < R$. Then, for every $|\rho| < R_0 < R$,

$$[z^n]F(z) = \rho^{-n} \sum_{j=1}^J c_j \frac{\Gamma(n + \alpha_j)}{\Gamma(\alpha_j)\Gamma(n+1)} + O(R_0^{-n}). \quad (31.15)$$

Moreover, for fixed α ,

$$\frac{\Gamma(n + \alpha)}{\Gamma(n + 1)} = n^{\alpha-1} \left(1 + \frac{\alpha(\alpha-1)}{2n} + \frac{\alpha(\alpha-1)(\alpha-2)(3\alpha-1)}{24n^2} + O(n^{-3}) \right). \quad (31.16)$$

Proof. The generalized binomial theorem gives the exact coefficient

$$[z^n](1 - z/\rho)^{-\alpha} = \rho^{-n} \frac{(\alpha)_n}{n!} = \rho^{-n} \frac{\Gamma(n + \alpha)}{\Gamma(\alpha)\Gamma(n+1)}.$$

Cauchy's estimate on the circle $|z| = R_0$ gives $[z^n]H = O(R_0^{-n})$, proving (31.15). Finally insert Stirling's expansion for the two gamma factors, subtract their logarithms, expand through n^{-2} , and exponentiate. The logarithmic coefficients are $\alpha(\alpha - 1)/2$ at order n^{-1} and $-\alpha(\alpha - 1)(2\alpha - 1)/12$ at order n^{-2} ; adding half the square of the first coefficient after exponentiation gives the second coefficient displayed in (31.16). $\square$

Example 31.8 (Catalan asymptotics from the square-root singularity). The Catalan generating function can be written

$$C(z) = \frac{1 - \sqrt{1 - 4z}}{2z} = \frac{2}{1 + \sqrt{1 - 4z}} = 2 - 2X + 2X^2 - 2X^3 + \dots, \quad X = \sqrt{1 - 4z}. \quad (31.17)$$

The integer powers of $X^2 = 1 - 4z$ contribute only finitely many coefficients; the successive half-integer powers predict the descending powers of n in the asymptotic expansion. For a completely explicit remainder proof, start from the exact coefficient formula $C_n = \binom{2n}{n}/(n + 1)$. Stirling's expansion gives

$$\binom{2n}{n} = \frac{4^n}{\sqrt{\pi n}} \left(1 - \frac{1}{8n} + \frac{1}{128n^2} + O(n^{-3}) \right),$$

while

$$\frac{1}{n + 1} = \frac{1}{n} \left(1 - \frac{1}{n} + \frac{1}{n^2} + O(n^{-3}) \right).$$

Multiplication yields

$$C_n = \frac{4^n}{\sqrt{\pi} n^{3/2}} \left(1 - \frac{9}{8n} + O(n^{-2}) \right) \quad (31.18)$$

and, one order more precisely,

$$C_n = \frac{4^n}{\sqrt{\pi} n^{3/2}} \left(1 - \frac{9}{8n} + \frac{145}{128n^2} + O(n^{-3}) \right). \quad (31.19)$$

Expanding the half-integer singular terms with (31.16) gives the same coefficients; the exact binomial calculation supplies the required control of the unshown local terms.

Chapter 32

Möbius inversion, explicit reversions, and coefficient algorithms

This final extension chapter collects material that was distributed among the five drafts: the incidence-algebra explanation of cumulants, inverse trigonometric coefficient evaluations, Kepler's equation, and triangular or Newton-style algorithms for all elementary formal-series operations.

32.1 Möbius inversion on the partition lattice

Let Π_n denote the lattice of set partitions of $[n]$, ordered by refinement, with bottom $\widehat{0}$ and top $\widehat{1}$.

Theorem 32.1 (Möbius function to the top). *For $n \geq 1$ and $\pi \in \Pi_n$,*

$$\mu_{\Pi_n}(\pi, \widehat{1}) = (-1)^{|\pi|-1}(|\pi| - 1)!. \quad (32.1)$$

More generally,

$$\mu_{\Pi_n}(\widehat{0}, \pi) = \prod_{B \in \pi} (-1)^{|B|-1}(|B| - 1)!. \quad (32.2)$$

Proof. The interval $[\pi, \widehat{1}]$ is canonically isomorphic to $\Pi_{|\pi|}$: coarsening π is the same as partitioning its set of blocks. It therefore suffices to compute $u_m = \mu_{\Pi_m}(\widehat{0}, \widehat{1})$. The interval $[\widehat{0}, \sigma]$ is the direct product $\prod_{B \in \sigma} \Pi_{|B|}$, and Möbius functions multiply on direct products; hence

$$\mu(\widehat{0}, \sigma) = \prod_{B \in \sigma} u_{|B|}. \quad (32.3)$$

The defining Möbius relation gives

$$\sum_{\sigma \in \Pi_m} \mu(\widehat{0}, \sigma) = \begin{cases} 1, & m = 0, 1, \\ 0, & m \geq 2. \end{cases}$$

By the exponential formula, the EGF of the left side is

$$\exp \left(\sum_{m \geq 1} u_m \frac{z^m}{m!} \right).$$

It must therefore equal $1 + z$. Taking the formal logarithm gives

$$\sum_{m \geq 1} u_m \frac{z^m}{m!} = \log(1 + z) = \sum_{m \geq 1} (-1)^{m-1} \frac{z^m}{m},$$

so $u_m = (-1)^{m-1}(m-1)!$. The interval isomorphism proves (32.1), and substituting the values of $u_{|B|}$ into (32.3) proves (32.2). $\square$

Corollary 32.2 (Moment–cumulant inversion). *If*

$$m_n = \sum_{\pi \in \Pi_n} \prod_{B \in \pi} \kappa_{|B|}, \quad (32.4)$$

then

$$\kappa_n = \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{B \in \pi} m_{|B|}. \quad (32.5)$$

Proof. For each partition σ , define $M(\sigma) = \prod_{B \in \sigma} m_{|B|}$ and $K(\sigma) = \prod_{B \in \sigma} \kappa_{|B|}$. Applying (32.4) independently inside every block of σ gives

$$M(\sigma) = \sum_{\pi \leq \sigma} K(\pi).$$

Möbius inversion on Π_n yields $K(\hat{1}) = \sum_{\pi \leq \hat{1}} \mu(\pi, \hat{1}) M(\pi)$. Since $K(\hat{1}) = \kappa_n$, substitution of (32.1) proves (32.5). $\square$

32.2 Inverse trigonometric coefficient identities

Theorem 32.3 (Inverse sine). *For $|y| < 1$,*

$$\arcsin y = \sum_{m \geq 0} \frac{1}{2m+1} \binom{2m}{m} \frac{y^{2m+1}}{4^m}. \quad (32.6)$$

Equivalently, Lagrange inversion gives the coefficient identity

$$[x^{2m}] \left(\frac{x}{\sin x} \right)^{2m+1} = \frac{1}{4^m} \binom{2m}{m}. \quad (32.7)$$

Proof. The generalized binomial theorem gives

$$\frac{d}{dy} \arcsin y = (1 - y^2)^{-1/2} = \sum_{m \geq 0} \binom{2m}{m} \frac{y^{2m}}{4^m}.$$

Integrate term by term from zero to y to obtain (32.6). On the other hand, the analytic inverse-function form (21.6), applied to $f(x) = \sin x$, gives

$$[y^{2m+1}] \arcsin y = \frac{1}{2m+1} [x^{2m}] \left(\frac{x}{\sin x} \right)^{2m+1}.$$

Comparing this with the coefficient in (32.6) proves (32.7). $\square$

Theorem 32.4 (Inverse tangent). *For $|y| < 1$,*

$$\arctan y = \sum_{m \geq 0} \frac{(-1)^m}{2m+1} y^{2m+1}, \quad (32.8)$$

and

$$[x^{2m}] \left(\frac{x}{\tan x} \right)^{2m+1} = (-1)^m. \quad (32.9)$$

Proof. Integrate the geometric series $(1 + y^2)^{-1} = \sum_{m \geq 0} (-1)^m y^{2m}$ to obtain (32.8). Lagrange inversion for $f(x) = \tan x$ gives

$$[y^{2m+1}] \arctan y = \frac{1}{2m+1} [x^{2m}] \left(\frac{x}{\tan x} \right)^{2m+1}.$$

Coefficient comparison proves (32.9). $\square$

32.3 Kepler's equation

For eccentricity $0 \leq e < 1$, Kepler's equation is

$$E = M + e \sin E. \quad (32.10)$$

Since $d(E - e \sin E)/dE = 1 - e \cos E > 0$, the map from eccentric anomaly E to mean anomaly M is globally invertible on the real line, with a 2π -periodic correction $E(M) - M$.

Theorem 32.5 (Reversion and Fourier–Bessel forms). *Formally in e ,*

$$E = M + \sum_{k \geq 1} \frac{e^k}{k!} \left(\frac{d}{dM} \right)^{k-1} \sin^k M. \quad (32.11)$$

For $0 \leq e < 1$, the convergent Fourier representation is

$$E = M + 2 \sum_{n \geq 1} \frac{J_n(ne)}{n} \sin(nM). \quad (32.12)$$

Proof. Equation (32.10) is the shifted implicit equation (23.1) with $x = M$, $y = e$, and $f(v) = \sin v$. Formula (23.3) gives (32.11).

For the Fourier form, write $E - M = \sum_{n \geq 1} c_n \sin(nM)$; oddness removes cosine terms. Changing variables from M to E gives

$$\begin{aligned} c_n &= \frac{1}{\pi} \int_0^{2\pi} (E - M) \sin(nM) dM \\ &= \frac{e}{\pi} \int_0^{2\pi} \sin E \sin(n(E - e \sin E)) (1 - e \cos E) dE. \end{aligned}$$

Since

$$\frac{d}{dE} \cos(n(E - e \sin E)) = -n(1 - e \cos E) \sin(n(E - e \sin E)),$$

integration by parts, with vanishing boundary term, gives

$$c_n = \frac{e}{\pi n} \int_0^{2\pi} \cos E \cos(nE - ne \sin E) dE.$$

Using

$$J_m(t) = \frac{1}{2\pi} \int_0^{2\pi} \cos(mE - t \sin E) dE$$

and the product-to-sum formula, the integral divided by π is $J_{n-1}(ne) + J_{n+1}(ne)$. The Bessel recurrence

$$J_{n-1}(t) + J_{n+1}(t) = \frac{2n}{t} J_n(t)$$

therefore gives $c_n = 2J_n(ne)/n$ when $e > 0$; the case $e = 0$ follows directly, or by continuity. This proves the stated value of every Fourier coefficient. For fixed $e < 1$, the map $E \mapsto E - e \sin E$ has nonzero derivative on the real axis. The analytic inverse-function theorem, applied uniformly on the compact quotient $E \bmod 2\pi$, extends its inverse to a strip $|\Im M| < \eta_e$. Hence the periodic analytic function $E(M) - M$ has Fourier coefficients $O(e^{-\eta_e n})$ by shifting the coefficient contour to the boundary of any smaller strip. Its Fourier series therefore converges absolutely and uniformly, proving (32.12) rather than merely identifying its formal coefficients. $\square$

32.4 Triangular coefficient recurrences

Let $A(z) = \sum_{n \geq 0} a_n z^n$, $B(z) = \sum_{n \geq 0} b_n z^n$, and $C(z) = \sum_{n \geq 0} c_n z^n$. The following recurrences determine the result through degree N using only lower coefficients.

Proposition 32.6 (Product, reciprocal, and quotient). *If $C = AB$, then*

$$c_n = \sum_{j=0}^n a_j b_{n-j}. \quad (32.13)$$

If $a_0 \neq 0$ and $C = 1/A$, then

$$c_0 = a_0^{-1}, \quad c_n = -a_0^{-1} \sum_{j=1}^n a_j c_{n-j}. \quad (32.14)$$

If $C = B/A$, then

$$c_n = a_0^{-1} \left(b_n - \sum_{j=1}^n a_j c_{n-j} \right). \quad (32.15)$$

Proof. Take the coefficient of z^n in $C = AB$, $AC = 1$, and $AC = B$, respectively, and isolate the term containing c_n . $\square$

Proposition 32.7 (Logarithm, exponential, and arbitrary powers). *Assume $a_0 = 1$. If $L = \log A = \sum_{n \geq 1} \ell_n z^n$, then*

$$\ell_n = a_n - \frac{1}{n} \sum_{j=1}^{n-1} j \ell_j a_{n-j}. \quad (32.16)$$

Conversely, if $A = e^L$, then

$$a_0 = 1, \quad a_n = \frac{1}{n} \sum_{j=1}^n j \ell_j a_{n-j}. \quad (32.17)$$

For $C = A^\alpha$,

$$c_0 = 1, \quad c_n = \frac{1}{n} \sum_{j=1}^n ((\alpha + 1)j - n) a_j c_{n-j}. \quad (32.18)$$

Proof. The identity $A' = AL'$ gives $na_n = \sum_{j=1}^n j \ell_j a_{n-j}$. Isolating ℓ_n proves (32.16); isolating a_n proves (32.17). For powers, logarithmic differentiation gives $AC' = \alpha A'C$. The coefficient of z^{n-1} is

$$\sum_{j=0}^{n-1} (n-j) a_j c_{n-j} = \alpha \sum_{j=1}^n j a_j c_{n-j}.$$

Isolating nc_n yields (32.18). $\square$

Proposition 32.8 (Composition and reversion). *Suppose $B(0) = 0$. Then*

$$[z^n]A(B(z)) = \sum_{k=0}^n a_k \widehat{\mathcal{B}}_{n,k}(b_1, \dots, b_{n-k+1}). \quad (32.19)$$

If $A(0) = 0$, $a_1 \neq 0$, and $B = A^{(-1)}$, then

$$b_1 = a_1^{-1}, \quad (32.20)$$

$$b_n = -a_1^{-1} \sum_{k=2}^n a_k \widehat{\mathcal{B}}_{n,k}(b_1, \dots, b_{n-k+1}), \quad n \geq 2. \quad (32.21)$$

Proof. Equation (32.19) is the ordinary composition theorem (20.3). Set $A(B(z)) = z$. The coefficient of z gives $a_1 b_1 = 1$. For $n \geq 2$, the coefficient vanishes and its $k = 1$ term is $a_1 b_n$; every term with $k \geq 2$ depends only on $b_1, \dots, b_{n-1}$. Isolating b_n proves (32.21). $\square$

32.5 Newton iteration and precision doubling

Triangular recurrences are transparent and excellent for symbolic work. With fast polynomial multiplication, Newton iteration is asymptotically faster at very high truncation orders.

Theorem 32.9 (Reciprocal Newton iteration). *Suppose $A(0)$ is invertible and C_m satisfies $AC_m \equiv 1 \pmod{z^m}$. Define*

$$C_{2m} = C_m(2 - AC_m) \pmod{z^{2m}}. \quad (32.22)$$

Then $AC_{2m} \equiv 1 \pmod{z^{2m}}$.

Proof. Write $AC_m = 1 - E$ with $E = O(z^m)$. Then

$$AC_m(2 - AC_m) = (1 - E)(1 + E) = 1 - E^2,$$

and $E^2 = O(z^{2m})$. $\square$

Theorem 32.10 (Newton reversion). *Let $A(0) = 0$, $A'(0)$ invertible, and suppose $A(B_m(z)) \equiv z \pmod{z^m}$. Put*

$$B_{2m} = B_m - \frac{A(B_m) - z}{A'(B_m)} \pmod{z^{2m}}. \quad (32.23)$$

Then $A(B_{2m}) \equiv z \pmod{z^{2m}}$.

Proof. Let $E = A(B_m) - z = O(z^m)$ and $\Delta = -E/A'(B_m) = O(z^m)$. Formal Taylor expansion gives

$$A(B_m + \Delta) = A(B_m) + A'(B_m)\Delta + O(\Delta^2) = z + O(z^{2m}).$$

The reciprocal of $A'(B_m)$ exists because its constant term is $A'(0)$. This proves precision doubling. $\square$

Remark 32.11 (Complexity). With schoolbook multiplication, filling all triangular recurrences through order N costs between $O(N^2)$ and $O(N^3)$, depending on whether Bell polynomials are reused. Newton iteration reduces reciprocal, logarithm, exponential, and reversion to a logarithmic number of truncated multiplications and compositions. The algebraic identities above are independent of any particular software package or coefficient representation.

Part VIII

Archive Concordance and Reference Record

Chapter 33

Concordance with the source archive

The supplied package contains five substantial LaTeX drafts in `articles/`, together with encyclopedic reference dossiers in `Wikipedia/` and `MathWorld/` [26, 27]. The drafts are not five successive chapters: they are parallel treatments with a large common core and different strengths. This synthesis therefore follows mathematical dependence rather than file order. Repeated definitions and proofs have been deduplicated; distinct examples, algorithms, geometric interpretations, and special-function extensions have been retained and reconciled. In the tables below, “normalized” means algebraically equivalent after applying the notation and boundary conventions fixed in the opening chapters; “corrected” means that an index, hypothesis, Jacobian, convergence domain, or undefined placeholder has been repaired and the repaired statement proved.

33.1 The five draft manuscripts

Draft	Principal strengths retained	Role in the unified article
<code>Combinatorial_Formulae_and_Inversion_Theorems.tex</code>	Foundational proofs for factorial bases, Stirling inversion, Bell-polynomial composition, Faà di Bruno, and Lagrange inversion; careful distinction between formal and analytic arguments.	Supplies proof details and normalization checks throughout Chapters 1–28.
<code>Formulae_Compendium.tex</code>	Compact cross-family formula tables, explicit reversion examples, higher product and quotient rules, and useful coefficient recurrences.	Its tables have been converted into derived theorems and algorithms rather than repeated as an independent catalogue; see especially Chapters 25, 28, and 32.
<code>Formulae_Monograph_Package/Formulae_Monograph.tex</code>	Broad special-function coverage, restricted partition variants, asymptotic methods, and extended worked examples.	Provides much of the breadth in Chapters 7, 8, 30, and 31.

Continued on the next page

Draft	Principal strengths retained	Role in the unified article
Formulae_Monograph_Source_and_PDF/Formulae_Monograph.tex	Geometric and polyhedral interpretations, computational observations, and a second independent audit of conventions and boundary cases.	Its nonduplicate material is incorporated in the Eulerian, Catalan, and associahedral chapters, while duplicated exposition has been collapsed into the common proofs.
combinatorial_coefficient_calculus.tex	The most coherent global organization: formal series, the Stirling–Bell–Eulerian chain, composition calculus, inversion, and an archive concordance.	Serves as the structural backbone. Every inherited result was nevertheless checked against the other four drafts, and the new factorial, umbral, multivariate, special-function, and algorithmic parts were integrated into its dependency order.

33.2 The supporting topic dossiers

The Wikipedia snapshots supply detailed topic inventories and historical or combinatorial interpretations; the MathWorld dossier adds Bernoulli, Euler, Genocchi, Cauchy, Pochhammer, gamma/Barnes, Darboux, Euler–Maclaurin, multivariate inversion, and associahedral material. These have not been copied as isolated encyclopedia entries. They have been recast in the common coefficient language and supplied with proofs in Chapters 25–32. The table that follows records the absorption of the principal Wikipedia topic files.

Source file	Topics represented	Location and status
Associahedron.txt/Associahedron.pdf	Stasheff polytopes, equivalent realizations, polygon dissections and triangulations, the Tamari order, face enumeration, and appearances in homotopy theory and moduli spaces.	Chapters 22 and 29. The polygon-dissection model is used to derive the complete face vector and the Narayana h -vector, with dimensions and index shifts stated explicitly.
Catalan_number.txt/Catalan_number.pdf	The Catalan recurrence and closed form; binary trees, Dyck paths, parenthesizations, triangulations, and noncrossing structures; generating functions, convolution, and asymptotic growth.	Chapters 22 and 29, together with the Darboux example in Chapter 31. Lagrange inversion proves the closed form, the dissection chapter organizes the bijections, and Stirling–Darboux analysis supplies a controlled full asymptotic expansion.

Continued on the next page

Source file	Topics represented	Location and status
Bell_number.txt	Set partitions and equivalence relations; squarefree factorizations; rhyme schemes and restricted-growth words; the card-shuffle and pattern-avoidance interpretations; the Bell–Aitken triangle; binomial and Stirling sums; Spivey-type identities; the exponential generating function and its differential equation; Dobiński’s formula and Poisson moments; Touchard congruences and modular periods; contour representation; inequalities, log-convexity, saddle-point growth, and Bell-prime computation.	Chapters 9–11. The many uses of the letter B are separated. Finite lists of computed primes and periods are treated as reproducible computations, while symbolic claims are proved. The source’s unnamed higher saddle coefficients are replaced by a defined recursive saddle expansion.
Bell_polynomials.txt	Exponential partial and complete Bell polynomials; ordinary Bell polynomials; weighted-partition interpretation and low-degree tables; generating functions; pointing and convolution recurrences; derivatives and scaling; Stirling, Lah, Touchard, and symmetric-function specializations; exponential–logarithmic inversion; Hessenberg determinants; moments and cumulants; cycle indices; Hermite polynomials; binomial-type sequences; Faà di Bruno, inverse-series, and Laplace applications.	Chapters 12–15 and 19–24. All conventions are reconciled through one coefficient formula. The quadratic differential recurrence is proved by translating every derivative back to the master generating function.
Eulerian_number.txt	Descents and ascents; recurrence, symmetry, explicit formula, and exponential generating function; Eulerian polynomials; Worpitzky’s identity and power sums; negative polylogarithms; conversion to the Stirling basis; the Irwin–Hall integral; alternating sums and Bernoulli numbers; cube, hypersimplex, and permutohedron interpretations; type-B and second-order Eulerian numbers.	Chapters 16–18. The type-B descent statistic is indexed at positions $0, 1, \dots, n-1$ with $\pi_0 = 0$; this corrects the one-step shift in the archived formula. The second-order interpolation formula is proved from its differential recurrence and generating function.

Continued on the next page

Source file	Topics represented	Location and status
Fa#U00e0_di_Bruno's_formula.txt	Set-partition, multiplicity, and Bell-polynomial forms of the higher chain rule; worked low-order derivatives; the exponential specialization; mixed partial derivatives and multivariate composition.	Chapter 19. All forms follow from the same labelled-partition count, and the multivariate formula is proved by distributing derivative labels among inner components. The filename's encoded accent is rendered throughout as "Faà di Bruno."
Lagrange_inversion_theorem.txt	Formal inversion by residues; the Lagrange–Bürmann coefficient formula; analytic local inversion; coefficient expressions through Bell polynomials; algebraic examples, Lambert W , Catalan and Fuss–Catalan series, and the rooted-tree/associahedral interpretation.	Chapters 21–22. Formal and analytic hypotheses are stated separately. Coefficient extraction, residues, and contour integrals are shown to be three realizations of the same identity.
Lagrange_reversion_theorem.txt	The shifted two-parameter implicit equation; reversion coefficients and their derivatives; the delta-function derivation.	Chapter 23. The delta argument is decoded as a change-of-variables statement: its Jacobian and orientation/absolute-value issue are made explicit rather than hidden in symbolic delta notation.
Stirling_number.txt	Unified first- and second-kind notation; falling and rising factorial bases; inverse triangular matrices; Lah numbers; Stirling transforms; finite differences; differential-operator normal ordering; symmetric-function descriptions; extensions to negative indices.	Chapters 2–6. Signed and unsigned first-kind numbers are never conflated, and the boundary convention makes every matrix identity valid without exceptional edge cases.
Stirling_numbers_of_the_first_kind.txt	Permutation cycles and recurrences; tables and fixed- k formulas; harmonic and near-diagonal identities; finite sums and convolutions; congruences; ordinary and exponential generating functions; explicit double sums; real-rootedness and asymptotics; zeta and Hurwitz-zeta sums; convergent expansions for $\log \Gamma$, the digamma function, and Stieltjes constants; generalized factorial coefficients.	Chapters 2–8. Near-diagonal partition formulas are derived by Lagrange inversion. The special-function series are supplied with their actual convergence domains and justified by Binet/Hermite integral transforms.

Continued on the next page

Source file	Topics represented	Location and status
Stirling_numbers_of_the_second_kind.txt	Set partitions, surjections, and recurrences; Bell-number relations; explicit inclusion–exclusion formula; parity; rational and exponential generating functions; finite identities, bounds, saddle asymptotics, row maxima and log-concavity; Poisson and fixed-point moments; rhyme schemes; r -Stirling, associated, and reduced variants.	Chapters 2–11. The two backward recurrences presented tentatively in the source are proved exactly by coefficient extraction. Strict log-concavity is obtained from real-rooted Touchard polynomials, and reduced numbers are proved through a chromatic-polynomial argument.

33.2.1 The MathWorld dossier

The following grouped inventory records every MathWorld topic file in the archive. A title followed by an accompanying .nb notebook is listed only once; the notebook is retained in the archive as provenance and as an independently inspectable symbolic source, while all identities used here are derived in the text.

Topic group	MathWorld files represented	Absorption into the monograph
Factorial and classical polynomial calculus	<i>Pochhammer Symbol</i> ; <i>Bernoulli Number</i> ; <i>Bernoulli Polynomial</i> ; <i>Bernoulli Polynomial of the Second Kind</i> ; <i>Euler Polynomial</i> ; <i>Genocchi Number</i> ; and <i>Modified Bernoulli Number</i> .	Chapters 25 and 26. Their generating functions, difference equations, reflection laws, multiplication formulas, and conversions to factorial bases are proved in one umbral framework.
Bell and Dobiński material	<i>Bell Number</i> ; <i>Bell Polynomial</i> ; <i>Dobiński's Formula</i> ; and <i>Complementary Bell Number</i> .	Chapters 9–15 and 26. Set-partition derivations, exponential generating functions, moment representations, Bell-polynomial recurrences, and the complementary sequence are connected without conflating their several conventional uses of B_n .
Stirling and Catalan families	<i>Stirling Number of the First Kind</i> ; <i>Stirling Number of the Second Kind</i> ; and <i>Catalan Number</i> .	Chapters 2–8, 21–22, and 29. The Stirling triangles are treated as inverse changes of factorial basis; Catalan numbers are then obtained by reversion and refined by Narayana and polygon-dissection counts.

Continued on the next page

Topic group	MathWorld files represented	Absorption into the monograph
Gamma hierarchy	<i>Digamma Function</i> ; <i>Polygamma Function</i> ; and <i>Barnes G-Function</i> .	Chapter 30. Weierstrass products and logarithmic differentiation yield the gamma hierarchy; Alexeiewsky's integral and Euler–Maclaurin summation yield the Barnes functional equation, integer values, and asymptotic expansion.
Summation and asymptotic transfer	<i>Euler–Maclaurin Integration Formulas</i> and <i>Darboux's Formula</i> .	Chapter 31. Periodic Bernoulli functions give an exact remainder, and local singular expansions are transferred to coefficients with explicit error control; harmonic and Catalan expansions serve as worked tests.

33.3 Formula-status ledger

The displayed identities audited across the five drafts and supporting topic files fall into four editorial classes.

1. *Exact identities* are reproduced, usually in a notation that prevents collisions, and proved by a bijection, coefficient extraction, finite differences, matrix inversion, or a generating-function calculation.
2. *Equivalent normal forms* are consolidated. For example, the multiplicity sum, Bell-polynomial identity, and set-partition form of Faà di Bruno's formula are one theorem viewed in three coordinate systems.
3. *Corrected statements* retain the source's intended content but repair an index shift, a missing range or analyticity hypothesis, or a suppressed Jacobian. The type-B Eulerian and delta-function reversion formulas are the main examples.
4. *Schematic or computational statements* are replaced by something checkable. Undefined asymptotic placeholders become recursively defined saddle coefficients; finite tables and modular searches come with exact recurrences or certificate procedures.

The resulting distinction is important: a theorem is not silently weakened into numerical evidence, and a numerical observation is not silently promoted to a theorem.

33.4 Implementation-neutral algorithms

The transient software lists in encyclopedic source material are best replaced by algorithms whose correctness follows from the proved recurrences. To compute all entries through row N , initialize the boundary values and fill either Stirling triangle by its two-term recurrence; this uses $O(N^2)$ arithmetic operations and $O(N)$ memory if only the preceding row is retained. Bell numbers are then row sums of the second-kind triangle, or may be produced by the binomial recurrence. Eulerian numbers admit the same triangular complexity. Partial Bell polynomials can be filled by the pointing

recurrence, with exact polynomial or symbolic-series arithmetic as appropriate. For inverse series, Lagrange–Bürmann gives each target coefficient independently from a truncated power of the defining series; Newton iteration is asymptotically faster for very high orders, while the Lagrange form is usually more transparent for symbolic coefficients. Every one of these algorithms is valid over any coefficient ring in which the displayed divisions make sense.

33.5 Dependency map

The logical flow of the monograph may be summarized as

factorial bases and finite differences $\longrightarrow$ Stirling transforms
 $\longrightarrow$ Bell partition calculus $\longrightarrow$ composition and differentiation.

with Eulerian numbers providing the parallel change of coordinates between powers and shifted binomial coefficients. Formal residues then turn implicit substitution into Lagrange inversion. This dependency map explains why identities that appear unrelated in the separate source files repeatedly reduce to the same three operations: change of basis, labelled decomposition, and coefficient extraction.

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- [28] *Formulae-2.zip*, the source package supplied for this project: five substantial LaTeX drafts in `articles/`, reference snapshots in `wikipedia/`, and MathWorld PDF/notebook materials in `MathWorld/`.

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